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(54) Title: METHODS OF PREDICTING CLINICAL COURSE AND TREATING MULTIPLE SCLEROSIS

(57) Abstract: Provided are methods and kits for classifying a subject as being more likely to have benign multiple sclerosis (BMS) or as being more likely to have typical relapsing remitting multiple sclerosis (RRMS). Classification of multiple sclerosis disease course is performed by comparing a level of expression of at least one gene involved in the RNA polymerase I pathway in a cell of the subject to a reference expression data of said at least one gene obtained from a cell of at least one subject pre-diagnosed as having BMS and/or from a cell of at least one subject pre-diagnosed as having typical RRMS, thereby classifying the subject as being more likely to have BMS or as being more likely to have typical RRMS. Also provided are methods of diagnosing and treating multiple sclerosis and methods of monitoring treatment efficiency.



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METHODS OF PREDICTING CLINICAL COURSE AND TREATING MULTIPLE SCLEROSIS

5 FIELD AND BACKGROUND OF THE INVENTION

The present invention, in some embodiments thereof, relates to methods of classifying a subject as having benign multiple sclerosis or typical relapsing remitting multiple sclerosis and, more particularly, but not exclusively, to methods of treating multiple sclerosis based on same.

10 Multiple sclerosis (MS) is the most common demyelinating disease of the central nervous system (CNS) affecting young adults (disease onset between 20 to 40 years of age) and is the third leading cause for disability after trauma and rheumatic diseases, with an estimated annual cost 34,000 USD per patient (total life time cost of 2.2 million USD per patient).

15 The disease is characterized by destruction of myelin, associated with death of oligodendrocytes and axonal loss. The main pathologic finding in MS is the presence of infiltrating mononuclear cells, predominantly T lymphocytes and macrophages, which surpass the blood brain barrier and induce an active inflammation within the brain and spinal cord. The neurological symptoms that characterize MS include complete or
20 partial vision loss, diplopia, sensory symptoms, motor weakness that can worsen to complete paralysis, bladder dysfunction and cognitive deficits, which eventually may lead to a significant disability. The associated multiple inflammatory foci lead to myelin destruction, plaques of demyelination, gliosis and axonal loss within the brain and spinal cord and are the reasons contribute to the clinical manifestations of
25 neurological disability.

The etiology of MS is not fully understood. The disease develops in genetically predisposed subjects exposed to yet undefined environmental factors and the pathogenesis involves autoimmune mechanisms associated with autoreactive T cells against myelin antigens. It is well established that not one dominant gene determines
30 genetic susceptibility to develop MS, but rather many genes, each with different influence, are involved.

Clinically, in 85 % of MS patients the illness is initiated with a relapsing-remitting course (RRMS), and in about 10-15 % of MS patients have an a-priori

primary progressive course (PPMS) without relapses. RRMS is characterized by inflammatory attacks associated with neurological deficits with periods of remissions between the relapses that vary in time. After a period of 10 years, about 50 % of RRMS patients will progress to a secondary progressive MS (SPMS) course, characterized by permanent neurological dysfunction, with or without relapses and progressive disability.

Benign MS (BMS) is a clinical variant of RRMS in which the patients develop low neurological disability if at all after a disease duration of at least 10 years. Accordingly, this group of MS patients do not experience devastating accumulating disability over-time and when these patients are examined neurologically and scored by the Expanded Disability Status Scale (EDSS) they receive a score that is equal to or lower than 3.0. This low EDSS score signifies mild disability and when this low disability occurs more than 10 years after disease onset, the course of MS is defined as benign (Pittock SJ and Rodriguez M, 2008; Costelloe, L., et al., 2008). Prediction of patients that will have BMS is currently impossible and the definition of these patients is retrospective. The molecular events accountable for the BMS variant of disease are not understood.

Diterpenoid triepoxide Triptolide (TPT), isolated from the Chinese herb *Tripterygium wilfordii* (Leuenroth SJ and Crews CM. Triptolide-induced transcriptional arrest is associated with changes in nuclear substructure. *Cancer Res.* 2008; 68:5257-5266) has various anti-inflammatory effects (Liu Y, et al. Triptolide, a component of Chinese herbal medicine, modulates the functional phenotype of dendritic cells. *Transplantation.* 2007; 84:1517-1526), it modulates T-cell inflammatory responses and ameliorates Experimental Autoimmune Encephalomyelitis (Wang Y, et al. Triptolide modulates T-cell inflammatory responses and ameliorates experimental autoimmune encephalomyelitis. *J Neurosci Res.* 2008; 86:2441-2449). Derivatives of TPT were suggested for treating autoimmune diseases (EP 0983256, PCT/US1998/008562; WO9852933A1).

Cycloheximide, inhibits the phosphorylation of RRN3 and causes its dissociation from RNA polymerase I. RRN3 interacts with the rpa43 subunit of RNA polymerase I, and treatment with cycloheximide inhibits the formation of a RRN3/rpa43 complex *in vivo* (Alice H. Cavanaugh, et al., 2002. Rrn3 Phosphorylation is a

regulatory checkpoint for ribosome biogenesis J. Biol. Chem., 2002; 277: 27423 – 27432).

PCT Application No. PCT/IL2007/32856 discloses methods and kits for predicting prognosis of multiple sclerosis.

5 PCT Application No. PCT/IL2007/001617 discloses methods and kits for predicting the prognosis of a subject diagnosed with multiple sclerosis and methods of selecting a treatment regimen of a subject diagnosed with multiple sclerosis.

Achiron A, et al., 2007 (Clinical and Experimental Immunology, 149: 235-242) describe genes of the zinc-ion binding and cytokine activity regulation pathways which
10 predict outcome in relapsing–remitting multiple sclerosis.

Additional background art includes PCT Pub. No. WO03081201A2.

SUMMARY OF THE INVENTION

According to an aspect of some embodiments of the present invention there is
15 provided a method of classifying a subject as being more likely to have benign multiple sclerosis (BMS) or as being more likely to have typical relapsing remitting multiple sclerosis (RRMS), the method comprising comparing a level of expression of at least one gene involved in the RNA polymerase I pathway in a cell of the subject to a
20 reference expression data of the at least one gene obtained from a cell of at least one subject pre-diagnosed as having BMS and/or from a cell of at least one subject pre-diagnosed as having typical RRMS, thereby classifying the subject as being more likely to have BMS or as being more likely to have typical RRMS.

According to an aspect of some embodiments of the present invention there is provided a method of diagnosing a subject pre-diagnosed with multiple sclerosis (MS)
25 as having benign multiple sclerosis (BMS) or typical relapsing remitting multiple sclerosis (RRMS), the method comprising:

- (a) classifying the subject as being more likely to have BMS or as being more likely to have typical RRMS according to the method of claim 1,
 - (i) wherein when the subject is classified as being more likely to have the BMS
30 then the subject is diagnosed as having BMS;
 - (ii) wherein when the subject is classified as being more likely to have the typical RRMS, then the subject is diagnosed as having typical RRMS; and

(c) informing the subject of the diagnosis,
thereby diagnosing the subject pre-diagnosed with the MS as having the BMS or the typical RRMS.

According to an aspect of some embodiments of the present invention there is provided a method of monitoring an efficiency of an anti multiple sclerosis (MS) drug in treating a subject diagnosed with a typical relapsing remitting multiple sclerosis (RRMS) course, the method comprising:

(a) treating the subject with the anti MS drug; and
(b) comparing a level of expression of least one gene involved in the RNA polymerase I pathway in a cell of the subject following the treating with the anti MS drug to a level of expression of the at least one gene in a cell of the subject prior to the treating the subject with the anti MS drug,

(i) wherein a decrease above a predetermined threshold in the level of expression of the at least one gene following the treating with the anti MS drug relative to the level of expression of the at least one gene prior to the treating with the anti MS drug indicates that the anti MS drug is efficient for treating the subject;

(ii) wherein an increase above a predetermined threshold in the level of expression of the at least one gene following the treating with the anti MS drug relative to the level of expression of the at least one gene prior to the treating with the anti MS drug indicates that the anti MS drug is not efficient for treating the subject; or

(iii) wherein when a level of expression of the at least one gene following the treating with the anti MS drug is identical or changed below a predetermined threshold as compared to prior to the treating with the anti MS drug then the treatment is not efficient for treating the subject.

thereby monitoring the efficiency of the anti multiple sclerosis (MS) drug in treating the subject diagnosed with the typical RRMS course.

According to an aspect of some embodiments of the present invention there is provided an *in vitro* method of predicting an efficiency of an anti multiple sclerosis (MS) drug for treatment of a subject diagnosed with a typical relapsing remitting multiple sclerosis (RRMS), the method comprising:

(a) contacting cells of the subject with a therapeutically effective amount of the anti MS drug; and

(b) comparing a level of expression in the cells of at least one gene involved in the RNA polymerase I pathway following the contacting with the anti MS drug to a level of expression of the at least one gene in the cells prior to the contacting with the anti MS drug,

5 (i) wherein a decrease above a predetermined threshold in the level of expression of the at least one gene following the contacting with the anti MS drug relative to the level of expression of the at least one gene prior to the contacting with the anti MS drug indicates that the treatment is efficient for treating the subject;

10 (ii) wherein an increase above a predetermined threshold in the level of expression of the at least one gene following the contacting with the anti MS drug relative to the level of expression of the at least one gene prior to the contacting with the anti MS drug indicates that the treatment is not efficient for treating the subject; or

(iii) wherein when a level of expression of the at least one gene following the contacting with the anti MS drug is identical or changed below a predetermined
15 threshold as compared to prior to the contacting with the anti MS drug then the treatment is not efficient for treating the subject.

thereby predicting the efficiency of the anti MS drug for treatment of the subject diagnosed with the typical RRMS.

According to an aspect of some embodiments of the present invention there is
20 provided a method of treating a subject diagnosed with multiple sclerosis, the method comprising

(a) classifying the subject as being more likely to have BMS or typical RRMS according to the method of claim 1,

(b) selecting a treatment regimen based on classification results of step (a);
25 thereby treating the subject diagnosed with multiple sclerosis.

According to an aspect of some embodiments of the present invention there is provided a method of treating a subject diagnosed with multiple sclerosis, the method comprising:

(a) diagnosing a typical relapsing remitting multiple sclerosis (RRMS) according
30 to the method of claim 2,

(b) administering to the subject a therapeutically effective amount of diterpenoid triepoxide Triptolide (TPT) or a derivative thereof, thereby treating the subject.

5 According to an aspect of some embodiments of the present invention there is provided a probeset comprising a plurality of oligonucleotides and no more than 50 oligonucleotides, wherein an oligonucleotide of the plurality of oligonucleotides specifically recognizes a polynucleotide of at least one gene involved in the RNA polymerase pathway.

10 According to an aspect of some embodiments of the present invention there is provided a kit for classifying a disease course in a subject diagnosed with multiple sclerosis (MS), comprising the probeset of claim 7.

According to an aspect of some embodiments of the present invention there is provided a method of selecting a drug for treating a typical relapsing remitting multiple sclerosis (RRMS) in a subject, the method comprising:

15 contacting cells of a subject classified as having a typical RRMS with a plurality of drug molecules,

identifying at least one drug molecule which downregulates a level of expression of at least one gene involved in the RNA polymerase I pathway, the at least one drug molecule is suitable for treating the typical RRMS in the subject,

20 thereby selecting the drug for treating the typical RRMS in the subject.

According to some embodiments of the invention, a decrease above a predetermined threshold in the level of expression of the at least one gene in the cell of the subject relative to the reference expression data of the at least one gene obtained from the at least one subject having the typical RRMS classifies the subject as being
25 more likely to have the BMS.

According to some embodiments of the invention, an increase above a predetermined threshold in the level of expression of the at least one gene in the cell of the subject relative to the reference expression data of the at least one gene obtained from the at least one subject having the BMS classifies the subject as being more likely
30 to have the typical RRMS.

According to some embodiments of the invention, a level of expression of the at least one gene in the cell of the subject is identical or changed below a predetermined

threshold as compared to the reference expression data of the at least one gene obtained from the at least one subject having the BMS, then the subject is classified as being more likely to have the BMS.

According to some embodiments of the invention, a level of expression of the at least one gene in the cell of the subject is identical or changed below a predetermined threshold as compared to the reference expression data of the at least one gene obtained from the at least one subject having the typical RRMS, then the subject is classified as being more likely to have the typical RRMS.

According to some embodiments of the invention, when the subject being more likely to have typical RRMS then the treatment regimen comprises administering to the subject an agent which downregulates the level of expression of the at least one gene involved in the RNA polymerase I pathway.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway is selected from the group consisting of POLR1D, LRPPRC, RRN3 and NCL.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises the POLR1D, LRPPRC, RRN3 and NCL genes.

According to some embodiments of the invention, the agent is selected from the group consisting of an siRNA, an antisense, an antibody and a small molecule.

According to some embodiments of the invention, the small molecule is Cycloheximide.

According to some embodiments of the invention, the at least one gene is RRN3, and whereas the downregulating is effected using diterpenoid triepoxide Triptolide (TPT) or a derivative thereof.

According to some embodiments of the invention, the at least one gene is RRN3, and whereas the downregulating is effected using Cycloheximide.

According to some embodiments of the invention, the kit further comprising a positive control for an expression level of the at least one gene involved in the RNA polymerase pathway.

According to some embodiments of the invention, each of the plurality of oligonucleotides is bound to a solid support.

According to some embodiments of the invention, the plurality of oligonucleotides are bound to the solid support in an addressable location.

According to some embodiments of the invention, the level of expression is determined using an RNA detection method.

5 According to some embodiments of the invention, the level of expression is determined using a protein detection method.

According to some embodiments of the invention, the cell is a blood cell.

According to some embodiments of the invention, the method further comprising administering to the subject a therapeutically effective amount of an anti
10 MS agent.

According to some embodiments of the invention, the anti-MS agent is selected from the group consisting of Diterpenoid triepoxide Triptolide (TPT), Adderall, Ambien, Avonex, Baclofen, Beta interferon, Betaseron, Celexa, Clonazepam, Copaxone, Corticosteroids, Cymbalta, Cytosan, Dexamethasone, Effexor, Elavil,
15 Gabapentin, Hydrocodone, Lexapro, Lyrica, Mitoxantrone, Naltrexone, Neurontin, Novantrone, Prednisone, Provigil, Rebif, Solumedrol, Symmetrel, Topamax, Tysabri, Wellbutrin, Xanax, Zanaflex, Zolof, fingolimod, laquinimod, Mylinax (cladribine), teriflunomide, BG-12 (Biogen Idec's), firtagrat (GSK/Mitsubishi Tanabe Pharma), ibudilast (MediciNova's), and CDP323 (Biogen/UCB).

20 According to an aspect of some embodiments of the present invention there is provided a method of predicting a benign multiple sclerosis course in a subject diagnosed with multiple sclerosis, the method comprising: (a) determining in a biological sample of the subject a level of expression of at least one polynucleotide selected from the group consisting of C22orf8, TLK1, HNRPH1, PLXDC1, TLK1,
25 PKN2, ALS2CR8, FLJ12547, ZNF238, PDPR, NT5E, PASK, HPGD, IL6ST, JARID1A, PASK, LEF1, FLJ10246, MTUS1, FLJ14011, VSIG4, MARCH-VI, FLJ10613, EWSR1, ATP8A1, SLC4A7, FLJ21127, HNRPH1, ABLIM1, ITGA6, ADCY9, CROCC, SH3YL1, SMA4, SPTBN1, DPEP3, PDE3B, AF5Q31, NRCAM, DOCK9, IPW, FLJ20152, SIRPB2, GALNT4, CD28, TXK, ETS1, DGCR5, ZNF192,
30 TCF7, CAMK4, SIM2, MGEA5, TGFBR2, RET, MAPK8IP3, RRN3, DKFZp547H025, FBXW11, ZNF423, DLG1, MGC17330, CD164L1, REPS1, ACHE, ITGB1BP2, LOC94431, LTK, RUNX1, EVER1, KIAA2010, CEACAM7, STX16, SLC4A5,

CRTAP, RECQL5, MAGEF1, VIPR1, FLJ10979, TTC3, CRSP2, BAZ2A, GTF2I, MGC50853, KIAA0508, BPHL, LTBP4, FN3KRP, SCARB1, MGC17330, HYAL4, DGKA, FLJ11196, DHRS6, EPHB4, IDI2/GTPBP4, SNTG2, SLC7A6, PMS2L2, KIAA0436, TOSO, THRAP3, T3JAM, LOC283232, LOC92482, PTER, ATM, NUCB2, PIK3R2, MGC1136, CD59, JARID1A, FLJ39616, ABLIM1, PBP, MAPK8IP3, FTS, LHX5, TNFRSF7, MYC, PBXIP1, DATF1, HTF9C, PUS1, KIAA0924, C6orf4, KIAA0372, WDR42A, CRYZL1, TERE1, LTBP4, TTC3, NFATC1, POM121/LOC340318, TOSO, LOC348926/MGC16279/SB153/FLJ10661, SPOCK2, KIAA0515, SLC37A4, CD44, SMARCA2, SPTBN1, C6orf130, TTC3, DLG1, SLC35E2, MCCC1, PMS2L11, RCN3, STX16, FLJ20618, STAT5B, SMARCA2, SATB1, POLR1D, ASXL1, REV1L, PMS2L2/PMS2L5, FLJ12355, CCNB1IP1, FLJ12270, KIAA0692, MCM7, GPSN2, STX16, MMS19L, GTF2I/GTF2IP1, AKAP7, ZNF444, SLC35A3, MGEA5, RUTBC3, C20orf36, RAD17, ALG12, LOC112869, C6orf48, CUTC, LGTN, DEF6, WAC, HNRPH3, NS, KIAA0892, LRPPRC, HMG20A, DDX42, TINP1, ZDHHC17, C19orf2, EIF4B, LOC376745, DKFZP434C171, TH1L, C19orf13, RPL22, PHF15, EWSR1, EIF4B, FAM48A, YT521, NEK9, EIF3S7, RPS6, RPL35A, EEF2, RPL3, RPS6, UBA52, RPL6, RPS6, RPL13, AL353949, AL580863, AF052160, AW128846, AW974481, N92920, BG178274, AW303460, BF057458, AL050035, M59917, AK025422, AI693985, AU158442, AK021460, AL023773, NM_003790, AC005011, M90355, AL353580, U38964, D50683 and BE967207, wherein downregulation below a predetermined threshold in the level of expression relative to a level of expression of the at least one polynucleotide in a biological sample of a subject diagnosed with typical relapsing-remitting multiple sclerosis (RRMS) is indicative of the prediction of the benign multiple sclerosis course of the subject; (b) informing the subject of the prediction of the benign multiple sclerosis course; thereby predicting the benign multiple sclerosis course in the subject diagnosed with the multiple sclerosis.

According to an aspect of some embodiments of the present invention there is provided a method of predicting a benign multiple sclerosis course in a subject diagnosed with multiple sclerosis, the method comprising: (a) determining in a biological sample of the subject a level of expression of at least one polynucleotide selected from the group consisting of YWHAB, ATP6V1E1, UBB, MRLC2/MRCL3,

UQCR, MRLC2, RTN4, UBE2A, RTN4, WDR1, PSMA6, C14orf123, PP1201, TBK1, CAST, CAST, RSN, PSME1, SDF2, GSTO1, CAST, DNCL1, SQRDL, ADIPOR2, ICMT, NDUFA6, NDUFA6, COX17, HIF1AN, FLJ20257, TBPL1, RAPGEF2, CRSP8, APOL1, PAOX, CNDP2, ETFA, DPP3, KPNA1, MGC3036, TUBB2, PDCL, CCL5, CDS2, RAP1GDS1, ATP6V1D, OBRGRP/LEPR, SF4, GCLC, MGST3, BICD2, BRF1, CHST12, EXOSC7, TOR1B, ZFP95, ILK, UNC13A, MTHFD2, CASP10, FLJ45850, CMRF-35H, ARF3, NDOR1, DUSP10, AP1M2, VRK2, GSN, PTRF, RBM19, RABGAP1L, ATP5S, STOM, TFPI2, SLCO3A1, PTPN12, CSF1, SIGLEC6, KIRREL, OBRGRP, TP53AP1, SUHW1, NUP98, IL15RA, MICB, CMRF-35H, SPHK1, TNFRSF6, FLJ11301, LRP5, STOM, EPHA2, SRC, FLJ11301, PSTPIP2, EBP, MCPH1, PTPRF, LIMK2, FSTL4, CBR1, MGC2654, MYCT1, NOL3, MITF, ATP10B, FBXO31, TBX21, LSS, SLC17A3, MNAB, CHPPR, GIF, VAMP5, ABCG2, KIF1B, LOH11CR2A, NID2, RBBP8, ETV7, CTSL, RUFY1, RSU1, PARD3, APOB, ACOX3, DAB2, LDLR, TJP2, GNAS, PARD3, NCKAP1, TAP2, HDGFRP3, LDLR, PIK3R3, HTR2B, GAS2L1, FER1L3, C3orf14, TP53TG3, LEPR, CLIC5, PDE4DIP, ATP9A, ITGB1BP1, INDO, SELP, FHL2, FER1L3, EGF, SIAT8A, HDGFRP3, LRAP, VWF, FLJ10134, IMP-3, DMN, MCTP1, FSTL1, CTNNAL1, RAB27B, THBS1, PROS1, MMRN1, CTTN, AL078596, AI148659, U00956 and M29383, wherein upregulation above a predetermined threshold in the level of expression relative to a level of expression of the at least one polynucleotide in a biological sample of a subject diagnosed with typical relapsing-remitting multiple sclerosis (RRMS) is indicative of the prediction of the benign multiple sclerosis course of the subject; (b) informing the subject of the prediction of the benign multiple sclerosis course; thereby predicting the benign multiple sclerosis course in the subject diagnosed with the multiple sclerosis.

According to an aspect of some embodiments of the present invention there is provided a method of monitoring treatment with an anti MS drug in a subject in need thereof, the method comprising: (a) treating the subject with the anti MS drug; and (b) determining in a biological sample of the subject a level of expression of at least one polynucleotide selected from the group consisting of YWHAB, ATP6V1E1, UBB, MRLC2/MRCL3, UQCR, MRLC2, RTN4, UBE2A, RTN4, WDR1, PSMA6, C14orf123, PP1201, TBK1, CAST, CAST, RSN, PSME1, SDF2, GSTO1, CAST, DNCL1, SQRDL, ADIPOR2, ICMT, NDUFA6, NDUFA6, COX17, HIF1AN,

FLJ20257, TBPL1, RAPGEF2, CRSP8, APOL1, PAOX, CNDP2, ETFA, DPP3, KPNA1, MGC3036, TUBB2, PDCL, CCL5, CDS2, RAP1GDS1, ATP6V1D, OBRGRP/LEPR, SF4, GCLC, MGST3, BICD2, BRF1, CHST12, EXOSC7, TOR1B, ZFP95, ILK, UNC13A, MTHFD2, CASP10, FLJ45850, CMRF-35H, ARF3, NDOR1, DUSP10, AP1M2, VRK2, GSN, PTRF, RBM19, RABGAP1L, ATP5S, STOM, TFPI2, SLCO3A1, PTPN12, CSF1, SIGLEC6, KIRREL, OBRGRP, TP53AP1, SUHW1, NUP98, IL15RA, MICB, CMRF-35H, SPHK1, TNFRSF6, FLJ11301, LRP5, STOM, EPHA2, SRC, FLJ11301, PSTPIP2, EBP, MCPH1, PTPRF, LIMK2, FSTL4, CBR1, MGC2654, MYCT1, NOL3, MITF, ATP10B, FBXO31, TBX21, LSS, SLC17A3, MNAB, CHPPR, GIF, VAMP5, ABCG2, KIF1B, LOH11CR2A, NID2, RBBP8, ETV7, CTSL, RUFY1, RSU1, PARD3, APOB, ACOX3, DAB2, LDLR, TJP2, GNAS, PARD3, NCKAP1, TAP2, HDGFRP3, LDLR, PIK3R3, HTR2B, GAS2L1, FER1L3, C3orf14, TP53TG3, LEPR, CLIC5, PDE4DIP, ATP9A, ITGB1BP1, INDO, SELP, FHL2, FER1L3, EGF, SIAT8A, HDGFRP3, LRAP, VWF, FLJ10134, IMP-3, DMN, MCTP1, FSTL1, CTNNAL1, RAB27B, THBS1, PROS1, MMRN1, CTTN, AL078596, AI148659, U00956 and M29383, wherein an upregulation above a predetermined threshold in the level of expression relative to a level of expression of the at least one polynucleotide in a biological sample of a subject diagnosed with typical relapsing-remitting multiple sclerosis (RRMS) is indicative of treatment efficacy.

According to some embodiments of the invention, step (b) is effected also prior to step (a) and wherein the upregulation is with respect to a level of the at least one polynucleotide prior to the treating.

According to an aspect of some embodiments of the present invention there is provided a method of monitoring treatment with an anti MS drug in a subject in need thereof, the method comprising: (a) treating the subject with the anti MS drug; and (b) determining in a biological sample of the subject a level of expression of at least one polynucleotide selected from the group consisting of C22orf8, TLK1, HNRPH1, PLXDC1, TLK1, PKN2, ALS2CR8, FLJ12547, ZNF238, PDPR, NT5E, PASK, HPGD, IL6ST, JARID1A, PASK, LEF1, FLJ10246, MTUS1, FLJ14011, VSIG4, MARCH-VI, FLJ10613, EWSR1, ATP8A1, SLC4A7, FLJ21127, HNRPH1, ABLIM1, ITGA6, ADCY9, CROCC, SH3YL1, SMA4, SPTBN1, DPEP3, PDE3B, AF5Q31, NRCAM, DOCK9, IPW, FLJ20152, SIRPB2, GALNT4, CD28, TXK, ETS1, DGCR5, ZNF192,

TCF7, CAMK4, SIM2, MGEA5, TGFBR2, RET, MAPK8IP3, RRN3, DKFZp547H025, FBXW11, ZNF423, DLG1, MGC17330, CD164L1, REPS1, ACHE, ITGB1BP2, LOC94431, LTK, RUNX1, EVER1, KIAA2010, CEACAM7, STX16, SLC4A5, CRTAP, RECQL5, MAGEF1, VIPR1, FLJ10979, TTC3, CRSP2, BAZ2A, GTF2I, 5 MGC50853, KIAA0508, BPHL, LTBP4, FN3KRP, SCARB1, MGC17330, HYAL4, DGKA, FLJ11196, DHRS6, EPHB4, IDI2/GTPBP4, SNTG2, SLC7A6, PMS2L2, KIAA0436, TOSO, THRAP3, T3JAM, LOC283232, LOC92482, PTER, ATM, NUCB2, PIK3R2, MGC1136, CD59, JARID1A, FLJ39616, ABLIM1, PBP, MAPK8IP3, FTS, LHX5, TNFRSF7, MYC, PBXIP1, DATF1, HTF9C, PUS1, 10 KIAA0924, C6orf4, KIAA0372, WDR42A, CRYZL1, TERE1, LTBP4, TTC3, NFATC1, POM121/LOC340318, TOSO, LOC348926/MGC16279/SB153/FLJ10661, SPOCK2, KIAA0515, SLC37A4, CD44, SMARCA2, SPTBN1, C6orf130, TTC3, DLG1, SLC35E2, MCCC1, PMS2L11, RCN3, STX16, FLJ20618, STAT5B, SMARCA2, SATB1, POLR1D, ASXL1, REV1L, PMS2L2/PMS2L5, FLJ12355, 15 CCNB1IP1, FLJ12270, KIAA0692, MCM7, GPSN2, STX16, MMS19L, GTF2I/GTF2IP1, AKAP7, ZNF444, SLC35A3, MGEA5, RUTBC3, C20orf36, RAD17, ALG12, LOC112869, C6orf48, CUTC, LGTN, DEF6, WAC, HNRPH3, NS, KIAA0892, LRPPRC, HMG20A, DDX42, TINP1, ZDHHC17, C19orf2, EIF4B, LOC376745, DKFZP434C171, TH1L, C19orf13, RPL22, PHF15, EWSR1, EIF4B, 20 FAM48A, YT521, NEK9, EIF3S7, RPS6, RPL35A, EEF2, RPL3, RPS6, UBA52, RPL6, RPS6, RPL13, AL353949, AL580863, AF052160, AW128846, AW974481, N92920, BG178274, AW303460, BF057458, AL050035, M59917, AK025422, AI693985, AU158442, AK021460, AL023773, NM_003790, AC005011, M90355, AL353580, U38964, D50683 and BE967207, wherein downregulation below a 25 predetermined threshold in the level of expression relative to a level of expression of the at least one polynucleotide in a biological sample of a subject diagnosed with typical relapsing-remitting multiple sclerosis (RRMS) is indicative of treatment efficacy.

According to some embodiments of the invention, step (b) is effected also prior to step (a) and wherein the downregulation is with respect to a level of the at least one 30 polynucleotide prior to the treating.

According to an aspect of some embodiments of the present invention there is provided a method of predicting a typical relapsing-remitting multiple sclerosis (RRMS)

course in a subject diagnosed with multiple sclerosis, the method comprising: (a) determining in a biological sample of the subject a level of expression of at least one polynucleotide selected from the group consisting of YWHAB, ATP6V1E1, UBB, MRCL2/MRCL3, UQCR, MRCL2, RTN4, UBE2A, RTN4, WDR1, PSMA6, C14orf123, PP1201, TBK1, CAST, CAST, RSN, PSME1, SDF2, GSTO1, CAST, DNCL1, SQRDL, ADIPOR2, ICMT, NDUFA6, NDUFA6, COX17, HIF1AN, FLJ20257, TBPL1, RAPGEF2, CRSP8, APOL1, PAOX, CNBP2, ETFA, DPP3, KPNA1, MGC3036, TUBB2, PDCL, CCL5, CDS2, RAP1GDS1, ATP6V1D, OBRGRP/LEPR, SF4, GCLC, MGST3, BICD2, BRF1, CHST12, EXOSC7, TOR1B, ZFP95, ILK, UNC13A, MTHFD2, CASP10, FLJ45850, CMRF-35H, ARF3, NDOR1, DUSP10, AP1M2, VRK2, GSN, PTRF, RBM19, RABGAP1L, ATP5S, STOM, TFPI2, SLCO3A1, PTPN12, CSF1, SIGLEC6, KIRREL, OBRGRP, TP53AP1, SUHW1, NUP98, IL15RA, MICB, CMRF-35H, SPHK1, TNFRSF6, FLJ11301, LRP5, STOM, EPHA2, SRC, FLJ11301, PSTPIP2, EBP, MCPH1, PTPRF, LIMK2, FSTL4, CBR1, MGC2654, MYCT1, NOL3, MTF, ATP10B, FBXO31, TBX21, LSS, SLC17A3, MNAB, CHPPR, GIF, VAMP5, ABCG2, KIF1B, LOH11CR2A, NID2, RBBP8, ETV7, CTSL, RUFY1, RSU1, PARD3, APOB, ACOX3, DAB2, LDLR, TJP2, GNAS, PARD3, NCKAP1, TAP2, HDGFRP3, LDLR, PIK3R3, HTR2B, GAS2L1, FER1L3, C3orf14, TP53TG3, LEPR, CLIC5, PDE4DIP, ATP9A, ITGB1BP1, INDO, SELP, FHL2, FER1L3, EGF, SIAT8A, HDGFRP3, LRAP, VWF, FLJ10134, IMP-3, DMN, MCTP1, FSTL1, CTNNA1, RAB27B, THBS1, PROS1, MMRN1, CTTN, AL078596, AI148659, U00956 and M29383, wherein downregulation below a predetermined threshold in the level of expression relative to a level of expression of the at least one polynucleotide in a biological sample of a subject diagnosed with benign multiple sclerosis (BMS) is indicative of the prediction of the relapsing-remitting multiple sclerosis course of the subject; (b) informing the subject of the prediction of the relapsing-remitting multiple sclerosis course; thereby predicting the relapsing-remitting multiple sclerosis course in the subject diagnosed with the multiple sclerosis.

According to an aspect of some embodiments of the present invention there is provided a method of predicting a typical relapsing-remitting multiple sclerosis (RRMS) course in a subject diagnosed with multiple sclerosis, the method comprising: (a) determining in a biological sample of the subject a level of expression of at least one

polynucleotide selected from the group consisting of C22orf8, TLK1, HNRPH1, PLXDC1, TLK1, PKN2, ALS2CR8, FLJ12547, ZNF238, PDPR, NT5E, PASK, HPGD, IL6ST, JARID1A, PASK, LEF1, FLJ10246, MTUS1, FLJ14011, VSIG4, MARCH-VI, FLJ10613, EWSR1, ATP8A1, SLC4A7, FLJ21127, HNRPH1, ABLIM1, ITGA6, ADCY9, CROCC, SH3YL1, SMA4, SPTBN1, DPEP3, PDE3B, AF5Q31, NRCAM, DOCK9, IPW, FLJ20152, SIRPB2, GALNT4, CD28, TXK, ETS1, DGCR5, ZNF192, TCF7, CAMK4, SIM2, MGEA5, TGFBR2, RET, MAPK8IP3, RRN3, DKFZp547H025, FBXW11, ZNF423, DLG1, MGC17330, CD164L1, REPS1, ACHE, ITGB1BP2, LOC94431, LTK, RUNX1, EVER1, KIAA2010, CEACAM7, STX16, SLC4A5, CRTAP, RECQL5, MAGEF1, VIPR1, FLJ10979, TTC3, CRSP2, BAZ2A, GTF2I, MGC50853, KIAA0508, BPHL, LTBP4, FN3KRP, SCARB1, MGC17330, HYAL4, DGKA, FLJ11196, DHRS6, EPHB4, IDI2/GTPBP4, SNTG2, SLC7A6, PMS2L2, KIAA0436, TOSO, THRAP3, T3JAM, LOC283232, LOC92482, PTER, ATM, NUCB2, PIK3R2, MGC1136, CD59, JARID1A, FLJ39616, ABLIM1, PBP, MAPK8IP3, FTS, LHX5, TNFRSF7, MYC, PBXIP1, DATF1, HTF9C, PUS1, KIAA0924, C6orf4, KIAA0372, WDR42A, CRYZL1, TERE1, LTBP4, TTC3, NFATC1, POM121/LOC340318, TOSO, LOC348926/MGC16279/SB153/FLJ10661, SPOCK2, KIAA0515, SLC37A4, CD44, SMARCA2, SPTBN1, C6orf130, TTC3, DLG1, SLC35E2, MCCC1, PMS2L11, RCN3, STX16, FLJ20618, STAT5B, SMARCA2, SATB1, POLR1D, ASXL1, REV1L, PMS2L2/PMS2L5, FLJ12355, CCNB1IP1, FLJ12270, KIAA0692, MCM7, GPSN2, STX16, MMS19L, GTF2I/GTF2IP1, AKAP7, ZNF444, SLC35A3, MGEA5, RUTBC3, C20orf36, RAD17, ALG12, LOC112869, C6orf48, CUTC, LGTN, DEF6, WAC, HNRPH3, NS, KIAA0892, LRPPRC, HMG20A, DDX42, TINP1, ZDHHC17, C19orf2, EIF4B, LOC376745, DKFZP434C171, TH1L, C19orf13, RPL22, PHF15, EWSR1, EIF4B, FAM48A, YT521, NEK9, EIF3S7, RPS6, RPL35A, EEF2, RPL3, RPS6, UBA52, RPL6, RPS6, RPL13, AL353949, AL580863, AF052160, AW128846, AW974481, N92920, BG178274, AW303460, BF057458, AL050035, M59917, AK025422, AI693985, AU158442, AK021460, AL023773, NM_003790, AC005011, M90355, AL353580, U38964, D50683 and BE967207, wherein upregulation above a predetermined threshold in the level of expression relative to a level of expression of the at least one polynucleotide in a biological sample of a subject diagnosed with benign

multiple sclerosis (BMS) is indicative of the prediction of the relapsing-remitting multiple sclerosis course of the subject; (b) informing the subject of the prediction of the relapsing-remitting multiple sclerosis course; thereby predicting the relapsing-remitting multiple sclerosis course in the subject diagnosed with the multiple sclerosis.

5 According to an aspect of some embodiments of the present invention there is provided a method of treating a subject diagnosed with multiple sclerosis, the method comprising: (a) determining if the subject is predicted to have a relapsing-remitting multiple sclerosis course according to the method of the invention, (b) selecting a treatment regimen based on the prediction of the relapsing-remitting multiple sclerosis
10 course; thereby treating the subject diagnosed with multiple sclerosis.

 According to an aspect of some embodiments of the present invention there is provided a method of treating a subject diagnosed with multiple sclerosis, the method comprising: (a) determining in a biological sample of the subject a level of expression of RRN3, wherein upregulation above a predetermined threshold in the level of expression
15 relative to a level of expression of the RRN3 in a biological sample of a subject diagnosed with benign multiple sclerosis (BMS) is indicative of the prediction of the relapsing-remitting multiple sclerosis course of the subject; (b) administering to the subject a therapeutically effective amount of diterpenoid triepoxide Triptolide (TPT) or a derivative thereof, thereby treating the subject.

20 According to an aspect of some embodiments of the present invention there is provided a method of treating a subject diagnosed with multiple sclerosis, the method comprising: administering to the subject an agent which downregulates the expression level and/or activity of at least one polynucleotide or polypeptide of the RNA polymerase 1 pathway, with the proviso that the agent is not diterpenoid triepoxide
25 Triptolide (TPT), thereby treating the subject.

 According to an aspect of some embodiments of the present invention there is provided a probeset comprising a plurality of oligonucleotides and no more than 500 oligonucleotides, the plurality of oligonucleotides specifically recognizing the polynucleotides of C22orf8, TLK1, HNRPH1, PLXDC1, TLK1, PKN2, ALS2CR8,
30 FLJ12547, ZNF238, PDPR, NT5E, PASK, HPGD, IL6ST, JARID1A, PASK, LEF1, FLJ10246, MTUS1, FLJ14011, VSIG4, MARCH-VI, FLJ10613, EWSR1, ATP8A1, SLC4A7, FLJ21127, HNRPH1, ABLIM1, ITGA6, ADCY9, CROCC, SH3YL1, SMA4,

SPTBN1, DPEP3, PDE3B, AF5Q31, NRCAM, DOCK9, IPW, FLJ20152, SIRPB2, GALNT4, CD28, TXK, ETS1, DGCR5, ZNF192, TCF7, CAMK4, SIM2, MGEA5, TGFBR2, RET, MAPK8IP3, RRN3, DKFZp547H025, FBXW11, ZNF423, DLG1, MGC17330, CD164L1, REPS1, ACHE, ITGB1BP2, LOC94431, LTK, RUNX1, EVER1, KIAA2010, CEACAM7, STX16, SLC4A5, CRTAP, RECQL5, MAGEF1, VIPR1, FLJ10979, TTC3, CRSP2, BAZ2A, GTF2I, MGC50853, KIAA0508, BPHL, LTBP4, FN3KRP, SCARB1, MGC17330, HYAL4, DGKA, FLJ11196, DHRS6, EPHB4, IDI2/GTPBP4, SNTG2, SLC7A6, PMS2L2, KIAA0436, TOSO, THRAP3, T3JAM, LOC283232, LOC92482, PTER, ATM, NUCB2, PIK3R2, MGC1136, CD59, JARID1A, FLJ39616, ABLIM1, PBP, MAPK8IP3, FTS, LHX5, TNFRSF7, MYC, PBXIP1, DATF1, HTF9C, PUS1, KIAA0924, C6orf4, KIAA0372, WDR42A, CRYZL1, TERE1, LTBP4, TTC3, NFATC1, POM121/LOC340318, TOSO, LOC348926/MGC16279/SB153/FLJ10661, SPOCK2, KIAA0515, SLC37A4, CD44, SMARCA2, SPTBN1, C6orf130, TTC3, DLG1, SLC35E2, MCCC1, PMS2L11, RCN3, STX16, FLJ20618, STAT5B, SMARCA2, SATB1, POLR1D, ASXL1, REV1L, PMS2L2/PMS2L5, FLJ12355, CCNB1IP1, FLJ12270, KIAA0692, MCM7, GPSN2, STX16, MMS19L, GTF2I/GTF2IP1, AKAP7, ZNF444, SLC35A3, MGEA5, RUTBC3, C20orf36, RAD17, ALG12, LOC112869, C6orf48, CUTC, LGTN, DEF6, WAC, HNRPH3, NS, KIAA0892, LRPPRC, HMG20A, DDX42, TINP1, ZDHHC17, C19orf2, EIF4B, LOC376745, DKFZP434C171, TH1L, C19orf13, RPL22, PHF15, EWSR1, EIF4B, FAM48A, YT521, NEK9, EIF3S7, RPS6, RPL35A, EEF2, RPL3, RPS6, UBA52, RPL6, RPS6, RPL13, AL353949, AL580863, AF052160, AW128846, AW974481, N92920, BG178274, AW303460, BF057458, AL050035, M59917, AK025422, AI693985, AU158442, AK021460, AL023773, NM_003790, AC005011, M90355, AL353580, U38964, D50683, BE967207, YWHAB, ATP6V1E1, UBB, MRCL2/MRCL3, UQCR, MRCL2, RTN4, UBE2A, RTN4, WDR1, PSMA6, C14orf123, PP1201, TBK1, CAST, CAST, RSN, PSME1, SDF2, GSTO1, CAST, DNCL1, SQRDL, ADIPOR2, ICMT, NDUFA6, NDUFA6, COX17, HIF1AN, FLJ20257, TBPL1, RAPGEF2, CRSP8, APOL1, PAOX, CNDP2, ETFA, DPP3, KPNA1, MGC3036, TUBB2, PDCL, CCL5, CDS2, RAP1GDS1, ATP6V1D, OBRGRP/LEPR, SF4, GCLC, MGST3, BICD2, BRF1, CHST12, EXOSC7, TOR1B, ZFP95, ILK, UNC13A, MTHFD2, CASP10, FLJ45850, CMRF-35H, ARF3, NDOR1,

DUSP10, AP1M2, VRK2, GSN, PTRF, RBM19, RABGAP1L, ATP5S, STOM, TFPI2, SLCO3A1, PTPN12, CSF1, SIGLEC6, KIRREL, OBRGRP, TP53AP1, SUHW1, NUP98, IL15RA, MICB, CMRF-35H, SPHK1, TNFRSF6, FLJ11301, LRP5, STOM, EPHA2, SRC, FLJ11301, PSTPIP2, EBP, MCPH1, PTPRF, LIMK2, FSTL4, CBR1,
 5 MGC2654, MYCT1, NOL3, MTF, ATP10B, FBXO31, TBX21, LSS, SLC17A3, MNAB, CHPPR, GIF, VAMP5, ABCG2, KIF1B, LOH11CR2A, NID2, RBBP8, ETV7, CTSL, RUFY1, RSU1, PARD3, APOB, ACOX3, DAB2, LDLR, TJP2, GNAS, PARD3, NCKAP1, TAP2, HDGFRP3, LDLR, PIK3R3, HTR2B, GAS2L1, FER1L3, C3orf14, TP53TG3, LEPR, CLIC5, PDE4DIP, ATP9A, ITGB1BP1, INDO, SELP, FHL2,
 10 FER1L3, EGF, SIAT8A, HDGFRP3, LRAP, VWF, FLJ10134, IMP-3, DMN, MCTP1, FSTL1, CTNNAL1, RAB27B, THBS1, PROS1, MMRN1, CTTN, AL078596, AI148659, U00956 and M29383.

According to an aspect of some embodiments of the present invention there is provided a kit for predicting a benign or relapsing-remitting course in a subject
 15 diagnosed with multiple sclerosis, comprising the probeset of the invention.

According to some embodiments of the invention, the kit further comprising a positive control for an expression level of at least one of the polynucleotides.

According to some embodiments of the invention, the at least one polynucleotide comprises the polynucleotides of RRN3, POLR1D and LRPPRC.

20 According to some embodiments of the invention, the treatment regimen comprises administering to the subject an agent which downregulates the expression level and/or activity of at least one polynucleotide or polypeptide of the RNA polymerase 1 pathway, thereby treating the subject.

25 According to some embodiments of the invention, the at least one polynucleotide or polypeptide of the RNA polymerase 1 pathway comprises RRN3.

According to some embodiments of the invention, the at least one polynucleotide or polypeptide of the RNA polymerase 1 pathway further comprises POLR1D and LRPPRC.

30 According to some embodiments of the invention, the method further comprising determining in the biological sample of the subject the level of expression of POLR1D and/or LRPPRC, wherein upregulation above a predetermined threshold in the level of expression relative to a level of expression of the POLR1D and/or LRPPRC in a

biological sample of a subject diagnosed with benign multiple sclerosis (BMS) is indicative of the prediction of the relapsing-remitting multiple sclerosis course of the subject.

Unless otherwise defined, all technical and/or scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the invention pertains. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of embodiments of the invention, exemplary methods and/or materials are described below. In case of conflict, the patent specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and are not intended to be necessarily limiting.

Implementation of the method and/or system of embodiments of the invention can involve performing or completing selected tasks manually, automatically, or a combination thereof. Moreover, according to actual instrumentation and equipment of embodiments of the method and/or system of the invention, several selected tasks could be implemented by hardware, by software or by firmware or by a combination thereof using an operating system.

For example, hardware for performing selected tasks according to embodiments of the invention could be implemented as a chip or a circuit. As software, selected tasks according to embodiments of the invention could be implemented as a plurality of software instructions being executed by a computer using any suitable operating system. In an exemplary embodiment of the invention, one or more tasks according to exemplary embodiments of method and/or system as described herein are performed by a data processor, such as a computing platform for executing a plurality of instructions. Optionally, the data processor includes a volatile memory for storing instructions and/or data and/or a non-volatile storage, for example, a magnetic hard-disk and/or removable media, for storing instructions and/or data. Optionally, a network connection is provided as well. A display and/or a user input device such as a keyboard or mouse are optionally provided as well.

BRIEF DESCRIPTION OF THE DRAWINGS

Some embodiments of the invention are herein described, by way of example only, with reference to the accompanying drawings. With specific reference now to the

drawings in detail, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of embodiments of the invention. In this regard, the description taken with the drawings makes apparent to those skilled in the art how embodiments of the invention may be practiced.

5 In the drawings:

FIG. 1 is a schematic illustration depicting the design of the study for identification of genes which predict benign multiple sclerosis;

FIG. 2 depicts Principal Component Analysis (PCA) based on 406 most informative genes (MIGs);

10

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

The present invention, in some embodiments thereof, relates to methods of classifying a subject as being more likely to have BMS or to be more likely to have typical RRMS and, more particularly, but not exclusively, to methods of diagnosing typical RRMS or BMS and treating a subject based on the diagnosis.

15 Before explaining at least one embodiment of the invention in detail, it is to be understood that the invention is not necessarily limited in its application to the details set forth in the following description or exemplified by the Examples. The invention is capable of other embodiments or of being practiced or carried out in various ways.

20 The present inventors have applied a high throughput gene expression technology to identify biomarkers for the diagnosis of benign multiple sclerosis (BMS) and for potential targets for therapeutic interventions in order switch RRMS course of disease to a BMS course of disease.

Thus, as shown in Example 1 of the Examples section which follows, the present
25 inventors have identified 406 genes which are differentially expressed between multiple sclerosis subjects having a benign MS course or an RRMS course (typical RRMS) (Tables 1 and 2). These genes can be used for classification of MS disease course, diagnosis of a typical RRMS course and selecting a suitable treatment regimen for a subject diagnosed with MS which will prevent deterioration in the subject's state while
30 avoiding unnecessary side effects. In addition, as shown in Table 3 (Example 1) and Tables 4A-C (Example 2) the present inventors have uncovered that the expression level of genes which are involved in the RNA polymerase I pathway such as POLR1D,

LRPPRC, RRN3 and NCL is downregulated in subjects having a BMS course of MS as compared to the expression level of these genes in subjects having an RRMS course of MS. Moreover, as shown in Tables 5 and 6 (Example 2) the present inventors identified MIGs (most informative genes) discriminating between BMS and typical RRMS and genes which can classify subjects as having a BMS or a typical RRMS. These results suggest the use of genes involved in the RNA polymerase I pathway as diagnostic markers and drug targets for treating and preventing a typical RRMS course in a subject diagnosed with MS.

Thus, according to an aspect of some embodiments of the invention there is provided a method of classifying a subject as being more likely to have benign multiple sclerosis (BMS) or as being more likely to have typical relapsing remitting multiple sclerosis (RRMS), the method comprising comparing a level of expression of at least one gene involved in the RNA polymerase I pathway in a cell of the subject to a reference expression data of the at least one gene obtained from a cell of at least one subject pre-diagnosed as having BMS and/or from a cell of at least one subject pre-diagnosed as having typical RRMS, thereby classifying the subject as being more likely to have BMS or as being more likely to have typical RRMS.

The term “subject” refers to mammal, preferably a human being.

According to some embodiments of the invention, the subject is diagnosed with multiple sclerosis.

As used herein, the phrase “diagnosed with multiple sclerosis” refers to a subject who experienced at least one neurological attack affecting the central nervous system (CNS) accompanied by demyelinating lesions within the brain or spinal cord, which may have, but not necessarily confirmed by magnetic resonance imaging (MRI). The neurological attack can involve acute or sub-acute neurological symptomatology (attack) manifested by various clinical presentations like unilateral loss of vision, vertigo, ataxia, incoordination, gait difficulties, sensory impairment characterized by paresthesia, dysesthesia, sensory loss, urinary disturbances until incontinence, diplopia, dysarthria, various degrees of motor weakness until paralysis, cognitive decline either as a monosymptomatic or in combination. The symptoms usually remain for several days to few weeks, and then partially or completely resolve.

The accepted diagnostic criteria of multiple sclerosis are presented in Hypertext Transfer Protocol:// World Wide Web (dot) mult-sclerosis (dot) org/DiagnosticCriteria (dot) html.

5 According to some embodiments of the invention, the subject is suspected of having multiple sclerosis.

According to some embodiments of the invention, the subject has probable multiple sclerosis.

According to some embodiments of the invention, the subject does not have a primary progressive course (PPMS).

10 According to some embodiments of the invention, the subject does not have a secondary progressive MS course (SPMS).

As used herein the term “classifying” refers to determining if the subject is more likely to have benign multiple sclerosis (BMS) or typical relapsing remitting multiple sclerosis (RRMS).

15 As used herein the phrase “being more likely to have” refers to having increased probability to have a certain disease course (classification of disease) than another disease course.

According to some embodiments of the invention, the phrase “being more likely to have” refers to a probability of at least about 51%, at least about 52%, at least about 20 53 %, at least about 54%, at least about 55 %, at least about 56 %, at least about 57%, at least about 58 %, at least about 59%, at least about 60%, at least about 61 %, at least about 62%, at least about 63 %, at least about 64%, at least about 65 %, at least about 66%, at least about 67 %, at least about 68%, at least about 69 %, at least about 70%, at least about 71 %, at least about 72%, at least about 73 %, at least about 74%, at least 25 about 75 %, at least about 76%, at least about 77 %, at least about 78%, at least about 79 %, at least about 80%, at least about 81 %, at least about 82%, at least about 83 %, at least about 84%, at least about 85 %, at least about 86%, at least about 87 %, at least about 88%, at least about 89 %, at least about 90%, at least about 91 %, at least about 92%, at least about 93 %, at least about 94%, at least about 95 %, at least about 96%, at 30 least about 97 %, at least about 98%, at least about 99 %, e.g., about 100% that the subject has a certain disease course and not the other disease course, *i.e.*, a BMS or typical RRMS.

As used herein, the phrase “benign multiple sclerosis” refers to a subject having MS and exhibiting an Expanded Disability Status Scale (EDSS) of less than 3.0 following at least 10 years from onset and/or diagnosis of MS.

As used herein, the phrase “typical RRMS” or a “relapsing-remitting multiple sclerosis course”, which is interchangeably used herein, refers to a subject having MS and exhibiting an Expanded Disability Status Scale (EDSS) higher than 3.0 within less than 10 years of disease onset and/or diagnosis.

The phrase “onset of multiple sclerosis (MS)” as used herein refers to the time of occurrence of the first clinical neurological symptomatology suggestive of MS.

The Kurtzke EDSS is a method scale of quantifying disability in MS by scoring eight Functional Systems (FS) (pyramidal, cerebellar, brainstem, sensory, bowel and bladder, visual, cerebral, and other) and allows neurologists to assign a Functional System Score (FSS) in each and to combine the FSS scores into the EDSS score as follows:.

- EDSS 0.0 - Normal neurological examination;
- EDSS 1.0 - No disability, minimal signs in one FS;
- EDSS 1.5 - No disability, minimal signs in more than one FS;
- EDSS 2.0 - Minimal disability in one FS;
- EDSS 2.5 - Mild disability in one FS or minimal disability in two FS;
- EDSS 3.0 - Moderate disability in one FS, or mild disability in three or four FS. Fully ambulatory;
- EDSS 3.5 - Fully ambulatory but with moderate disability in one FS and more than minimal disability in several others;
- EDSS 4.0 - Fully ambulatory without aid, self-sufficient, up and about some 12 hours a day despite relatively severe disability; able to walk without aid or rest some 500 meters;
- EDSS 4.5 - Fully ambulatory without aid, up and about much of the day, able to work a full day, may otherwise have some limitation of full activity or require minimal assistance; characterized by relatively severe disability; able to walk without aid or rest some 300 meters;
- EDSS 5.0 - Ambulatory without aid or rest for about 200 meters; disability severe enough to impair full daily activities (work a full day without special provisions);

EDSS 5.5 - Ambulatory without aid or rest for about 100 meters; disability severe enough to preclude full daily activities;

EDSS 6.0 - Intermittent or unilateral constant assistance (cane, crutch, brace) required to walk about 100 meters with or without resting;

5 EDSS 6.5 - Constant bilateral assistance (canes, crutches, braces) required to walk about 20 meters without resting;

EDSS 7.0 - Unable to walk beyond approximately five meters even with aid, essentially restricted to wheelchair; wheels self in standard wheelchair and transfers alone; up and about in wheelchair some 12 hours a day;

10 EDSS 7.5 - Unable to take more than a few steps; restricted to wheelchair; may need aid in transfer; wheels self but cannot carry on in standard wheelchair a full day, May require motorized wheelchair;

EDSS 8.0 - Essentially restricted to bed or chair or perambulated in wheelchair, but may be out of bed itself much of the day; retains many self-care functions; generally
15 has effective use of arms;

EDSS 8.5 - Essentially restricted to bed much of day, has some effective use of arms retains some self care functions;

EDSS 9.0 - Confined to bed; can still communicate and eat;

EDSS 9.5 - Totally helpless bed patient; unable to communicate effectively or
20 eat/swallow;

EDSS 10.0 - Death due to MS;

As mentioned, the diagnosis of MS is performed by clinical neurological symptoms and/or findings such as laboratory tests involving evaluation of IgG synthesis and oligoclonal bands (immunoglobulins) in the cerebrospinal fluid (CSF) which
25 provide evidence of chronic inflammation of the central nervous system, and brain or spinal cord MRI according to the McDonald criteria [McDonald WI, Compston A, Edan G, et al., 2001, "Recommended diagnostic criteria for multiple sclerosis: guidelines from the International Panel on the diagnosis of multiple sclerosis". Ann. Neurol. 50 (1): 121–7; Polman CH, Reingold SC, Edan G, et al., 2005, "Diagnostic criteria for
30 multiple sclerosis: 2005 revisions to the "McDonald Criteria"". Ann. Neurol. 58 (6): 840–6].

It should be noted that onset of multiple sclerosis and the diagnosis of multiple sclerosis could occur on the same time.

As used herein, the phrase “level of expression” refers to the degree of gene expression and/or gene product activity in a specific cell. For example, up-regulation or
5 down-regulation of various genes can affect the level of the gene product (*i.e.*, RNA and/or protein) in a specific cell.

Sequence information regarding gene products (*i.e.*, RNA transcripts and polypeptide sequences) of the genes of the polynucleotides of the invention such as the genes of RNA polymerase I pathway and of probes which can be used to detect thereof
10 can be found in Tables 1, 2, 3, 5 and 6 of the Examples section which follows.

It should be noted that the level of expression can be determined in arbitrary absolute units, or in normalized units (relative to known expression levels of a control reference). For example, when using DNA chips, the expression levels are normalized according to the chips’ internal controls or by using quantile normalization such as
15 RMA.

As used herein the phrase “a cell of the subject” refers to at least one cell (e.g., an isolated cell), cell culture, cell content and/or cell secreted content which contains RNA and/or proteins of the subject. Examples include a blood cell, a cell obtained from any tissue biopsy [e.g., cerebrospinal fluid, (CSF), brain biopsy], a bone marrow
20 cell, body fluids such as plasma, serum, saliva, spinal fluid, lymph fluid, the external sections of the skin, respiratory, intestinal, and genitourinary tracts, tears, saliva, sputum and milk. According to an embodiment of the invention, the cell is a blood cell (e.g., white blood cells, macrophages, B- and T-lymphocytes, monocytes, neutrophils, eosinophiles, and basophiles) which can be obtained using a syringe needle from a vein
25 of the subject. It should be noted that the cell may be isolated from the subject (e.g., for *in vitro* detection) or may optionally comprise a cell that has not been physically removed from the subject (e.g., *in vivo* detection).

According to some embodiments of the invention, the white blood cell comprises peripheral blood mononuclear cells (PBMC). The phrase, “peripheral blood
30 mononuclear cells (PBMCs)” as used herein, refers to a mixture of monocytes and lymphocytes. Several methods for isolating white blood cells are known in the art. For example, PBMCs can be isolated from whole blood samples using density gradient

centrifugation procedures. Typically, anticoagulated whole blood is layered over the separating medium. At the end of the centrifugation step, the following layers are visually observed from top to bottom: plasma/platelets, PBMCs, separating medium and erythrocytes/granulocytes. The PBMC layer is then removed and washed to
5 remove contaminants (e.g., red blood cells) prior to determining the expression level of the polynucleotide(s) therein.

The cell or the biological sample comprising same can be obtained from the subject at any time, e.g., immediately after an attack or at any time during remission.

According to some embodiments of the invention, the level of expression of the
10 gene(s) of the invention is determined using an RNA and/or a protein detection method.

According to some embodiments of the invention, the RNA or protein molecules are extracted from the cell of the subject.

Methods of extracting RNA or protein molecules from cells of a subject are well known in the art. Once obtained, the RNA or protein molecules can be characterized
15 for the expression and/or activity level of various RNA and/or protein molecules using methods known in the arts.

Non-limiting examples of methods of detecting RNA molecules in a cell sample include Northern blot analysis, RT-PCR, RNA *in situ* hybridization (using e.g., DNA or RNA probes to hybridize RNA molecules present in the cells or tissue sections), *in situ*
20 RT-PCR (e.g., as described in Nuovo GJ, et al. Am J Surg Pathol. 1993, 17: 683-90; Komminoth P, et al. Pathol Res Pract. 1994, 190: 1017-25), and oligonucleotide microarray (e.g., by hybridization of polynucleotide sequences derived from a sample to oligonucleotides attached to a solid surface [e.g., a glass wafer) with addressable location, such as Affymetrix microarray (Affymetrix®, Santa Clara, CA)].

25 For example, the level of RRN3 in a sample can be determined by RT-PCR using primers available from Santa Cruz Biotechnology Inc. (sc-106866-PR), or Taqman Gene Expression Assay HS00607907_ml (Applied Biosystems, Foster City, CA, USA), according to manufacturer's recommendation.

Non-limiting examples of methods of detecting the level and/or activity of
30 specific protein molecules in a cell sample include Enzyme linked immunosorbent assay (ELISA), Western blot analysis, radio-immunoassay (RIA), Fluorescence activated cell sorting (FACS), immunohistochemical analysis, *in situ* activity assay (using e.g., a

chromogenic substrate applied on the cells containing an active enzyme), in vitro activity assays (in which the activity of a particular enzyme is measured in a protein mixture extracted from the cells). For example, in case the detection of the expression level of a secreted protein is desired, ELISA assay may be performed on a sample of fluid obtained from the subject (e.g., serum), which contains cell-secreted content.

As used herein the phrase “reference expression data” refers to the expression level of the gene in a cell of at least one subject who is pre-diagnosed as having BMS or typical RRMS. Such as an expression level can be known from the literature, from the database, or from biological samples comprising RNA or protein molecules obtained from a reference subject who is already diagnosed as having BMS or typical RRMS.

As used herein the phrase “pre-diagnosed” refers to being diagnosed based on the acceptable clinical tools/markers as described above (e.g., by evaluating the EDSS score after 10 years from onset or diagnosis of MS).

According to some embodiments of the invention, the reference expression data is obtained from at least subject who is pre-diagnosed as having BMS, e.g., from at least 2, from at least 3, from at least 4, from at least 5, from at least 6, from at least 7, from at least 8, from at least 9, from at least 10, from at least 20, from at least 30, from at least 40, from at least 50, from at least 100 or more subjects who are pre-diagnosed as having BMS.

According to some embodiments of the invention, the reference expression data is obtained from at least one subject who is pre-diagnosed as having typical RRMS, e.g., from at least 2, from at least 3, from at least 4, from at least 5, from at least 6, from at least 7, from at least 8, from at least 9, from at least 10, from at least 20, from at least 30, from at least 40, from at least 50, from at least 100 or more subjects who are pre-diagnosed as having typical RRMS.

It should be noted that when more than one reference subjects (*i.e.*, a subject who is pre-diagnosed as having BMS or typical RRMS) is used, the reference expression data may comprise an average of the expression level of several or all subjects, and those of skills in the art are capable of averaging expression levels from 2 or more subject, using e.g., normalized expression values.

According to some embodiments of the invention, a decrease above a predetermined threshold in the level of expression of the at least one gene in the cell of

the subject relative to the reference expression data of the at least one gene obtained from a cell of the at least one subject having the typical RRMS classifies the subject as being more likely to have the BMS.

As used herein the phrase “a decrease above a predetermined threshold” refers to a decrease in the level of expression in the cell of the subject relative to the reference expression data obtained from a cell of the at least one subject having the typical RRMS which is higher than a predetermined threshold such as a about 10 %, e.g., higher than about 20 %, e.g., higher than about 30 %, e.g., higher than about 40 %, e.g., higher than about 50 %, e.g., higher than about 60 %, higher than about 70 %, higher than about 80 %, higher than about 90 %, higher than about 2 times, higher than about three times, higher than about four time, higher than about five times, higher than about six times, higher than about seven times, higher than about eight times, higher than about nine times, higher than about 20 times, higher than about 50 times, higher than about 100 times, higher than about 200 times, higher than about 350, higher than about 500 times, higher than about 1000 times, or more relative to the reference expression data obtained from a cell of the at least subject having the typical RRMS.

According to some embodiments of the invention, an increase above a predetermined threshold in the level of expression of the at least one gene in the cell of the subject relative to the reference expression data of the at least one gene obtained from a cell of the at least one subject having the BMS classifies the subject as being more likely to have the typical RRMS.

As used herein the phrase “an increase above a predetermined threshold” refers to an increase in the level of expression in the cell of the subject relative to the reference expression data obtained from a cell of the at least one subject having the BMS which is higher than a predetermined threshold such as a about 10 %, e.g., higher than about 20 %, e.g., higher than about 30 %, e.g., higher than about 40 %, e.g., higher than about 50 %, e.g., higher than about 60 %, higher than about 70 %, higher than about 80 %, higher than about 90 %, higher than about 2 times, higher than about three times, higher than about four time, higher than about five times, higher than about six times, higher than about seven times, higher than about eight times, higher than about nine times, higher than about 20 times, higher than about 50 times, higher than about 100 times, higher than about 200 times, higher than about 350, higher than about 500 times, higher than

about 1000 times, or more relative to the reference expression data obtained from a cell of the at least one subject having the BMS.

According to some embodiments of the invention, when a level of expression of the at least one gene in the cell of the subject is identical or changed below a predetermined threshold as compared to the reference expression data of the at least one gene obtained from a cell of the at least one subject having the BMS, then the subject is classified as being more likely to have the BMS.

As used herein the phrase “changed below a predetermined threshold as compared to the reference expression data ... subject having the BMS” refers to an increase or a decrease in the level of expression in the cell of the subject relative to the reference expression data obtained from a cell of the at least one subject having the BMS which is lower than a predetermined threshold, such as lower than about 10 times, e.g., lower than about 9 times, e.g., lower than about 8 times, e.g., lower than about 7 times, e.g., lower than about 6 times, e.g., lower than about 5 times, e.g., lower than about 4 times, e.g., lower than about 3 times, e.g., lower than about 2 times, e.g., lower than about 90%, e.g., lower than about 80%, e.g., lower than about 70%, e.g., lower than about 60%, e.g., lower than about 50%, e.g., lower than about 40%, e.g., lower than about 30%, e.g., lower than about 20%, e.g., lower than about 10%, e.g., lower than about 9%, e.g., lower than about 8%, e.g., lower than about 7%, e.g., lower than about 6%, e.g., lower than about 5%, e.g., lower than about 4%, e.g., lower than about 3%, e.g., lower than about 2%, e.g., lower than about 1% relative to the reference expression data obtained from a cell of the at least one subject having the BMS.

According to some embodiments of the invention, when a level of expression of the at least one gene in the cell of the subject is identical or changed below a predetermined threshold as compared to the reference expression data of the at least one gene obtained from a cell of the at least one subject having the typical RRMS, then the subject is classified as being more likely to have the typical RRMS.

As used herein the phrase “changed below a predetermined threshold as compared to the reference expression data ... subject having the typical RRMS” refers to an increase or a decrease in the level of expression in the cell of the subject relative to the reference expression data obtained from a cell of the one subject having the typical RRMS which is lower than a predetermined threshold, such as lower than about 10

times, e.g., lower than about 9 times, e.g., lower than about 8 times, e.g., lower than about 7 times, e.g., lower than about 6 times, e.g., lower than about 5 times, e.g., lower than about 4 times, e.g., lower than about 3 times, e.g., lower than about 2 times, e.g., lower than about 90%, e.g., lower than about 80%, e.g., lower than about 70%, e.g., lower than about 60%, e.g., lower than about 50%, e.g., lower than about 40%, e.g., lower than about 30%, e.g., lower than about 20%, e.g., lower than about 10%, e.g., lower than about 9%, e.g., lower than about 8%, e.g., lower than about 7%, e.g., lower than about 6%, e.g., lower than about 5%, e.g., lower than about 4%, e.g., lower than about 3%, e.g., lower than about 2%, e.g., lower than about 1% relative to the reference expression data obtained from the at least one subject having the typical RRMS.

Non-limiting examples of genes involved in the RNA polymerase I pathway which can be used according to the method of the invention are provided in Table 3 along with representative polynucleotides thereof and probes which can be used to detect thereof (Example 1 of the Examples section which follows; e.g., RRN3, LRPPRC, POLR1B, POLR1C, POLR1D, POLR2A, POLR2B, POLR2C, POLR2D, POLR2E, POLR2F, POLR2G, POLR2H, POLR2I, POLR2J, POLR2J2, MGC13098, POLR2K, POLR2L, POLR3B, POLR3C, POLR3D, POLR3E, POLR3F, POLR3G, POLR3K, POLRMT, POLRMT and POLS).

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway is selected from the group consisting of POLR1D, LRPPRC, RRN3 and NCL.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway is RRN3.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway is LRPPRC.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway is POLR1D.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises RRN3 and POLR1D.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises RRN3 and LRPPRC.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises POLR1D and LRPPRC.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises RRN3, LRPPRC and POLR1D.

5 According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway is RRN3 and NCL.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway is LRPPRC and NCL.

10 According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway is POLR1D and NCL.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises RRN3, POLR1D and NCL.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises RRN3, LRPPRC and NCL.

15 According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises POLR1D, LRPPRC and NCL.

According to some embodiments of the invention, the at least one gene involved in the RNA polymerase 1 pathway comprises RRN3, LRPPRC, POLR1D and NCL.

20 Tables 4A-C in the Examples section which follows demonstrate exemplary combinations of genes of the RNA polymerase I pathway along with their classification rates for BMS and typical RRMS.

25 The prediction of the MS course is important in terms of monitoring the clinical state of the subject (e.g., how often does the patient need to be evaluated for the disease progression in terms of neurological evaluation and EDSS), planning of subject's future life (e.g., making decisions regarding marriage, having children, being involved in high risk activities, getting a life-insurance, etc.) and planning the treatment regimen of the subject.

For example, a subject who is more likely to have BMS may be advised to reduce the frequency of neurological clinical evaluations to no more than once per year; to avoid frequent MRI examinations; to not be included in treatment schedule of MS; and/or to avoid receiving immunomodulatory drugs which have side effects or adverse events that can be even life-threatening [e.g., progressive multifocal

30

leukoencephalopathy (PML) in MS patients treated with natalizumab (Tysabri®, Biogen-Idec); Hypertext Transfer Protocol:// World Wide Web (dot) va (dot) gov/MS/pressreleases/Treating_Natalizumab_and_Risk_of_PML (dot) asp].

On the other hand, a subject who is more likely to have typical RRMS may be
5 advised to have neurological clinical evaluations at a higher frequency, e.g., about 3-4 times per year; to have frequent MRI examinations; to be included in treatment schedule of MS; and/or to receive immunomodulatory drugs.

It should be noted that the classification of the subject as being more likely to have BMS or typical RRMS can be used to diagnose the subject as having BMS or
10 typical RRMS.

As used herein the term “diagnosing” refers to determining presence or absence of a pathology (e.g., a disease, disorder, condition or syndrome) and/or likelihood of same, classifying a pathology or a symptom, determining a severity of the pathology, monitoring pathology progression, forecasting an outcome of a pathology and/or
15 prospects of recovery and screening of a subject for a specific disease.

According to some embodiments of the invention the method of diagnosing is effected by (a) classifying the subject as being more likely to have BMS or as being more likely to have typical RRMS according to the method of the invention,

(i) wherein when the subject is classified as being more likely to have the BMS
20 then the subject is diagnosed as having BMS;

(ii) wherein when the subject is classified as being more likely to have the typical RRMS, then the subject is diagnosed as having typical RRMS; and

(c) informing the subject of the diagnosis,
thereby diagnosing the subject pre-diagnosed with the MS as having the BMS or
25 the typical RRMS.

According to some embodiments of the invention, the subject is pre-diagnosed with multiple sclerosis (MS), *i.e.*, has a confirmed diagnosis of MS without knowing the disease course, e.g., typical RRMS, BMS.

As used herein the term “informing” refers to providing to the subject the results
30 of the diagnosis of the disease sub-class (*i.e.*, BMS or typical RRMS). The results may be provided as a computer output and/or oral conversation with the subject.

The teachings of the invention can be also used to determine efficiency anti

multiple sclerosis drugs by determining the effect of the drug(s) on the expression level of the at least one gene of the RNA polymerase I pathway.

Thus, according to an aspect of some embodiments of the invention, there is provided a method of monitoring an efficiency of an anti multiple sclerosis (MS) drug in treating a subject diagnosed with a typical relapsing remitting multiple sclerosis (RRMS) course, the method is effected by:

(a) treating the subject with the anti MS drug; and

(b) comparing a level of expression of least one gene involved in the RNA polymerase I pathway in a cell of the subject following treating with the anti MS drug to a level of expression of the at least one gene in a cell of the subject prior to the treating the subject with the anti MS drug,

(i) wherein a decrease above a predetermined threshold in the level of expression of the at least one gene following the treating with the anti MS drug relative to the level of expression of the at least one gene prior to the treating with the anti MS drug indicates that the anti MS drug is efficient for treating the subject;

(ii) wherein an increase above a predetermined threshold in the level of expression of the at least one gene following the treating with the anti MS drug relative to the level of expression of the at least one gene prior to the treating with the anti MS drug indicates that the anti MS drug is not efficient for treating the subject; or

(iii) wherein when a level of expression of the at least one gene following the treating with the anti MS drug is identical or changed below a predetermined threshold as compared to prior to the treating with the anti MS drug then the treatment is not efficient for treating the subject.

thereby monitoring the efficiency of the anti multiple sclerosis (MS) drug in treating the subject diagnosed with the typical RRMS course.

As used herein the phrase “treating” refers to inhibiting or arresting the development of pathology [multiple sclerosis, e.g., typical RRMS] and/or causing the reduction, remission, or regression of a pathology and/or optimally curing the pathology. Those of skill in the art will understand that various methodologies and assays can be used to assess the development of pathology, and similarly, various methodologies and assays may be used to assess the reduction, remission or regression of the pathology.

According to some embodiments of the invention, treating the subject refers to changing the disease course of the subject from a typical RRMS course to a BMS course.

5 According to some embodiments of the invention, treating the subject refers to preventing a typical RRMS course.

As used herein the phrase “following treating with the anti MS drug” refers to any time period after administering the anti MS drug to the subject, e.g., from a few minutes to hours, or from a few days to weeks or months after drug administration.

10 According to some embodiments of the invention the level of expression is determined following the first dose of the anti MS drug.

According to some embodiments of the invention the level of expression is determined following any dose of the anti MS drug.

15 As used herein the phrase “prior to treating with the anti MS drug” refers to any time period prior administering the anti MS drug to the subject, e.g., from a few minutes to hours, or from a few days to weeks or months prior to drug administration.

According to some embodiments of the invention the level of expression is determined prior any dose of the anti MS drug (e.g., when the subject is naïve to treatment).

20 According to some embodiments of the invention prior to treating refers to when the subject is first diagnosed with multiple sclerosis.

According to some embodiments of the invention prior to treating refers to when the subject is suspected of having multiple sclerosis, or diagnosed with probable multiple sclerosis.

25 According to some embodiments of the invention prior to treating refers to upon MS disease onset.

According to some embodiments of the invention the effect of the treatment on the subject can be evaluated by monitoring the level of expression of at least one of the polynucleotides described hereinabove. For example, downregulation in the level of RRN3 in the subject following treatment can be indicative of the positive effect of the
30 treatment on the subject, *i.e.*, switching from a typical RRMS to a BMS course of disease.

The teachings of the invention can be also used to predict efficiency of a drug *in vitro*.

Thus, according to an aspect of some embodiments of the invention there is provided an *in vitro* method of predicting an efficiency of an anti multiple sclerosis (MS) drug for treatment of a subject diagnosed with a typical relapsing remitting multiple sclerosis (RRMS), the method is effected by:

(a) contacting cells of the subject with a therapeutically effective amount of the anti MS drug; and

(b) comparing a level of expression in the cells of at least one gene involved in the RNA polymerase I pathway following the contacting with the anti MS drug to a level of expression of the at least one gene in the cells prior to the contacting with the anti MS drug,

(i) wherein a decrease above a predetermined threshold in the level of expression of the at least one gene following the contacting with the anti MS drug relative to the level of expression of the at least one gene prior to the contacting with the anti MS drug indicates that the treatment is efficient for treating the subject;

(ii) wherein an increase above a predetermined threshold in the level of expression of the at least one gene following the contacting with the anti MS drug relative to the level of expression of the at least one gene prior to the contacting with the anti MS drug indicates that the treatment is not efficient for treating the subject; or

(iii) wherein when a level of expression of the at least one gene following the contacting with the anti MS drug is identical or changed below a predetermined threshold as compared to prior to the contacting with the anti MS drug then the treatment is not efficient for treating the subject.

thereby predicting the efficiency of the anti MS drug for treatment of the subject diagnosed with the typical RRMS.

Contacting cells with the anti MS drug can be performed by any *in vitro* conditions including for example, adding the anti MS drug to cells derived from a subject (e.g., a primary cell culture, a cell line) or to a biological sample comprising same (e.g., a fluid, liquid which comprises the cells) such that the drug is in direct contact with the cells. According to some embodiments of the invention, the cells of the subject are incubated with the anti MS drug. The conditions used for incubating the

cells are selected for a time period/concentration of cells/concentration of drug/ratio between cells and drug and the like which enable the drug to induce cellular changes, such as changes in transcription and/or translation rate of specific genes, proliferation rate, differentiation, cell death, necrosis, apoptosis and the like.

5 Methods of monitoring cellular changes induced by the drugs are known in the art and include for example, the MTT test which is based on the selective ability of living cells to reduce the yellow salt MTT (3-(4, 5- dimethylthiazolyl-2)-2, 5- diphenyltetrazolium bromide) (Sigma, Aldrich St Louis, MO, USA) to a purple-blue insoluble formazan precipitate; the BrDu assay [Cell Proliferation ELISA BrdU
10 colorimetric kit (Roche, Mannheim, Germany); the TUNEL assay [Roche, Mannheim, Germany]; the Annexin V assay [ApoAlert® Annexin V Apoptosis Kit (Clontech Laboratories, Inc., CA, USA)]; the Senescence associated- β -galactosidase assay (Dimri GP, Lee X, et al. 1995. A biomarker that identifies senescent human cells in culture and in aging skin in vivo. Proc Natl Acad Sci U S A 92:9363-9367); as well as various RNA
15 and protein detection methods (which detect level of expression and/or activity) which are further described hereinabove.

According to some embodiments of the invention, the cells are incubated under conditions which enable the effect of the drug on cellular processes such as downregulation of the at least one gene of the RNA polymerase I pathway.

20 According to an aspect of some embodiments of the invention there is provided a method of treating a subject diagnosed with multiple sclerosis, the method is effected by: (a) classifying the subject as being more likely to have BMS or typical RRMS according to the method of the invention, (b) selecting a treatment regimen based on classification results of step (a); thereby treating the subject diagnosed with multiple
25 sclerosis.

As used herein the phrase "treatment regimen" refers to a treatment plan that specifies the type of treatment, dosage, schedule and/or duration of a treatment provided to a subject in need thereof (e.g., a subject diagnosed with MS). The selected treatment regimen can be an aggressive one which is expected to result in the best clinical
30 outcome (e.g., complete cure of the pathology) or a more moderate one which may relieve symptoms of the pathology yet results in incomplete cure of the pathology. It will be appreciated that in certain cases the more aggressive treatment regimen may be

associated with some discomfort to the subject or adverse side effects (e.g., a damage to healthy cells or tissue). The dosage, schedule and duration of treatment can vary, depending on the severity of pathology and the selected type of treatment, and those of skills in the art are capable of adjusting the type of treatment with the dosage, schedule and duration of treatment.

For example, when the subject is classified as being more likely to have typical RRMS (or being diagnosed with typical RRMS) then the treatment regimen is an aggressive therapy such as an immunomodulation therapy, e.g., a high dosage of interferon beta 1a [Rebif, which can be administered subcutaneously, at a dosage of e.g., 44 µg, three times a week].

MS drugs which can be administered to a subject predicted to have an RRMS course of disease according to the present teachings include, but are not limited to Diterpenoid triepoxide Triptolide (TPT), Adderall; Ambien; Avonex; Baclofen; Beta interferon; Betaseron; Celexa; Clonazepam; Copaxone; Corticosteroids; Cymbalta; Cytoxin; Dexamethasone; Effexor; Elavil; Gabapentin; Hydrocodone; Lexapro; Lyrica; Mitoxantrone; Naltrexone; Neurontin; Novantrone; Prednisone; Provigil; Rebif; Solumedrol; Symmetrel; Topamax; Tysabri; Wellbutrin; Xanax; Zanaflex; Zolof; Novartis' fingolimod [sphingosine 1-phosphate receptor (S1P-R) modulator]; Teva's laquinimod; Merck KGaA's Mylinax (cladribine); Sanofi-aventis' teriflunomide; Biogen Idec's BG-12 (Phase III); GSK/Mitsubishi Tanabe Pharma's firategrast; MediciNova's ibudilast; Biogen/UCB's CDP323 (Phase II).

One the other hand, when the subject is classified as being more likely to have BMS (or is diagnosed with BMS) then the aggressive treatment is not recommended, and these patients would not be treated or treatment can be delayed.

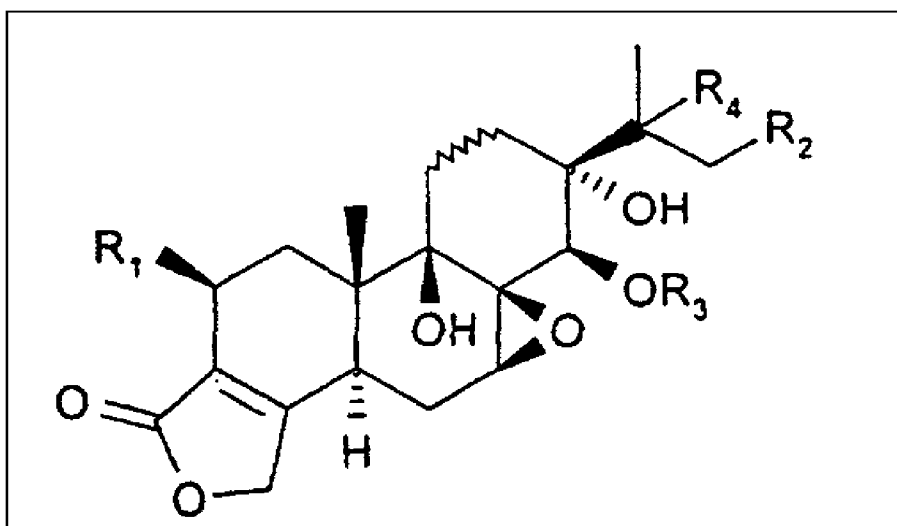
In addition, knowing the prediction or classification of MS disease course (BMS or typical RRMS) is highly beneficial in terms of saving un-necessary costs to the health system.

According to an aspect of some embodiments of the invention there is provided a method of treating a subject diagnosed with multiple sclerosis, the method is effected by (a) diagnosing a typical relapsing remitting multiple sclerosis (RRMS) according to the method of the invention, and (b) administering to the subject a therapeutically

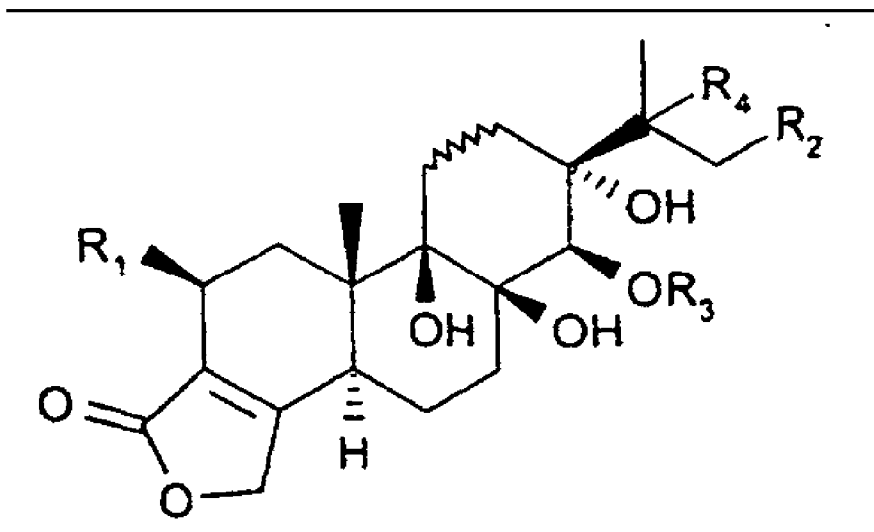
effective amount of diterpenoid triepoxide Triptolide (TPT) or a derivative thereof, thereby treating the subject

TPT derivatives and preparation thereof are described in WO9852933A1, which is fully incorporated herein by reference. Non-limiting examples of TPT derivatives
5 include, compounds of the general formulas.

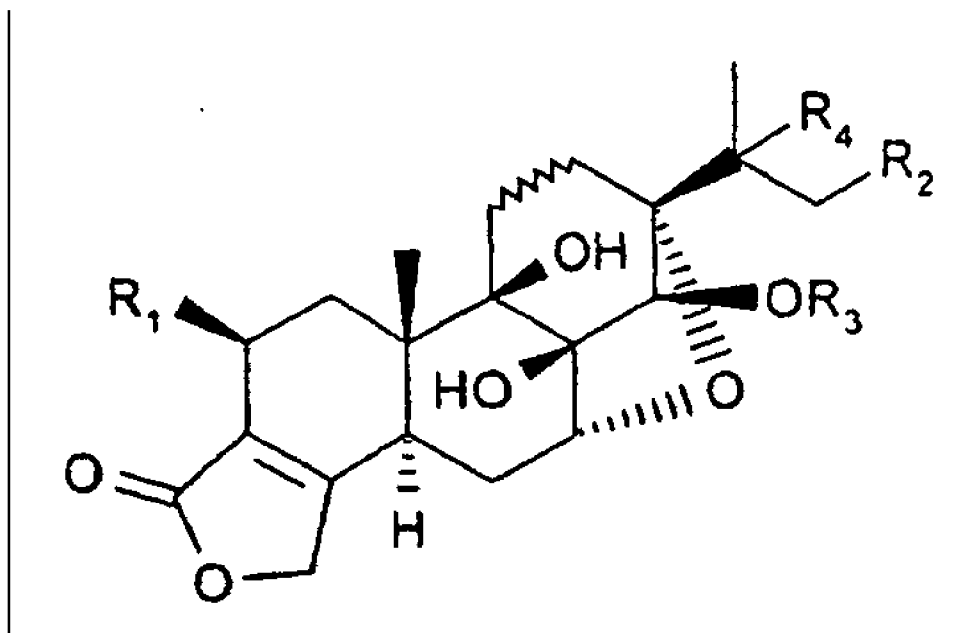
Formula (1),



10 formula (2),



or formula (3),



5 Wherein



represents a single or double bond;

R_1 and R_2 each independently is H or $-OR_5$;

R_3 is H, $-C(=O)(CH_2)_nCO_2H$ or a suitable amino acid; R_4 is H or $-OH$; R_5 is H, -

10 $C(=O)(CH_2)_nCO_2H$ or a suitable amino acid;

n is the integer 2, 3, 4, 5 or 6;

and the stereoisomers, enantiomers and pharmaceutically acceptable salts thereof; provided that R_1 and R_2 are H when R_4 is other than H (for further details see WO9852933A1).

15 A commercially available preparation of Triptolide which can be used according to the teachings of the invention is Trisoxireno(4b,5:6,7:8a,9)phenanthro(1,2-c)furan-1(3H)-one, 3b,4,4a,6,6a,7a,7b,8b,9,10-decahydro-6-hydroxy-8b-methyl-6a-(1-methylethyl)-, (3bS,4aS,5aS,6R,6aR,7aS,7bS,8aS,8bS)- [CAS No.: 38748-32-2; PG490, Chengdu Biopurify Phytochemicals Ltd. Chengdu, Sichuan, China].

20 According to some embodiments of the invention, when the subject is more

likely to have typical RRMS then the treatment regimen comprises administering to the subject an agent which downregulates the level of expression of the at least one gene involved in the RNA polymerase I pathway.

According to some embodiments of the invention, treating the subject is effected by downregulating the expression level and/or activity (RNA and/or polypeptide encoded thereby) of at least one polynucleotide of the polymerase I pathway (for details see the list of genes/polynucleotide in Table 3 in Example 1 of the Examples section which follows).

Following is a list of downregulating agents which can decrease the expression level of the gene product (RNA or protein molecules) of at least one of the polynucleotides of the polymerase I pathway.

Downregulation can be effected on the genomic and/or the transcript level using a variety of molecules which interfere with transcription and/or translation (e.g., RNA silencing agents, Ribozyme, DNAzyme and antisense), or on the protein level using e.g., an antibody, antagonists, enzymes that cleave the polypeptide and the like.

One example, of an agent capable of downregulating a polypeptide-of-interest is an antibody or antibody fragment capable of specifically binding the polypeptide-of-interest. Preferably, the antibody specifically binds at least one epitope of the polypeptide-of-interest. As used herein, the term "epitope" refers to any antigenic determinant on an antigen to which the paratope of an antibody binds.

Epitopic determinants usually consist of chemically active surface groupings of molecules such as amino acids or carbohydrate side chains and usually have specific three dimensional structural characteristics, as well as specific charge characteristics.

The term "antibody" as used in this invention includes intact molecules as well as functional fragments thereof, such as Fab, F(ab')₂, and Fv that are capable of binding to macrophages. These functional antibody fragments are defined as follows: (1) Fab, the fragment which contains a monovalent antigen-binding fragment of an antibody molecule, can be produced by digestion of whole antibody with the enzyme papain to yield an intact light chain and a portion of one heavy chain; (2) Fab', the fragment of an antibody molecule that can be obtained by treating whole antibody with pepsin, followed by reduction, to yield an intact light chain and a portion of the heavy chain; two Fab' fragments are obtained per antibody molecule; (3) (Fab')₂, the fragment of the

antibody that can be obtained by treating whole antibody with the enzyme pepsin without subsequent reduction; F(ab')₂ is a dimer of two Fab' fragments held together by two disulfide bonds; (4) Fv, defined as a genetically engineered fragment containing the variable region of the light chain and the variable region of the heavy chain expressed as two chains; and (5) Single chain antibody ("SCA"), a genetically engineered molecule containing the variable region of the light chain and the variable region of the heavy chain, linked by a suitable polypeptide linker as a genetically fused single chain molecule.

Methods of producing polyclonal and monoclonal antibodies as well as fragments thereof are well known in the art (See for example, Harlow and Lane, *Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory, New York, 1988, incorporated herein by reference).

Antibody fragments according to the present invention can be prepared by proteolytic hydrolysis of the antibody or by expression in *E. coli* or mammalian cells (e.g. Chinese hamster ovary cell culture or other protein expression systems) of DNA encoding the fragment. Antibody fragments can be obtained by pepsin or papain digestion of whole antibodies by conventional methods. For example, antibody fragments can be produced by enzymatic cleavage of antibodies with pepsin to provide a 5S fragment denoted F(ab')₂. This fragment can be further cleaved using a thiol reducing agent, and optionally a blocking group for the sulfhydryl groups resulting from cleavage of disulfide linkages, to produce 3.5S Fab' monovalent fragments. Alternatively, an enzymatic cleavage using pepsin produces two monovalent Fab' fragments and an Fc fragment directly. These methods are described, for example, by Goldenberg, U.S. Pat. Nos. 4,036,945 and 4,331,647, and references contained therein, which patents are hereby incorporated by reference in their entirety. See also Porter, R. R. [Biochem. J. 73: 119-126 (1959)]. Other methods of cleaving antibodies, such as separation of heavy chains to form monovalent light-heavy chain fragments, further cleavage of fragments, or other enzymatic, chemical, or genetic techniques may also be used, so long as the fragments bind to the antigen that is recognized by the intact antibody.

Fv fragments comprise an association of VH and VL chains. This association may be noncovalent, as described in Inbar et al. [Proc. Nat'l Acad. Sci. USA 69:2659-62

(19720]. Alternatively, the variable chains can be linked by an intermolecular disulfide bond or cross-linked by chemicals such as glutaraldehyde. Preferably, the Fv fragments comprise VH and VL chains connected by a peptide linker. These single-chain antigen binding proteins (sFv) are prepared by constructing a structural gene comprising DNA sequences encoding the VH and VL domains connected by an oligonucleotide. The structural gene is inserted into an expression vector, which is subsequently introduced into a host cell such as *E. coli*. The recombinant host cells synthesize a single polypeptide chain with a linker peptide bridging the two V domains. Methods for producing sFvs are described, for example, by [Whitlow and Filpula, *Methods* 2: 97-105 (1991); Bird et al., *Science* 242:423-426 (1988); Pack et al., *Bio/Technology* 11:1271-77 (1993); and U.S. Pat. No. 4,946,778, which is hereby incorporated by reference in its entirety.

Another form of an antibody fragment is a peptide coding for a single complementarity-determining region (CDR). CDR peptides ("minimal recognition units") can be obtained by constructing genes encoding the CDR of an antibody of interest. Such genes are prepared, for example, by using the polymerase chain reaction to synthesize the variable region from RNA of antibody-producing cells. See, for example, Larrick and Fry [*Methods*, 2: 106-10 (1991)].

Humanized forms of non-human (e.g., murine) antibodies are chimeric molecules of immunoglobulins, immunoglobulin chains or fragments thereof (such as Fv, Fab, Fab', F(ab').sub.2 or other antigen-binding subsequences of antibodies) which contain minimal sequence derived from non-human immunoglobulin. Humanized antibodies include human immunoglobulins (recipient antibody) in which residues form a complementary determining region (CDR) of the recipient are replaced by residues from a CDR of a non-human species (donor antibody) such as mouse, rat or rabbit having the desired specificity, affinity and capacity. In some instances, Fv framework residues of the human immunoglobulin are replaced by corresponding non-human residues. Humanized antibodies may also comprise residues which are found neither in the recipient antibody nor in the imported CDR or framework sequences. In general, the humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the CDR regions correspond to those of a non-human immunoglobulin and all or substantially all of the FR regions are

those of a human immunoglobulin consensus sequence. The humanized antibody optimally also will comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin [Jones et al., *Nature*, 321:522-525 (1986); Riechmann et al., *Nature*, 332:323-329 (1988); and Presta, *Curr. Op. Struct. Biol.*, 2:593-596 (1992)].

Methods for humanizing non-human antibodies are well known in the art. Generally, a humanized antibody has one or more amino acid residues introduced into it from a source which is non-human. These non-human amino acid residues are often referred to as import residues, which are typically taken from an import variable domain. Humanization can be essentially performed following the method of Winter and co-workers [Jones et al., *Nature*, 321:522-525 (1986); Riechmann et al., *Nature* 332:323-327 (1988); Verhoeyen et al., *Science*, 239:1534-1536 (1988)], by substituting rodent CDRs or CDR sequences for the corresponding sequences of a human antibody. Accordingly, such humanized antibodies are chimeric antibodies (U.S. Pat. No. 4,816,567), wherein substantially less than an intact human variable domain has been substituted by the corresponding sequence from a non-human species. In practice, humanized antibodies are typically human antibodies in which some CDR residues and possibly some FR residues are substituted by residues from analogous sites in rodent antibodies.

Human antibodies can also be produced using various techniques known in the art, including phage display libraries [Hoogenboom and Winter, *J. Mol. Biol.*, 227:381 (1991); Marks et al., *J. Mol. Biol.*, 222:581 (1991)]. The techniques of Cole et al. and Boerner et al. are also available for the preparation of human monoclonal antibodies (Cole et al., *Monoclonal Antibodies and Cancer Therapy*, Alan R. Liss, p. 77 (1985) and Boerner et al., *J. Immunol.*, 147(1):86-95 (1991)]. Similarly, human antibodies can be made by introduction of human immunoglobulin loci into transgenic animals, e.g., mice in which the endogenous immunoglobulin genes have been partially or completely inactivated. Upon challenge, human antibody production is observed, which closely resembles that seen in humans in all respects, including gene rearrangement, assembly, and antibody repertoire. This approach is described, for example, in U.S. Pat. Nos. 5,545,807; 5,545,806; 5,569,825; 5,625,126; 5,633,425; 5,661,016, and in the following scientific publications: Marks et al., *Bio/Technology* 10,: 779-783 (1992); Lonberg et

al., Nature 368: 856-859 (1994); Morrison, Nature 368 812-13 (1994); Fishwild et al., Nature Biotechnology 14, 845-51 (1996); Neuberger, Nature Biotechnology 14: 826 (1996); and Lonberg and Huszar, Intern. Rev. Immunol. 13, 65-93 (1995).

In case the target antigen (the protein which is detected by the antibody) is presented within the cell, the antibody of the invention can be expressed within the cell intracellular antibodies (also known as "intrabodies") or a particular compartment thereof. Intrabodies are essentially SCA to which intracellular localization signals have been added (e.g., ER, mitochondrial, nuclear, cytoplasmic). This technology has been successfully applied in the art (for review, see Richardson and Marasco, 1995, TIBTECH vol. 13). Intrabodies have been shown to virtually eliminate the expression of otherwise abundant cell surface receptors and to inhibit a protein function within a cell (See, for example, Richardson et al., 1995, Proc. Natl. Acad. Sci. USA 92: 3137-3141; Deshane et al., 1994, Gene Ther. 1: 332-337; Marasco et al., 1998 Human Gene Ther 9: 1627-42; Shaheen et al., 1996 J. Virol. 70: 3392-400; Werge, T. M. et al., 1990, FEBS Letters 274:193-198; Carlson, J.R. 1993 Proc. Natl. Acad. Sci. USA 90:7427-7428; Biocca, S. et al., 1994, Bio/Technology 12: 396-399; Chen, S-Y. et al., 1994, Human Gene Therapy 5:595-601; Duan, L et al., 1994, Proc. Natl. Acad. Sci. USA 91:5075-5079; Chen, S-Y. et al., 1994, Proc. Natl. Acad. Sci. USA 91:5932-5936; Beerli, R.R. et al., 1994, J. Biol. Chem. 269:23931-23936; Mhashilkar, A.M. et al., 1995, EMBO J. 14:1542-1551; PCT Publication No. WO 94/02610 by Marasco et al.; and PCT Publication No. WO 95/03832 by Duan et al.).

To prepare an intracellular antibody expression vector, the cDNA encoding the antibody light and heavy chains specific for the target protein of interest are isolated, typically from a hybridoma that secretes a monoclonal antibody specific for the marker. Hybridomas secreting anti-marker monoclonal antibodies, or recombinant monoclonal antibodies, can be prepared using methods known in the art. Once a monoclonal antibody specific for the marker protein is identified (e.g., either a hybridoma-derived monoclonal antibody or a recombinant antibody from a combinatorial library), DNAs encoding the light and heavy chains of the monoclonal antibody are isolated by standard molecular biology techniques. For hybridoma derived antibodies, light and heavy chain cDNAs can be obtained, for example, by PCR amplification or cDNA library screening. For recombinant antibodies, such as from a phage display library, cDNA encoding the

light and heavy chains can be recovered from the display package (e.g., phage) isolated during the library screening process and the nucleotide sequences of antibody light and heavy chain genes are determined. For example, many such sequences are disclosed in Kabat, E. A., et al. (1991) Sequences of Proteins of Immunological Interest, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242 and in the "Vbase" human germline sequence database. Once obtained, the antibody light and heavy chain sequences are cloned into a recombinant expression vector using standard methods.

For cytoplasmic expression of the light and heavy chains, the nucleotide sequences encoding the hydrophobic leaders of the light and heavy chains are removed. An intracellular antibody expression vector can encode an intracellular antibody in one of several different forms. For example, in one embodiment, the vector encodes full-length antibody light and heavy chains such that a full-length antibody is expressed intracellularly. In another embodiment, the vector encodes a full-length light chain but only the VH/CH1 region of the heavy chain such that a Fab fragment is expressed intracellularly. In another embodiment, the vector encodes a single chain antibody (scFv) wherein the variable regions of the light and heavy chains are linked by a flexible peptide linker [e.g., (Gly₄Ser)₃] and expressed as a single chain molecule. To inhibit marker activity in a cell, the expression vector encoding the intracellular antibody is introduced into the cell by standard transfection methods, as discussed hereinbefore.

Once antibodies are obtained, they may be tested for activity, for example via ELISA.

Downregulation of the polynucleotide-of-interest can be also achieved by RNA silencing. As used herein, the phrase "RNA silencing" refers to a group of regulatory mechanisms [e.g. RNA interference (RNAi), transcriptional gene silencing (TGS), post-transcriptional gene silencing (PTGS), quelling, co-suppression, and translational repression] mediated by RNA molecules which result in the inhibition or "silencing" of the expression of a corresponding protein-coding gene. RNA silencing has been observed in many types of organisms, including plants, animals, and fungi.

As used herein, the term "RNA silencing agent" refers to an RNA which is capable of inhibiting or "silencing" the expression of a target gene. In certain embodiments, the RNA silencing agent is capable of preventing complete processing

(e.g., the full translation and/or expression) of an mRNA molecule through a post-transcriptional silencing mechanism. RNA silencing agents include noncoding RNA molecules, for example RNA duplexes comprising paired strands, as well as precursor RNAs from which such small non-coding RNAs can be generated. Exemplary RNA
5 silencing agents include dsRNAs such as siRNAs, miRNAs and shRNAs. In one embodiment, the RNA silencing agent is capable of inducing RNA interference. In another embodiment, the RNA silencing agent is capable of mediating translational repression.

The RNA silencing agent can be directed to a specific compartment within the
10 cells, such as to the nucleus (see e.g., Shim MS and Kwon YJ. 2009, "Controlled cytoplasmic and nuclear localization of plasmid DNA and siRNA by differentially tailored polyethylenimine"; J. Control Release. 133:206-13, Epub 2008 Nov 1), nucleoli, and the like.

RNA interference refers to the process of sequence-specific post-transcriptional
15 gene silencing in animals mediated by short interfering RNAs (siRNAs). The corresponding process in plants is commonly referred to as post-transcriptional gene silencing or RNA silencing and is also referred to as quelling in fungi. The process of post-transcriptional gene silencing is thought to be an evolutionarily-conserved cellular defense mechanism used to prevent the expression of foreign genes and is commonly
20 shared by diverse flora and phyla. Such protection from foreign gene expression may have evolved in response to the production of double-stranded RNAs (dsRNAs) derived from viral infection or from the random integration of transposon elements into a host genome via a cellular response that specifically destroys homologous single-stranded RNA or viral genomic RNA.

The presence of long dsRNAs in cells stimulates the activity of a ribonuclease
25 III enzyme referred to as dicer. Dicer is involved in the processing of the dsRNA into short pieces of dsRNA known as short interfering RNAs (siRNAs). Short interfering RNAs derived from dicer activity are typically about 21 to about 23 nucleotides in length and comprise about 19 base pair duplexes. The RNAi response also features an
30 endonuclease complex, commonly referred to as an RNA-induced silencing complex (RISC), which mediates cleavage of single-stranded RNA having sequence complementary to the antisense strand of the siRNA duplex. Cleavage of the target

RNA takes place in the middle of the region complementary to the antisense strand of the siRNA duplex.

Accordingly, the present invention contemplates use of dsRNA to downregulate protein expression from mRNA.

5 According to one embodiment, the dsRNA is greater than 30 bp. The use of long dsRNAs (i.e. dsRNA greater than 30 bp) has been very limited owing to the belief that these longer regions of double stranded RNA will result in the induction of the interferon and PKR response. However, the use of long dsRNAs can provide numerous advantages in that the cell can select the optimal silencing sequence alleviating the need
10 to test numerous siRNAs; long dsRNAs will allow for silencing libraries to have less complexity than would be necessary for siRNAs; and, perhaps most importantly, long dsRNA could prevent viral escape mutations when used as therapeutics.

Various studies demonstrate that long dsRNAs can be used to silence gene expression without inducing the stress response or causing significant off-target effects -
15 see for example [Strat et al., *Nucleic Acids Research*, 2006, Vol. 34, No. 13 3803–3810; Bhargava A et al. *Brain Res. Protoc.* 2004;13:115–125; Diallo M., et al., *Oligonucleotides*. 2003;13:381–392; Paddison P.J., et al., *Proc. Natl Acad. Sci. USA*. 2002;99:1443–1448; Tran N., et al., *FEBS Lett.* 2004;573:127–134].

In particular, the present invention also contemplates introduction of long
20 dsRNA (over 30 base transcripts) for gene silencing in cells where the interferon pathway is not activated (e.g. embryonic cells and oocytes) see for example Billy et al., *PNAS* 2001, Vol 98, pages 14428-14433. and Diallo et al, *Oligonucleotides*, October 1, 2003, 13(5): 381-392. doi:10.1089/154545703322617069.

The present invention also contemplates introduction of long dsRNA
25 specifically designed not to induce the interferon and PKR pathways for down-regulating gene expression. For example, Shinagwa and Ishii [*Genes & Dev.* 17 (11): 1340-1345, 2003] have developed a vector, named pDECAP, to express long double-strand RNA from an RNA polymerase II (Pol II) promoter. Because the transcripts from pDECAP lack both the 5'-cap structure and the 3'-poly(A) tail that facilitate ds-
30 RNA export to the cytoplasm, long ds-RNA from pDECAP does not induce the interferon response.

Another method of evading the interferon and PKR pathways in mammalian systems is by introduction of small inhibitory RNAs (siRNAs) either via transfection or endogenous expression.

The term "siRNA" refers to small inhibitory RNA duplexes (generally between
5 18-30 basepairs) that induce the RNA interference (RNAi) pathway. Typically, siRNAs are chemically synthesized as 21mers with a central 19 bp duplex region and symmetric 2-base 3'-overhangs on the termini, although it has been recently described that chemically synthesized RNA duplexes of 25-30 base length can have as much as a 100-fold increase in potency compared with 21mers at the same location. The observed
10 increased potency obtained using longer RNAs in triggering RNAi is theorized to result from providing Dicer with a substrate (27mer) instead of a product (21mer) and that this improves the rate or efficiency of entry of the siRNA duplex into RISC.

It has been found that position of the 3'-overhang influences potency of an siRNA and asymmetric duplexes having a 3'-overhang on the antisense strand are
15 generally more potent than those with the 3'-overhang on the sense strand (Rose et al., 2005). This can be attributed to asymmetrical strand loading into RISC, as the opposite efficacy patterns are observed when targeting the antisense transcript.

The strands of a double-stranded interfering RNA (e.g., an siRNA) may be connected to form a hairpin or stem-loop structure (e.g., an shRNA). Thus, as
20 mentioned the RNA silencing agent of the present invention may also be a short hairpin RNA (shRNA).

A non-limiting example for an siRNA which can be used to down regulate RRN3 (RNA polymerase I transcription factor homolog) expression level in a cell of a subject is Rrn3 siRNA (h): sc-106866 (Santa Cruz Biotechnology, Inc. Santa Cruz, CA, USA). In addition, downregulation of RRN3 can be achieved by Rrn3 shRNA plasmid
25 (h): sc-106866-SH and Rrn3 shRNA (h) Lentiviral Particles: sc-106866-V ((Santa Cruz Biotechnology, Inc. Santa Cruz, CA, USA).

The term "shRNA", as used herein, refers to an RNA agent having a stem-loop structure, comprising a first and second region of complementary sequence, the degree
30 of complementarity and orientation of the regions being sufficient such that base pairing occurs between the regions, the first and second regions being joined by a loop region, the loop resulting from a lack of base pairing between nucleotides (or nucleotide

analogues) within the loop region. The number of nucleotides in the loop is a number between and including 3 to 23, or 5 to 15, or 7 to 13, or 4 to 9, or 9 to 11. Some of the nucleotides in the loop can be involved in base-pair interactions with other nucleotides in the loop. Examples of oligonucleotide sequences that can be used to form the loop
5 include 5'-UUCAAGAGA-3' (Brummelkamp, T. R. et al. (2002) *Science* 296: 550) and 5'-UUUGUGUAG-3' (Castanotto, D. et al. (2002) *RNA* 8:1454). It will be recognized by one of skill in the art that the resulting single chain oligonucleotide forms a stem-loop or hairpin structure comprising a double-stranded region capable of interacting with the RNAi machinery.

10 According to another embodiment the RNA silencing agent may be a miRNA. miRNAs are small RNAs made from genes encoding primary transcripts of various sizes. They have been identified in both animals and plants. The primary transcript (termed the "pri-miRNA") is processed through various nucleolytic steps to a shorter precursor miRNA, or "pre-miRNA." The pre-miRNA is present in a folded form so that
15 the final (mature) miRNA is present in a duplex, the two strands being referred to as the miRNA (the strand that will eventually basepair with the target) The pre-miRNA is a substrate for a form of dicer that removes the miRNA duplex from the precursor, after which, similarly to siRNAs, the duplex can be taken into the RISC complex. It has been demonstrated that miRNAs can be transgenically expressed and be effective through
20 expression of a precursor form, rather than the entire primary form (Parizotto et al. (2004) *Genes & Development* 18:2237-2242 and Guo et al. (2005) *Plant Cell* 17:1376-1386).

Unlike, siRNAs, miRNAs bind to transcript sequences with only partial complementarity (Zeng et al., 2002, *Molec. Cell* 9:1327-1333) and repress translation
25 without affecting steady-state RNA levels (Lee et al., 1993, *Cell* 75:843-854; Wightman et al., 1993, *Cell* 75:855-862). Both miRNAs and siRNAs are processed by Dicer and associate with components of the RNA-induced silencing complex (Hutvagner et al., 2001, *Science* 293:834-838; Grishok et al., 2001, *Cell* 106: 23-34; Ketting et al., 2001, *Genes Dev.* 15:2654-2659; Williams et al., 2002, *Proc. Natl. Acad. Sci. USA* 99:6889-
30 6894; Hammond et al., 2001, *Science* 293:1146-1150; Moulatos et al., 2002, *Genes Dev.* 16:720-728). A recent report (Hutvagner et al., 2002, *Scienceexpress* 297:2056-2060) hypothesizes that gene regulation through the miRNA pathway versus the siRNA

pathway is determined solely by the degree of complementarity to the target transcript. It is speculated that siRNAs with only partial identity to the mRNA target will function in translational repression, similar to an miRNA, rather than triggering RNA degradation.

5 Synthesis of RNA silencing agents suitable for use with the present invention can be effected as follows. First, the mRNA sequence of the polynucleotide-of-interest is scanned downstream of the AUG start codon for AA dinucleotide sequences. Occurrence of each AA and the 3' adjacent 19 nucleotides is recorded as potential siRNA target sites. Preferably, siRNA target sites are selected from the open reading
10 frame, as untranslated regions (UTRs) are richer in regulatory protein binding sites. UTR-binding proteins and/or translation initiation complexes may interfere with binding of the siRNA endonuclease complex [Tuschl ChemBiochem. 2:239-245]. It will be appreciated though, that siRNAs directed at untranslated regions may also be effective, as demonstrated for GAPDH wherein siRNA directed at the 5' UTR mediated
15 about 90 % decrease in cellular GAPDH mRNA and completely abolished protein level (www.ambion.com/techlib/tn/91/912.html).

Second, potential target sites are compared to an appropriate genomic database (e.g., human, mouse, rat etc.) using any sequence alignment software, such as the BLAST software available from the NCBI server (www.ncbi.nlm.nih.gov/BLAST/).

20 Putative target sites which exhibit significant homology to other coding sequences are filtered out.

Qualifying target sequences are selected as template for siRNA synthesis. Preferred sequences are those including low G/C content as these have proven to be more effective in mediating gene silencing as compared to those with G/C content
25 higher than 55 %. Several target sites are preferably selected along the length of the target gene for evaluation. For better evaluation of the selected siRNAs, a negative control is preferably used in conjunction. Negative control siRNA preferably include the same nucleotide composition as the siRNAs but lack significant homology to the genome. Thus, a scrambled nucleotide sequence of the siRNA is preferably used,
30 provided it does not display any significant homology to any other gene.

It will be appreciated that the RNA silencing agent of the present invention need not be limited to those molecules containing only RNA, but further encompasses chemically-modified nucleotides and non-nucleotides.

In some embodiments, the RNA silencing agent provided herein can be functionally associated with a cell-penetrating peptide. As used herein, a "cell-penetrating peptide" is a peptide that comprises a short (about 12-30 residues) amino acid sequence or functional motif that confers the energy-independent (i.e., non-endocytotic) translocation properties associated with transport of the membrane-permeable complex across the plasma and/or nuclear membranes of a cell. The cell-penetrating peptide used in the membrane-permeable complex of the present invention preferably comprises at least one non-functional cysteine residue, which is either free or derivatized to form a disulfide link with a double-stranded ribonucleic acid that has been modified for such linkage. Representative amino acid motifs conferring such properties are listed in U.S. Pat. No. 6,348,185, the contents of which are expressly incorporated herein by reference. The cell-penetrating peptides of the present invention preferably include, but are not limited to, penetratin, transportan, pIsl, TAT(48-60), pVEC, MTS, and MAP.

mRNAs to be targeted using RNA silencing agents include, but are not limited to, those whose expression is correlated with an undesired phenotypic trait. Exemplary mRNAs that may be targeted are those that encode truncated proteins i.e. comprise deletions. Accordingly the RNA silencing agent of the present invention may be targeted to a bridging region on either side of the deletion. Introduction of such RNA silencing agents into a cell would cause a down-regulation of the mutated protein while leaving the non-mutated protein unaffected.

Another agent capable of downregulating the polynucleotide-of-interest is a DNAzyme molecule capable of specifically cleaving an mRNA transcript or DNA sequence of the polynucleotide-of-interest. DNAzymes are single-stranded polynucleotides which are capable of cleaving both single and double stranded target sequences (Breaker, R.R. and Joyce, G. Chemistry and Biology 1995;2:655; Santoro, S.W. & Joyce, G.F. Proc. Natl. Acad. Sci. USA 1997;94:4262) A general model (the "10-23" model) for the DNAzyme has been proposed. "10-23" DNAzymes have a catalytic domain of 15 deoxyribonucleotides, flanked by two substrate-recognition

domains of seven to nine deoxyribonucleotides each. This type of DNAzyme can effectively cleave its substrate RNA at purine:pyrimidine junctions (Santoro, S.W. & Joyce, G.F. Proc. Natl. Acad. Sci. USA 199; for rev of DNAzymes see Khachigian, LM [Curr Opin Mol Ther 4:119-21 (2002)]).

5 Examples of construction and amplification of synthetic, engineered DNAzymes recognizing single and double-stranded target cleavage sites have been disclosed in U.S. Pat. No. 6,326,174 to Joyce et al. DNAzymes of similar design directed against the human Urokinase receptor were recently observed to inhibit Urokinase receptor expression, and successfully inhibit colon cancer cell metastasis in vivo (Itoh et al ,
10 20002, Abstract 409, Ann Meeting Am Soc Gen Ther World Wide Web (dot) asgt (dot) org). In another application, DNAzymes complementary to bcr-abl oncogenes were successful in inhibiting the oncogenes expression in leukemia cells, and lessening relapse rates in autologous bone marrow transplant in cases of CML and ALL.

15 Downregulation of the polynucleotide-of-interest can also be effected by using an antisense polynucleotide capable of specifically hybridizing with an mRNA transcript encoding the polynucleotide-of-interest.

20 Design of antisense molecules which can be used to efficiently downregulate the polynucleotide-of-interest must be effected while considering two aspects important to the antisense approach. The first aspect is delivery of the oligonucleotide into the cytoplasm of the appropriate cells, while the second aspect is design of an oligonucleotide which specifically binds the designated mRNA within cells in a way which inhibits translation thereof.

25 The prior art teaches of a number of delivery strategies which can be used to efficiently deliver oligonucleotides into a wide variety of cell types [see, for example, Luft J Mol Med 76: 75-6 (1998); Kronenwett et al. Blood 91: 852-62 (1998); Rajur et al. Bioconjug Chem 8: 935-40 (1997); Lavigne et al. Biochem Biophys Res Commun 237: 566-71 (1997) and Aoki et al. (1997) Biochem Biophys Res Commun 231: 540-5 (1997)].

30 In addition, algorithms for identifying those sequences with the highest predicted binding affinity for their target mRNA based on a thermodynamic cycle that accounts for the energetics of structural alterations in both the target mRNA and the

oligonucleotide are also available [see, for example, Walton et al. *Biotechnol Bioeng* 65: 1-9 (1999)].

Such algorithms have been successfully used to implement an antisense approach in cells. For example, the algorithm developed by Walton et al. enabled scientists to successfully design antisense oligonucleotides for rabbit beta-globin (RBG) and mouse tumor necrosis factor-alpha (TNF alpha) transcripts. The same research group has more recently reported that the antisense activity of rationally selected oligonucleotides against three model target mRNAs (human lactate dehydrogenase A and B and rat gp130) in cell culture as evaluated by a kinetic PCR technique proved effective in almost all cases, including tests against three different targets in two cell types with phosphodiester and phosphorothioate oligonucleotide chemistries.

In addition, several approaches for designing and predicting efficiency of specific oligonucleotides using an in vitro system were also published (Matveeva et al., *Nature Biotechnology* 16: 1374 - 1375 (1998)).

Several clinical trials have demonstrated safety, feasibility and activity of antisense oligonucleotides. For example, antisense oligonucleotides suitable for the treatment of cancer have been successfully used [Holmund et al., *Curr Opin Mol Ther* 1:372-85 (1999)], while treatment of hematological malignancies via antisense oligonucleotides targeting c-myc gene, p53 and Bcl-2 had entered clinical trials and had been shown to be tolerated by patients [Gerwitz *Curr Opin Mol Ther* 1:297-306 (1999)].

More recently, antisense-mediated suppression of human heparanase gene expression has been reported to inhibit pleural dissemination of human cancer cells in a mouse model [Uno et al., *Cancer Res* 61:7855-60 (2001)].

Thus, the current consensus is that recent developments in the field of antisense technology which, as described above, have led to the generation of highly accurate antisense design algorithms and a wide variety of oligonucleotide delivery systems, enable an ordinarily skilled artisan to design and implement antisense approaches suitable for downregulating expression of known sequences without having to resort to undue trial and error experimentation.

Another agent capable of downregulating the polynucleotide-of-interest is a ribozyme molecule capable of specifically cleaving an mRNA transcript encoding the polynucleotide-of-interest. Ribozymes are being increasingly used for the sequence-

specific inhibition of gene expression by the cleavage of mRNAs encoding proteins of interest [Welch et al., *Curr Opin Biotechnol.* 9:486-96 (1998)]. The possibility of designing ribozymes to cleave any specific target RNA has rendered them valuable tools in both basic research and therapeutic applications. In the therapeutics area, ribozymes have been exploited to target viral RNAs in infectious diseases, dominant oncogenes in cancers and specific somatic mutations in genetic disorders [Welch et al., *Clin Diagn Virol.* 10:163-71 (1998)]. Most notably, several ribozyme gene therapy protocols for HIV patients are already in Phase 1 trials. More recently, ribozymes have been used for transgenic animal research, gene target validation and pathway elucidation. Several ribozymes are in various stages of clinical trials. ANGIOZYME was the first chemically synthesized ribozyme to be studied in human clinical trials. ANGIOZYME specifically inhibits formation of the VEGF-r (Vascular Endothelial Growth Factor receptor), a key component in the angiogenesis pathway. Ribozyme Pharmaceuticals, Inc., as well as other firms have demonstrated the importance of anti-angiogenesis therapeutics in animal models. HEPTAZYME, a ribozyme designed to selectively destroy Hepatitis C Virus (HCV) RNA, was found effective in decreasing Hepatitis C viral RNA in cell culture assays (Ribozyme Pharmaceuticals, Incorporated - WEB home page).

An additional method of regulating the expression of the polynucleotide-of-interest in cells is via triplex forming oligonucleotides (TFOs). Recent studies have shown that TFOs can be designed which can recognize and bind to polypurine/polypirimidine regions in double-stranded helical DNA in a sequence-specific manner. These recognition rules are outlined by Maher III, L. J., et al., *Science*, 1989;245:725-730; Moser, H. E., et al., *Science*, 1987;238:645-630; Beal, P. A., et al, *Science*, 1992;251:1360-1363; Cooney, M., et al., *Science*, 1988;241:456-459; and Hogan, M. E., et al., EP Publication 375408. Modification of the oligonucleotides, such as the introduction of intercalators and backbone substitutions, and optimization of binding conditions (pH and cation concentration) have aided in overcoming inherent obstacles to TFO activity such as charge repulsion and instability, and it was recently shown that synthetic oligonucleotides can be targeted to specific sequences (for a recent review see Seidman and Glazer, *J Clin Invest* 2003;112:487-94).

Another agent capable of downregulating the expression level of the polynucleotide-of-interest is a small molecule which inhibits the activity, level and/or interactions of the gene product of polynucleotide-of-interest, such as by interfering with the pathway in which the gene product of the polynucleotide-of-interest is involved.

Non-limiting examples of small molecules which can be used along with the method of the invention to treat the subject include Cycloheximide and diterpenoid triepoxide Triptolide (TPT) or a derivative thereof.

For example, when RRN3 is upregulated in the typical RRMS subject then the treatment can be with diterpenoid triepoxide Triptolide (TPT) or a derivative thereof; and/or with Cycloheximide or a derivative thereof.

Any of the downregulating agents described hereinabove (e.g., the agent which downregulates the gene of the RNA polymerase I pathway, e.g., siRNA, antibody) can be provided to the subject in need thereof along with any of the known multiple sclerosis therapies (e.g., the anti MS drugs) described hereinabove (combination therapy) and/or with Triptolide or a derivative thereof.

Any of the downregulating agents described above can be administered to the subject per se, or in a pharmaceutical composition where it is mixed with suitable carriers or excipients.

As used herein a "pharmaceutical composition" refers to a preparation of one or more of the active ingredients described herein with other chemical components such as physiologically suitable carriers and excipients. The purpose of a pharmaceutical composition is to facilitate administration of a compound to an organism.

Herein the term "active ingredient" refers to the downregulating agent accountable for the biological effect.

Hereinafter, the phrases "physiologically acceptable carrier" and "pharmaceutically acceptable carrier" which may be interchangeably used refer to a carrier or a diluent that does not cause significant irritation to an organism and does not abrogate the biological activity and properties of the administered compound. An adjuvant is included under these phrases.

Herein the term "excipient" refers to an inert substance added to a pharmaceutical composition to further facilitate administration of an active ingredient.

Examples, without limitation, of excipients include calcium carbonate, calcium phosphate, various sugars and types of starch, cellulose derivatives, gelatin, vegetable oils and polyethylene glycols.

Techniques for formulation and administration of drugs may be found in
5 “Remington’s Pharmaceutical Sciences,” Mack Publishing Co., Easton, PA, latest edition, which is incorporated herein by reference.

Suitable routes of administration may, for example, include oral, rectal, transmucosal, especially transnasal, intestinal or parenteral delivery, including intramuscular, subcutaneous and intramedullary injections as well as intrathecal, direct
10 intraventricular, intracardiac, e.g., into the right or left ventricular cavity, into the common coronary artery, intravenous, intraperitoneal, intranasal, or intraocular injections.

For example, the pharmaceutical composition may be administered using a dermal patch which releases the active ingredient [e.g., Diterpenoid triepoxide
15 Triptolide (TPT) has a molecular weight of 360.40 and it will thus be suitable to be used in a dermal patch].

Conventional approaches for drug delivery to the central nervous system (CNS) include: neurosurgical strategies (e.g., intracerebral injection or intracerebroventricular infusion); molecular manipulation of the agent (e.g., production of a chimeric fusion
20 protein that comprises a transport peptide that has an affinity for an endothelial cell surface molecule in combination with an agent that is itself incapable of crossing the BBB) in an attempt to exploit one of the endogenous transport pathways of the BBB; pharmacological strategies designed to increase the lipid solubility of an agent (e.g., conjugation of water-soluble agents to lipid or cholesterol carriers); and the transitory
25 disruption of the integrity of the BBB by hyperosmotic disruption (resulting from the infusion of a mannitol solution into the carotid artery or the use of a biologically active agent such as an angiotensin peptide). However, each of these strategies has limitations, such as the inherent risks associated with an invasive surgical procedure, a size limitation imposed by a limitation inherent in the endogenous transport systems, potentially undesirable biological side effects associated with the systemic
30 administration of a chimeric molecule comprised of a carrier motif that could be active

outside of the CNS, and the possible risk of brain damage within regions of the brain where the BBB is disrupted, which renders it a suboptimal delivery method.

Alternately, one may administer the pharmaceutical composition in a local rather than systemic manner, for example, via injection of the pharmaceutical composition directly into a tissue region of a patient.

Pharmaceutical compositions of the present invention may be manufactured by processes well known in the art, e.g., by means of conventional mixing, dissolving, granulating, dragee-making, levigating, emulsifying, encapsulating, entrapping or lyophilizing processes.

Pharmaceutical compositions for use in accordance with the present invention thus may be formulated in conventional manner using one or more physiologically acceptable carriers comprising excipients and auxiliaries, which facilitate processing of the active ingredients into preparations which, can be used pharmaceutically. Proper formulation is dependent upon the route of administration chosen.

For injection, the active ingredients of the pharmaceutical composition may be formulated in aqueous solutions, preferably in physiologically compatible buffers such as Hank's solution, Ringer's solution, or physiological salt buffer. For transmucosal administration, penetrants appropriate to the barrier to be permeated are used in the formulation. Such penetrants are generally known in the art.

For oral administration, the pharmaceutical composition can be formulated readily by combining the active compounds with pharmaceutically acceptable carriers well known in the art. Such carriers enable the pharmaceutical composition to be formulated as tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspensions, and the like, for oral ingestion by a patient. Pharmacological preparations for oral use can be made using a solid excipient, optionally grinding the resulting mixture, and processing the mixture of granules, after adding suitable auxiliaries if desired, to obtain tablets or dragee cores. Suitable excipients are, in particular, fillers such as sugars, including lactose, sucrose, mannitol, or sorbitol; cellulose preparations such as, for example, maize starch, wheat starch, rice starch, potato starch, gelatin, gum tragacanth, methyl cellulose, hydroxypropylmethyl-cellulose, sodium carbomethylcellulose; and/or physiologically acceptable polymers such as polyvinylpyrrolidone (PVP). If desired,

disintegrating agents may be added, such as cross-linked polyvinyl pyrrolidone, agar, or alginic acid or a salt thereof such as sodium alginate.

Dragee cores are provided with suitable coatings. For this purpose, concentrated sugar solutions may be used which may optionally contain gum arabic, talc, polyvinyl pyrrolidone, carbopol gel, polyethylene glycol, titanium dioxide, lacquer solutions and suitable organic solvents or solvent mixtures. Dyestuffs or pigments may be added to the tablets or dragee coatings for identification or to characterize different combinations of active compound doses.

Pharmaceutical compositions which can be used orally, include push-fit capsules made of gelatin as well as soft, sealed capsules made of gelatin and a plasticizer, such as glycerol or sorbitol. The push-fit capsules may contain the active ingredients in admixture with filler such as lactose, binders such as starches, lubricants such as talc or magnesium stearate and, optionally, stabilizers. In soft capsules, the active ingredients may be dissolved or suspended in suitable liquids, such as fatty oils, liquid paraffin, or liquid polyethylene glycols. In addition, stabilizers may be added. All formulations for oral administration should be in dosages suitable for the chosen route of administration.

For buccal administration, the compositions may take the form of tablets or lozenges formulated in conventional manner.

For administration by nasal inhalation, the active ingredients for use according to the present invention are conveniently delivered in the form of an aerosol spray presentation from a pressurized pack or a nebulizer with the use of a suitable propellant, e.g., dichlorodifluoromethane, trichlorofluoromethane, dichloro-tetrafluoroethane or carbon dioxide. In the case of a pressurized aerosol, the dosage unit may be determined by providing a valve to deliver a metered amount. Capsules and cartridges of, e.g., gelatin for use in a dispenser may be formulated containing a powder mix of the compound and a suitable powder base such as lactose or starch.

The pharmaceutical composition described herein may be formulated for parenteral administration, e.g., by bolus injection or continuous infusion. Formulations for injection may be presented in unit dosage form, e.g., in ampoules or in multidose containers with optionally, an added preservative. The compositions may be suspensions, solutions or emulsions in oily or aqueous vehicles, and may contain formulatory agents such as suspending, stabilizing and/or dispersing agents.

Pharmaceutical compositions for parenteral administration include aqueous solutions of the active preparation in water-soluble form. Additionally, suspensions of the active ingredients may be prepared as appropriate oily or water based injection suspensions. Suitable lipophilic solvents or vehicles include fatty oils such as sesame oil, or synthetic fatty acids esters such as ethyl oleate, triglycerides or liposomes. Aqueous injection suspensions may contain substances, which increase the viscosity of the suspension, such as sodium carboxymethyl cellulose, sorbitol or dextran. Optionally, the suspension may also contain suitable stabilizers or agents which increase the solubility of the active ingredients to allow for the preparation of highly concentrated solutions.

Alternatively, the active ingredient may be in powder form for constitution with a suitable vehicle, e.g., sterile, pyrogen-free water based solution, before use.

The pharmaceutical composition of the present invention may also be formulated in rectal compositions such as suppositories or retention enemas, using, e.g., conventional suppository bases such as cocoa butter or other glycerides.

Pharmaceutical compositions suitable for use in context of the present invention include compositions wherein the active ingredients are contained in an amount effective to achieve the intended purpose. More specifically, a therapeutically effective amount means an amount of active ingredients (the downregulating agent) effective to prevent, alleviate or ameliorate symptoms of a disorder (e.g., RRMS) or prolong the survival of the subject being treated.

Determination of a therapeutically effective amount is well within the capability of those skilled in the art, especially in light of the detailed disclosure provided herein.

For any preparation used in the methods of the invention, the therapeutically effective amount or dose can be estimated initially from in vitro and cell culture assays. For example, a dose can be formulated in animal models to achieve a desired concentration or titer. Such information can be used to more accurately determine useful doses in humans.

Toxicity and therapeutic efficacy of the active ingredients described herein can be determined by standard pharmaceutical procedures in vitro, in cell cultures or experimental animals. The data obtained from these in vitro and cell culture assays and animal studies can be used in formulating a range of dosage for use in human. The

dosage may vary depending upon the dosage form employed and the route of administration utilized. The exact formulation, route of administration and dosage can be chosen by the individual physician in view of the patient's condition. (See e.g., Fingl, et al., 1975, in "The Pharmacological Basis of Therapeutics", Ch. 1 p.1).

5 Dosage amount and interval may be adjusted individually to provide levels of the active ingredient are sufficient to induce or suppress the biological effect (minimal effective concentration, MEC). The MEC will vary for each preparation, but can be estimated from in vitro data. Dosages necessary to achieve the MEC will depend on individual characteristics and route of administration. Detection assays can be used to
10 determine plasma concentrations.

Depending on the severity and responsiveness of the condition to be treated, dosing can be of a single or a plurality of administrations, with course of treatment lasting from several days to several weeks or until cure is effected or diminution of the disease state is achieved.

15 The amount of a composition to be administered will, of course, be dependent on the subject being treated, the severity of the affliction, the manner of administration, the judgment of the prescribing physician, etc.

Compositions of the present invention may, if desired, be presented in a pack or dispenser device, such as an FDA approved kit, which may contain one or more unit
20 dosage forms containing the active ingredient. The pack may, for example, comprise metal or plastic foil, such as a blister pack. The pack or dispenser device may be accompanied by instructions for administration. The pack or dispenser may also be accommodated by a notice associated with the container in a form prescribed by a governmental agency regulating the manufacture, use or sale of pharmaceuticals, which
25 notice is reflective of approval by the agency of the form of the compositions or human or veterinary administration. Such notice, for example, may be of labeling approved by the U.S. Food and Drug Administration for prescription drugs or of an approved product insert. Compositions comprising a preparation of the invention formulated in a compatible pharmaceutical carrier may also be prepared, placed in an appropriate
30 container, and labeled for treatment of an indicated condition, as is further detailed above.

According to an aspect of some embodiments of the invention, there is provided

a method of selecting a drug for treating a typical relapsing remitting multiple sclerosis (RRMS) in a subject, the method is effected by (a) contacting cells of a subject classified as having a typical RRMS with a plurality of drug molecules, (b) identifying at least one drug molecule which downregulates a level of expression of at least one gene involved in the RNA polymerase I pathway, the at least one drug molecule is suitable for treating the typical RRMS in the subject, thereby selecting the drug for treating the typical RRMS in the subject.

The plurality of drug molecules can be peptides, RNA, DNA, aptamers and small molecules.

According to some embodiments of the invention the polynucleotides described hereinabove (e.g., oligonucleotides) can form a part of a probeset.

According to some embodiments of the invention, the probeset comprises a plurality of oligonucleotides and no more than 50 oligonucleotides (e.g., no more than about 40, no more than about 30, e.g., no more than about 20, e.g., no more than about 15, e.g., no more than about 10 oligonucleotides) wherein an oligonucleotide of the plurality of oligonucleotides specifically recognizes a polynucleotide of at least one gene involved in the RNA polymerase pathway. For example, each of the oligonucleotides can specifically recognize a polynucleotide of the RNA polymerase I pathway.

According to some embodiments of the invention the probeset comprises a plurality of oligonucleotides and no more than 500 oligonucleotides wherein each of the plurality of oligonucleotides is capable of specifically recognizing at least one polynucleotide sequence selected from the group consisting of C22orf8, TLK1, HNRPH1, PLXDC1, TLK1, PKN2, ALS2CR8, FLJ12547, ZNF238, PDPR, NT5E, PASK, HPGD, IL6ST, JARID1A, PASK, LEF1, FLJ10246, MTUS1, FLJ14011, VSIG4, MARCH-VI, FLJ10613, EWSR1, ATP8A1, SLC4A7, FLJ21127, HNRPH1, ABLIM1, ITGA6, ADCY9, CROCC, SH3YL1, SMA4, SPTBN1, DPEP3, PDE3B, AF5Q31, NRCAM, DOCK9, IPW, FLJ20152, SIRPB2, GALNT4, CD28, TXK, ETS1, DGCR5, ZNF192, TCF7, CAMK4, SIM2, MGEA5, TGFBR2, RET, MAPK8IP3, RRN3, DKFZp547H025, FBXW11, ZNF423, DLG1, MGC17330, CD164L1, REPS1, ACHE, ITGB1BP2, LOC94431, LTK, RUNX1, EVER1, KIAA2010, CEACAM7, STX16, SLC4A5, CRTAP, RECQL5, MAGEF1, VIPR1, FLJ10979, TTC3, CRSP2,

BAZ2A, GTF2I, MGC50853, KIAA0508, BPHL, LTBP4, FN3KRP, SCARB1, MGC17330, HYAL4, DGKA, FLJ11196, DHRS6, EPHB4, IDI2/GTPBP4, SNTG2, SLC7A6, PMS2L2, KIAA0436, TOSO, THRAP3, T3JAM, LOC283232, LOC92482, PTER, ATM, NUCB2, PIK3R2, MGC1136, CD59, JARID1A, FLJ39616, ABLIM1, 5 PBP, MAPK8IP3, FTS, LHX5, TNFRSF7, MYC, PBXIP1, DATF1, HTF9C, PUS1, KIAA0924, C6orf4, KIAA0372, WDR42A, CRYZL1, TERE1, LTBP4, TTC3, NFATC1, POM121/LOC340318, TOSO, LOC348926/MGC16279/SB153/FLJ10661, SPOCK2, KIAA0515, SLC37A4, CD44, SMARCA2, SPTBN1, C6orf130, TTC3, DLG1, SLC35E2, MCCC1, PMS2L11, RCN3, STX16, FLJ20618, STAT5B, 10 SMARCA2, SATB1, POLR1D, ASXL1, REV1L, PMS2L2/PMS2L5, FLJ12355, CCNB1IP1, FLJ12270, KIAA0692, MCM7, GPSN2, STX16, MMS19L, GTF2I/GTF2IP1, AKAP7, ZNF444, SLC35A3, MGEA5, RUTBC3, C20orf36, RAD17, ALG12, LOC112869, C6orf48, CUTC, LGTN, DEF6, WAC, HNRPH3, NS, KIAA0892, LRPPRC, HMG20A, DDX42, TINP1, ZDHHC17, C19orf2, EIF4B, 15 LOC376745, DKFZP434C171, TH1L, C19orf13, RPL22, PHF15, EWSR1, EIF4B, FAM48A, YT521, NEK9, EIF3S7, RPS6, RPL35A, EEF2, RPL3, RPS6, UBA52, RPL6, RPS6, RPL13, AL353949, AL580863, AF052160, AW128846, AW974481, N92920, BG178274, AW303460, BF057458, AL050035, M59917, AK025422, AI693985, AU158442, AK021460, AL023773, NM_003790, AC005011, M90355, 20 AL353580, U38964, D50683, BE967207, YWHAB, ATP6V1E1, UBB, MRCL2/MRCL3, UQCR, MRCL2, RTN4, UBE2A, RTN4, WDR1, PSMA6, C14orf123, PP1201, TBK1, CAST, CAST, RSN, PSME1, SDF2, GSTO1, CAST, DNCL1, SQRDL, ADIPOR2, ICMT, NDUFA6, NDUFA6, COX17, HIF1AN, FLJ20257, TBPL1, RAPGEF2, CRSP8, APOL1, PAOX, CNBP2, ETFB, DPP3, 25 KPNA1, MGC3036, TUBB2, PDCL, CCL5, CDS2, RAP1GDS1, ATP6V1D, OBRGRP/LEPR, SF4, GCLC, MGST3, BICD2, BRF1, CHST12, EXOSC7, TOR1B, ZFP95, ILK, UNC13A, MTHFD2, CASP10, FLJ45850, CMRF-35H, ARF3, NDOR1, DUSP10, AP1M2, VRK2, GSN, PTRF, RBM19, RABGAP1L, ATP5S, STOM, TFPI2, SLC3A1, PTPN12, CSF1, SIGLEC6, KIRREL, OBRGRP, TP53AP1, SUHW1, 30 NUP98, IL15RA, MICB, CMRF-35H, SPHK1, TNFRSF6, FLJ11301, LRP5, STOM, EPHA2, SRC, FLJ11301, PSTPIP2, EBP, MCPH1, PTPRF, LIMK2, FSTL4, CBR1, MGC2654, MYCT1, NOL3, MTF, ATP10B, FBXO31, TBX21, LSS, SLC17A3,

MNAB, CHPPR, GIF, VAMP5, ABCG2, KIF1B, LOH11CR2A, NID2, RBBP8, ETV7, CTSL, RUFY1, RSU1, PARD3, APOB, ACOX3, DAB2, LDLR, TJP2, GNAS, PARD3, NCKAP1, TAP2, HDGFRP3, LDLR, PIK3R3, HTR2B, GAS2L1, FER1L3, C3orf14, TP53TG3, LEPR, CLIC5, PDE4DIP, ATP9A, ITGB1BP1, INDO, SELP, FHL2, FER1L3, EGF, SIAT8A, HDGFRP3, LRAP, VWF, FLJ10134, IMP-3, DMN, MCTP1, FSTL1, CTNNAL1, RAB27B, THBS1, PROS1, MMRN1, CTTN, AL078596, AI148659, U00956 and M29383.

It will be appreciated that the isolated nucleic acid sequences included in the kit or the probeset of the present invention can be bound to a solid support e.g., a glass wafer in a specific order, *i.e.*, in the form of a microarray. Alternatively, isolated nucleic acid sequences can be synthesized directly on the solid support using well known prior art approaches (Seo TS, et al., 2004, Proc. Natl. Acad. Sci. USA, 101: 5488-93.). In any case, the isolated nucleic acid sequences are attached to the support in a location specific manner such that each specific isolated nucleic acid sequence has a specific address on the support (*i.e.*, an addressable location) which denotes the identity (*i.e.*, the sequence) of that specific isolated nucleic acid sequence.

The kit may further include a positive control for an expression level of at least one of the polynucleotides of the invention (e.g., which involves in the RNA polymerase I pathway). The positive control can be any biological sample derived from a reference subject (*i.e.*, a subject with a known course of MS, *i.e.*, BMS or typical RRMS), a biological sample with known amount/concentration of the gene product (*i.e.*, RNA or protein) of at least one of the polynucleotides of the invention; or a pre-determined level (amount/concentration) of purified, chemically synthesized or recombinantly generated RNA or protein molecules (gene products) of the at least one polynucleotide of the invention. The kit may further comprise instructions for use in classifying a subject as being more like to have BMS or typical RRMS, to diagnose BMS or typical RRMS, to monitor treatment efficiency, to select a treatment regimen, to treat a subject having multiple sclerosis and/or to select for drugs suitable for treating multiple sclerosis.

As used herein the term “about” refers to $\pm 10\%$.

The terms "comprises", "comprising", "includes", "including", "having" and their conjugates mean "including but not limited to".

The term "consisting of means "including and limited to".

5 The term "consisting essentially of" means that the composition, method or structure may include additional ingredients, steps and/or parts, but only if the additional ingredients, steps and/or parts do not materially alter the basic and novel characteristics of the claimed composition, method or structure.

10 As used herein, the singular form "a", "an" and "the" include plural references unless the context clearly dictates otherwise. For example, the term "a compound" or "at least one compound" may include a plurality of compounds, including mixtures thereof.

15 Throughout this application, various embodiments of this invention may be presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the invention. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual numerical values within that range. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed subranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6 etc., as well as individual numbers within that range, for example, 1, 2, 3, 4, 5, and 6. This applies
20 regardless of the breadth of the range.

Whenever a numerical range is indicated herein, it is meant to include any cited numeral (fractional or integral) within the indicated range. The phrases "ranging/ranges between" a first indicate number and a second indicate number and "ranging/ranges from" a first indicate number "to" a second indicate number are used herein
25 interchangeably and are meant to include the first and second indicated numbers and all the fractional and integral numerals therebetween.

As used herein the term "method" refers to manners, means, techniques and procedures for accomplishing a given task including, but not limited to, those manners, means, techniques and procedures either known to, or readily developed from known manners, means, techniques and procedures by practitioners of the chemical, pharmacological, biological, biochemical and medical arts.
30

It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable subcombination or as suitable in any other described embodiment of the invention. Certain features described in the context of various embodiments are not to be considered essential features of those embodiments, unless the embodiment is inoperative without those elements.

Various embodiments and aspects of the present invention as delineated hereinabove and as claimed in the claims section below find experimental support in the following examples.

EXAMPLES

Reference is now made to the following examples, which together with the above descriptions illustrate some embodiments of the invention in a non limiting fashion.

Generally, the nomenclature used herein and the laboratory procedures utilized in the present invention include molecular, biochemical, microbiological and recombinant DNA techniques. Such techniques are thoroughly explained in the literature. See, for example, "Molecular Cloning: A laboratory Manual" Sambrook et al., (1989); "Current Protocols in Molecular Biology" Volumes I-III Ausubel, R. M., ed. (1994); Ausubel et al., "Current Protocols in Molecular Biology", John Wiley and Sons, Baltimore, Maryland (1989); Perbal, "A Practical Guide to Molecular Cloning", John Wiley & Sons, New York (1988); Watson et al., "Recombinant DNA", Scientific American Books, New York; Birren et al. (eds) "Genome Analysis: A Laboratory Manual Series", Vols. 1-4, Cold Spring Harbor Laboratory Press, New York (1998); methodologies as set forth in U.S. Pat. Nos. 4,666,828; 4,683,202; 4,801,531; 5,192,659 and 5,272,057; "Cell Biology: A Laboratory Handbook", Volumes I-III Cellis, J. E., ed. (1994); "Current Protocols in Immunology" Volumes I-III Coligan J. E., ed. (1994); Stites et al. (eds), "Basic and Clinical Immunology" (8th Edition), Appleton & Lange, Norwalk, CT (1994); Mishell and Shiigi (eds), "Selected Methods in Cellular Immunology", W. H. Freeman and Co., New York (1980); available immunoassays are extensively described in the patent and scientific literature, see, for example, U.S. Pat.

Nos. 3,791,932; 3,839,153; 3,850,752; 3,850,578; 3,853,987; 3,867,517; 3,879,262; 3,901,654; 3,935,074; 3,984,533; 3,996,345; 4,034,074; 4,098,876; 4,879,219; 5,011,771 and 5,281,521; "Oligonucleotide Synthesis" Gait, M. J., ed. (1984); "Nucleic Acid Hybridization" Hames, B. D., and Higgins S. J., eds. (1985); "Transcription and Translation" Hames, B. D., and Higgins S. J., Eds. (1984); "Animal Cell Culture" Freshney, R. I., ed. (1986); "Immobilized Cells and Enzymes" IRL Press, (1986); "A Practical Guide to Molecular Cloning" Perbal, B., (1984) and "Methods in Enzymology" Vol. 1-317, Academic Press; "PCR Protocols: A Guide To Methods And Applications", Academic Press, San Diego, CA (1990); Marshak et al., "Strategies for Protein Purification and Characterization - A Laboratory Course Manual" CSHL Press (1996); all of which are incorporated by reference as if fully set forth herein. Other general references are provided throughout this document. The procedures therein are believed to be well known in the art and are provided for the convenience of the reader. All the information contained therein is incorporated herein by reference.

GENERAL MATERIALS AND EXPERIMENTAL METHODS

Subjects – 31 patients (age 44.5 ± 1.5 ; female to male ratio 24:7) with BMS were characterized by mean EDSS 1.95 ± 0.15 , disease duration 17.0 ± 1.3 years, annual EDSS rate 0.13 ± 0.01 , annual relapse rate 0.23 ± 0.04 . 36 patients (age 40.3 ± 1.8 ; female to male ratio 8:3) with typical RRMS were characterized by mean EDSS 3.54 ± 0.23 , disease duration 10.9 ± 1.4 years, annual EDSS rate 0.45 ± 0.06 , annual relapse rate 0.64 ± 0.09 .

RNA isolation and microarray expression profiling - Peripheral blood mononuclear cells (PBMC) were separated on ficoll-hypaque gradient. Total RNA was isolated using the TRIzol Reagent (Invitrogen, Carlsbad, CA), and cDNA was synthesized, labeled and hybridized to HG-U133A-2 array (Affymetrix, Inc, Santa Clara, CA) containing 22,215 gene-transcripts, washed and scanned (Hewlett Packard, GeneArray-TM scanner G2500A) according to manufacturer's protocol Affymetrix (Inc, Santa Clara, CA).

Data Analysis - Data analysis was performed using the Partek Genomics Solution software [World Wide Web (dot) partek (dot) com]. Expression values were computed from raw CEL (cell) files by applying the Robust Multi-Chip Average

(RMA) background correction algorithm. The RMA correction included: 1) values background correction; 2) quantile normalization; 3) log2 transformation; 4) median polish summarization. In order to avoid the noise caused by variable set effects each set was normalized to pre-saved distribution pattern of a well balanced set used as a reference distribution. To reduce batch effect ANOVA multiple model analysis was applied. Source of variation was analyzed; nuisance batches effects such as working batch, patient age, gender and treatment were eliminated. Most informative genes (MIGs) were defined as genes with $p < 0.01$ by ANOVA linear contrasts model. For samples classification, principal component analyses (PCA) were performed.

Gene functional annotation, enrichment and pathway analysis were performed using functional classification tools, David Bioinformatics Resources [Hypertext Transfer Protocol:// david (dot) abcc (dot) ncifcrf (dot) gov/home (dot) jsp], and Ingenuity Pathways Analysis web-software [World Wide Web (dot) ingenuity (dot) com]. Enrichment was defined as significantly ($p < 0.05$) higher proportion of genes in a given gene set than expected by chance analysis. The study design is demonstrated in Figure 1.

EXAMPLE 1

Experimental Results

Identification of differentiating genes between patients with BMS and patients with typical RRMS – BMS patients differentiated from typical RRMS by 406 MIGs (most informative genes), 171 genes were over-expressed (upregulated) and 235 were down-expressed (downregulated), with the log fold change ranged from -3.1 to 3.3 (Figure 2).

Table 1 hereinbelow provides the differentiating genes between BMS and typical RRMS patients.

Table 1

Genes which are differentially expressed in blood samples of benign multiple sclerosis (BMS) and typical relapsing-remitting multiple sclerosis (RRMS) subjects

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
216683_a t	1	AL353949	407	6.85E-04	-1.129	---	---
219629_a t	2	NM_01791 1	408	5.51E-03	-1.117	C22orf8	chromosome 22 open reading frame 8
210379_s _at	3	AI469203	409	7.42E-03	-1.083	TLK1	tousled-like kinase 1
213472_a t	4	AI022387	410	4.25E-03	-1.081	HNRPH1	heterogeneous nuclear ribonucleoprotein H1 (H)
219700_a t	5	NM_02040 5	411	5.56E-05	-1.081	PLXDC1	plexin domain containing 1
211077_s _at	6	Z25421	412	9.89E-04	-1.080	TLK1	tousled-like kinase 1 /// tousled-like kinase 1
210969_a t	7	AF118089	413	4.20E-04	-1.078	PKN2	protein kinase N2
216298_a t	8	AL580863	414	3.57E-03	-1.075	---	Similar to T-cell receptor gamma chain V region PT- gamma-1/2 precursor /// Simil
219834_a t	9	NM_02474 4	415	1.20E-03	-1.069	ALS2CR8	amyotrophic lateral sclerosis 2 (juvenile) chromosome region, candidate 8
215262_a t	10	AF052160	416	2.29E-04	-1.068	---	Clone 24629 mRNA sequence
220715_a t	11	NM_02499 2	417	1.53E-04	-1.067	FLJ12547	hypothetical protein FLJ12547
207164_s _at	12	NM_00635 2	418	4.17E-03	-1.067	ZNF238	zinc finger protein 238
220236_a t	13	NM_01799 0	419	7.14E-04	-1.067	PDPR	pyruvate dehydrogenase phosphatase regulatory subunit
203939_a t	14	NM_00252 6	420	1.84E-03	-1.065	NT5E	5'-nucleotidase, ecto (CD73)
216945_x _at	15	U79240	421	2.45E-03	-1.063	PASK	PAS domain containing serine/threonine kinase
203913_s _at	16	NM_00086 0	422	4.09E-03	-1.062	HPGD	hydroxyprostaglandi n dehydrogenase 15- (NAD)

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
204864_s _at	17	BE856546	423	9.98E-03	-1.060	IL6ST	interleukin 6 signal transducer (gp130, oncostatin M receptor)
215698_a _t	18	AF007135	424	3.73E-03	-1.057	JARID1A	Jumonji, AT rich interactive domain 1A (RBBP2-like)
213534_s _at	19	D50925	425	3.86E-03	-1.056	PASK	PAS domain containing serine/threonine kinase
210948_s _at	20	AF294627	426	1.98E-03	-1.056	LEF1	lymphoid enhancer- binding factor 1
220458_a _t	21	NM_01803 8	427	8.30E-03	-1.055	FLJ10246	hypothetical protein FLJ10246
212093_s _at	22	AI695017	428	8.37E-03	-1.052	MTUS1	mitochondrial tumor suppressor 1
207120_a _t	23	NM_02210 3	429	1.75E-03	-1.050	FLJ14011	hypothetical zinc finger protein FLJ14011
204787_a _t	24	NM_00726 8	430	3.61E-03	-1.050	VSIG4	V-set and immunoglobulin domain containing 4
215512_a _t	25	AK000970	431	1.74E-03	-1.049	MARCH-VI	membrane- associated RING- CH protein VI
46947_at	26	T87245	432	1.23E-03	-1.049	FLJ10613	Hypothetical protein FLJ10613
210012_s _at	27	BC000527	433	7.10E-03	-1.048	EWSR1	Ewing sarcoma breakpoint region 1
210192_a _t	28	AB013452	434	7.32E-03	-1.048	ATP8A1	ATPase, aminophospholipid transporter (APLT), Class I, type 8A, member 1
207603_a _t	29	NM_00361 5	435	2.30E-03	-1.047	SLC4A7	solute carrier family 4, sodium bicarbonate cotransporter, member 7
218584_a _t	30	NM_02454 9	436	7.79E-03	-1.047	FLJ21127	hypothetical protein FLJ21127
213470_s _at	31	BF983406	437	8.91E-03	-1.047	HNRPH1	heterogeneous nuclear ribonucleoprotein H1 (H)
210461_s _at	32	BC002448	438	1.63E-03	-1.047	ABLIM1	actin binding LIM protein 1
217656_a _t	33	AW128846	439	5.27E-04	-1.045	---	---

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
215177_s _at	34	AV733308	440	4.01E-04	-1.045	ITGA6	integrin, alpha 6
204498_s _at	35	NM_00111 6	441	3.72E-04	-1.043	ADCY9	adenylate cyclase 9
216419_a _t	36	AK026910	442	7.82E-03	-1.043	CROCC	ciliary rootlet coiled-coil, rootletin
204019_s _at	37	NM_01567 7	443	1.11E-03	-1.043	SH3YL1	SH3 domain containing, Ysc84- like 1 (S. cerevisiae)
214850_a _t	38	X75940	444	4.48E-03	-1.043	SMA4	SMA4
200672_x _at	39	NM_00312 8	445	1.59E-03	-1.043	SPTBN1	spectrin, beta, non- erythrocytic 1
220179_a _t	40	NM_02235 7	446	1.40E-03	-1.042	DPEP3	dipeptidase 3
208591_s _at	41	NM_00092 2	447	6.54E-05	-1.042	PDE3B	phosphodiesterase 3B, cGMP-inhibited
219199_a _t	42	NM_01442 3	448	5.72E-03	-1.042	AF5Q31	ALL1 fused gene from 5q31
216959_x _at	43	U55258	449	1.65E-03	-1.041	NRCAM	neuronal cell adhesion molecule
215041_s _at	44	BE259050	450	6.36E-03	-1.040	DOCK9	dedicator of cytokinesis 9
213447_a _t	45	AI672541	451	5.32E-05	-1.040	IPW	imprinted in Prader- Willi syndrome
218532_s _at	46	NM_01900 0	452	4.80E-03	-1.040	FLJ20152	hypothetical protein FLJ20152
220485_s _at	47	NM_01855 6	453	1.56E-03	-1.039	SIRPB2	signal-regulatory protein beta 2
220442_a _t	48	NM_00377 4	454	8.94E-03	-1.039	GALNT4	UDP-N-acetyl- alpha-D- galactosamine:polyp eptide N- acetylgalactosaminy ltransferase
211856_x _at	49	AF222341	455	9.84E-03	-1.039	CD28	CD28 antigen (Tp44)
206828_a _t	50	NM_00332 8	456	1.62E-03	-1.039	TXK	TXK tyrosine kinase
214447_a _t	51	NM_00523 8	457	3.28E-03	-1.038	ETS1	v-ets erythroblastosis virus E26 oncogene homolog 1 (avian)
215244_a _t	52	AI479306	458	6.97E-03	-1.038	DGCR5	DiGeorge syndrome critical region gene 5 (non-coding)
206579_a _t	53	NM_00629 8	459	5.64E-03	-1.038	ZNF192	zinc finger protein 192
205254_x _at	54	AW027359	460	3.05E-03	-1.038	TCF7	transcription factor 7 (T-cell specific, HMG-box)

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
210349_a t	55	L24959	461	3.07E-03	-1.037	CAMK4	calcium/calmodulin- dependent protein kinase IV
208157_a t	56	NM_00958 6	462	9.47E-03	-1.037	SIM2	single-minded homolog 2 (Drosophila)
214972_a t	57	AU144791	463	4.25E-03	-1.037	MGEA5	Meningioma expressed antigen 5 (hyaluronidase)
207334_s _at	58	NM_00324 2	464	2.70E-03	-1.037	TGFBR2	transforming growth factor, beta receptor II (70/80kDa)
217666_a t	59	AW974481	465	1.06E-03	-1.037	---	---
215771_x _at	60	X15786	466	1.14E-03	-1.036	RET	ret proto-oncogene (multiple endocrine neoplasia and medullary thyroid carcinoma)
216139_s _at	61	AL031718	467	7.40E-03	-1.036	MAPK8IP3	mitogen-activated protein kinase 8 interacting protein 3
216902_s _at	62	AF001549	468	4.86E-05	-1.036	RRN3	RRN3 RNA polymerase I transcription factor homolog (yeast)
208265_a t	63	NM_02016 1	469	6.06E-04	-1.036	DKFZp547 H025	hypothetical protein DKFZp547H025
209456_s _at	64	AB033281	470	3.49E-03	-1.036	FBXW11	F-box and WD-40 domain protein 11
217237_a t	65	Y10615	471	7.72E-03	-1.035	ZNF423	Zinc finger protein 423
217208_s _at	66	AL121981	472	2.91E-03	-1.035	DLG1	discs, large homolog 1 (Drosophila)
221757_a t	67	BE042976	473	1.83E-04	-1.035	MGC17330	HGFL gene /// HGFL gene
213481_a t	68	N92920	474	4.47E-03	-1.035	---	---
219025_a t	69	NM_02040 4	475	2.84E-03	-1.035	CD164L1	CD164 sialomucin- like 1
215201_a t	70	AW166925	476	9.15E-03	-1.034	REPS1	RALBP1 associated Eps domain containing 1
205378_s _at	71	NM_01583 1	477	6.79E-03	-1.034	ACHE	acetylcholinesterase (YT blood group)
219829_a t	72	NM_01227 8	478	6.24E-03	-1.033	ITGB1BP2	integrin beta 1 binding protein (melusin) 2

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
216908_x _at	73	AF001549	468	6.22E-04	-1.033	LOC94431	similar to RNA polymerase I transcription factor RRN3
217184_s _at	74	X52213	479	4.40E-03	-1.033	LTK	leukocyte tyrosine kinase
211181_x _at	75	AF312386	480	4.63E-03	-1.033	RUNX1	runt-related transcription factor 1 (acute myeloid leukemia 1; aml1 oncogene)
214958_s _at	76	AK021738	481	1.62E-03	-1.033	EVER1	epidermodysplasia verruciformis 1
220369_a _t	77	NM_01793 6	482	7.90E-03	-1.033	KIAA2010	KIAA2010
211848_s _at	78	AF006623	483	8.77E-04	-1.033	CEACAM7	carcinoembryonic antigen-related cell adhesion molecule 7
221638_s _at	79	AF008937	484	8.81E-03	-1.033	STX16	syntaxin 16
221723_s _at	80	AF243499	485	8.08E-03	-1.033	SLC4A5	solute carrier family 4, sodium bicarbonate cotransporter, member 5
201380_a _t	81	NM_00637 1	486	8.45E-04	-1.032	CRTAP	cartilage associated protein
34063_at	82	AB006533	487	1.03E-03	-1.032	RECQL5	RecQ protein-like 5
214757_a _t	83	BG178274	488	5.20E-03	-1.032	---	Hypothetical gene supported by AK024602
218176_a _t	84	NM_02214 9	489	4.04E-03	-1.032	MAGEF1	melanoma antigen, family F, 1
221977_a _t	85	AW303460	490	7.35E-03	-1.032	---	---
214161_a _t	86	BF057458	491	9.56E-03	-1.031	---	---
214857_a _t	87	AL050035	492	6.21E-03	-1.031	---	MRNA; cDNA DKFZp566H0124 (from clone DKFZp566H0124)
216230_x _at	88	M59917	493	1.74E-03	-1.031	---	---
205019_s _at	89	NM_00462 4	494	3.56E-03	-1.031	VIPR1	vasoactive intestinal peptide receptor 1
221707_s _at	90	BC006116	495	7.60E-03	-1.031	FLJ10979	hypothetical protein FLJ10979 /// hypothetical protein FLJ10979
208664_s _at	91	AU131711	496	1.15E-04	-1.030	TTC3	tetratricopeptide repeat domain 3

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
215167_a t	92	BE567032	497	8.69E-03	-1.030	CRSP2	cofactor required for Sp1 transcriptional activation, subunit 2, 150kDa
215437_x _at	93	BE513659	498	4.06E-03	-1.030	BAZ2A	bromodomain adjacent to zinc finger domain, 2A
210892_s _at	94	BC004472	499	3.77E-03	-1.029	GTF2I	general transcription factor II, i
212400_a t	95	AL043266	500	1.57E-03	-1.029	MGC50853	hypothetical protein MGC50853
215137_a t	96	H92070	501	5.46E-03	-1.029	KIAA0508	KIAA0508 protein
205750_a t	97	NM_00433 2	502	1.95E-03	-1.028	BPHL	biphenyl hydrolase- like (serine hydrolase; breast epithelial mucin- associated an
210628_x _at	98	AF051344	503	1.93E-03	-1.028	LTBP4	latent transforming growth factor beta binding protein 4
218210_a t	99	NM_02461 9	504	1.42E-03	-1.028	FN3KRP	fructosamine-3- kinase-related protein
216784_a t	100	AK025422	505	2.68E-03	-1.028	---	Transcribed locus, weakly similar to XP_375174.1 hypothetical gene supported by
201819_a t	101	NM_00550 5	506	7.48E-03	-1.028	SCARB1	scavenger receptor class B, member 1
214312_a t	102	AI693985	507	2.93E-03	-1.028	---	---
215556_a t	103	AU158442	508	3.90E-03	-1.028	---	---
221756_a t	104	BE042976	473	2.52E-03	-1.027	MGC17330	HGFL gene /// HGFL gene
216909_a t	105	AK021460	509	5.34E-03	-1.027	---	---
220249_a t	106	NM_01226 9	510	8.17E-03	-1.027	HYAL4	hyaluronoglucosami nidase 4
211272_s _at	107	AF064771	511	3.46E-03	-1.027	DGKA	diacylglycerol kinase, alpha 80kDa
218651_s _at	108	NM_01835 7	512	7.30E-03	-1.027	FLJ11196	acheron
218285_s _at	109	NM_02013 9	513	2.93E-03	-1.027	DHRS6	dehydrogenase/redu ctase (SDR family) member 6
217385_a t	110	AL023773	514	2.79E-03	-1.027	---	---

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
202894_a t	111	NM_00444 4	515	1.52E-03	-1.026	EPHB4	EPH receptor B4
217631_a t	112	AI081107	516	6.97E-03	-1.026	IDI2 /// GTPBP4	isopentenyl- diphosphate delta isomerase 2 /// GTP binding protein 4
220487_a t	113	NM_01896 8	517	9.34E-03	-1.026	SNTG2	syntrophin, gamma 2
203579_s _at	114	AI660619	518	1.98E-03	-1.026	SLC7A6	solute carrier family 7 (cationic amino acid transporter, y+ system), member 6
215412_x _at	115	AB017007	519	7.10E-04	-1.026	PMS2L2	postmeiotic segregation increased 2-like 2
212217_a t	116	AU154782	520	1.32E-03	-1.026	KIAA0436	putative prolyl oligopeptidase
221602_s _at	117	AI084226	521	3.58E-04	-1.026	TOSO	regulator of Fas- induced apoptosis /// regulator of Fas- induced apoptosis
217847_s _at	118	NM_00511 9	522	8.43E-03	-1.025	THRAP3	thyroid hormone receptor associated protein 3
215275_a t	119	AW963138	523	9.31E-03	-1.025	T3JAM	TRAF3-interacting Jun N-terminal kinase (JNK)- activating modulator
221951_a t	120	AI739035	524	2.65E-03	-1.025	LOC283232	hypothetical protein LOC283232
213224_s _at	121	AK025724	525	7.84E-05	-1.024	LOC92482	Hypothetical protein LOC92482
218967_s _at	122	BF112019	526	5.08E-03	-1.024	PTER	phosphotriesterase related
208442_s _at	123	NM_00005 1	527	9.87E-04	-1.024	ATM	ataxia telangiectasia mutated (includes complementation groups A, C and D)
203675_a t	124	NM_00501 3	528	5.38E-03	-1.024	NUCB2	nucleobindin 2
207105_s _at	125	NM_00502 7	529	2.15E-03	-1.023	PIK3R2	phosphoinositide-3- kinase, regulatory subunit 2 (p85 beta)
219144_a t	126	NM_02402 5	530	3.23E-03	-1.023	MGC1136	hypothetical protein MGC1136
212463_a t	127	BE379006	531	9.58E-03	-1.023	CD59	CD59 antigen p18- 20 (antigen identified by monoclonal antibodies 16.3A5, EJ16, E

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
202040_s_at	128	NM_005056	532	8.73E-03	-1.023	JARID1A	Jumonji, AT rich interactive domain 1A (RBBP2-like)
64432_at	129	W05463	533	4.53E-03	-1.023	FLJ39616	apoptosis-related protein PNAS-1
200965_s_at	130	NM_006720	534	2.98E-03	-1.022	ABLIM1	actin binding LIM protein 1
205353_s_at	131	NM_002567	535	9.67E-03	-1.022	PBP	prostatic binding protein
213177_at	132	AB028989	536	9.13E-03	-1.022	MAPK8IP3	mitogen-activated protein kinase 8 interacting protein 3
218373_at	133	NM_022476	537	6.35E-03	-1.022	FTS	fused toes homolog (mouse)
208333_at	134	NM_022363	538	4.11E-03	-1.022	LHX5	LIM homeobox 5
206150_at	135	NM_001242	539	8.17E-03	-1.022	TNFRSF7	tumor necrosis factor receptor superfamily, member 7 /// tumor necrosis factor r
202431_s_at	136	NM_002467	540	7.39E-03	-1.022	MYC	v-myc myelocytomatosis viral oncogene homolog (avian)
207838_x_at	137	NM_020524	541	3.63E-03	-1.021	PBXIP1	pre-B-cell leukemia transcription factor interacting protein 1
218325_s_at	138	NM_022105	542	1.46E-03	-1.021	DATF1	death associated transcription factor 1
218475_at	139	NM_022727	543	1.53E-03	-1.021	HTF9C	HpaII tiny fragments locus 9C
218670_at	140	NM_025215	544	1.05E-03	-1.021	PUS1	pseudouridylate synthase 1
205594_at	141	NM_014897	545	8.79E-03	-1.021	KIAA0924	KIAA0924 protein
215411_s_at	142	AL008730	546	7.21E-03	-1.021	C6orf4	chromosome 6 open reading frame 4
203048_s_at	143	NM_014639	547	2.14E-03	-1.020	KIAA0372	KIAA0372
216885_s_at	144	AK026481	548	4.07E-04	-1.020	WDR42A	WD repeat domain 42A
219767_s_at	145	NM_005111	549	3.79E-03	-1.020	CRYZL1	crystallin, zeta (quinone reductase)-like 1
219131_at	146	NM_013319	550	3.21E-03	-1.020	TERE1	transitional epithelia response protein
213176_s_at	147	AI910869	551	6.21E-03	-1.020	LTBP4	latent transforming growth factor beta binding protein 4

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208661_s_at	148	D84294	552	4.85E-03	-1.020	TTC3	tetratricopeptide repeat domain 3
208196_x_at	149	NM_006162	553	9.44E-03	-1.020	NFATC1	nuclear factor of activated T-cells, cytoplasmic, calcineurin-dependent 1
212178_s_at	150	AK022555	554	2.95E-03	-1.020	POM121 /// LOC340318	POM121 membrane glycoprotein (rat) /// hypothetical protein LOC340318
221601_s_at	151	AI084226	521	2.54E-03	-1.020	TOSO	regulator of Fas-induced apoptosis /// regulator of Fas-induced apoptosis
222013_x_at	152	BE348837	555	5.16E-03	-1.019	LOC348926 /// MGC16279 /// SB153 /// FLJ10661	hypothetical protein LOC348926 /// hypothetical protein MGC16279 /// hypotheticala
202524_s_at	153	AI952009	556	5.43E-03	-1.019	SPOCK2	sparc/osteonectin, cwcv and kazal-like domains proteoglycan (testican) 2
212068_s_at	154	AB011087	557	9.23E-03	-1.019	KIAA0515	KIAA0515
202830_s_at	155	NM_001467	558	5.82E-03	-1.019	SLC37A4	solute carrier family 37 (glycerol-6-phosphate transporter), member 4
209835_x_at	156	BC004372	559	5.79E-03	-1.019	CD44	CD44 antigen (homing function and Indian blood group system)
212257_s_at	157	AW131754	560	1.51E-03	-1.019	SMARCA2	SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subf
212071_s_at	158	BE968833	561	2.68E-03	-1.019	SPTBN1	spectrin, beta, non-erythrocytic 1
213322_a_t	159	AL031778	562	8.31E-04	-1.019	C6orf130	chromosome 6 open reading frame 130
210645_s_at	160	D83077	563	2.20E-03	-1.019	TTC3	tetratricopeptide repeat domain 3
202514_a_t	161	AW139131	564	4.70E-03	-1.018	DLG1	DKFZP586B0319 protein
217122_s_at	162	AL031282	565	4.05E-03	-1.018	SLC35E2	solute carrier family 35, member E2

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218440_at	163	NM_020166	566	9.21E-05	-1.018	MCCC1	methylcrotonoyl-Coenzyme A carboxylase 1 (alpha)
210707_x_at	164	U38980	567	3.76E-04	-1.018	PMS2L11	postmeiotic segregation increased 2-like 11
61734_at	165	AI797684	568	3.19E-03	-1.018	RCN3	reticulocalbin 3, EF-hand calcium binding domain
221500_s_at	166	AK026970	569	1.50E-04	-1.018	STX16	syntaxin 16
219422_at	167	NM_003790	570	6.24E-03	-1.018	---	---
222244_s_at	168	AK000749	571	5.14E-03	-1.018	FLJ20618	hypothetical protein FLJ20618
212550_at	169	AI149535	572	1.46E-03	-1.018	STAT5B	signal transducer and activator of transcription 5B
206544_x_at	170	NM_003070	573	5.64E-03	-1.018	SMARCA2	SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subf
216380_x_at	171	AC005011	574	3.97E-03	-1.018	---	---
203408_s_at	172	NM_002971	575	6.31E-03	-1.018	SATB1	special AT-rich sequence binding protein 1 (binds to nuclear matrix/scaffold-ass
218258_at	173	NM_015972	576	2.71E-03	-1.017	POLR1D	polymerase (RNA) I polypeptide D, 16kDa
212234_at	174	AL034550	577	5.94E-03	-1.017	ASXL1	additional sex combs like 1 (Drosophila)
217461_x_at	175	M90355	578	7.56E-03	-1.017	---	---
218428_s_at	176	NM_016316	579	8.87E-03	-1.017	REV1L	REV1-like (yeast)
215667_x_at	177	AI375694	580	1.00E-03	-1.017	PMS2L2 /// PMS2L5	postmeiotic segregation increased 2-like 2 /// postmeiotic segregation increased
220465_at	178	NM_024988	581	2.08E-03	-1.017	FLJ12355	hypothetical protein FLJ12355
217988_at	179	NM_021178	582	9.05E-03	-1.017	CCNB1IP1	cyclin B1 interacting protein 1

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221981_s_at	180	AA702154	583	3.45E-03	-1.017	FLJ12270	hypothetical protein FLJ12270
212201_at	181	AW274877	584	5.96E-03	-1.017	KIAA0692	KIAA0692 protein
208795_s_at	182	D55716	585	1.81E-03	-1.017	MCM7	MCM7 minichromosome maintenance deficient 7 (S. cerevisiae)
208336_s_at	183	NM_004868	586	5.33E-03	-1.017	GPSN2	glycoprotein, synaptic 2
221499_s_at	184	AK026970	569	8.49E-03	-1.017	STX16	syntaxin 16
202167_s_at	185	NM_022362	587	8.82E-03	-1.017	MMS19L	MMS19-like (MET18 homolog, S. cerevisiae)
201065_s_at	186	NM_001518	588	4.02E-03	-1.017	GTF2I /// GTF2IP1	general transcription factor II, i /// general transcription factor II, i, pseud
211172_x_at	187	AF161075	589	2.00E-03	-1.017	AKAP7	A kinase (PRKA) anchor protein 7
218707_at	188	NM_018337	590	7.16E-03	-1.016	ZNF444	zinc finger protein 444
206770_s_at	189	NM_012243	591	9.43E-03	-1.016	SLC35A3	solute carrier family 35 (UDP-N-acetylglucosamine (UDP-GlcNAc) transporter), mem
200898_s_at	190	AK002091	592	2.97E-03	-1.016	MGEA5	meningioma expressed antigen 5 (hyaluronidase)
215519_x_at	191	AI081779	593	3.72E-03	-1.016	RUTBC3	RUN and TBC1 domain containing 3
212406_s_at	192	AB028973	594	4.48E-03	-1.016	C20orf36	chromosome 20 open reading frame 36
210826_x_at	193	AF098533	595	4.28E-03	-1.015	RAD17	RAD17 homolog (S. pombe)
218444_at	194	NM_024105	596	6.99E-03	-1.015	ALG12	asparagine-linked glycosylation 12 homolog (yeast, alpha-1,6-mannosyltransferase
221822_at	195	BE544663	597	4.91E-03	-1.015	LOC112869	hypothetical protein BC011981
220755_s_at	196	NM_016947	598	7.95E-04	-1.015	C6orf48	chromosome 6 open reading frame 48
218970_s_at	197	NM_015960	599	9.75E-03	-1.015	CUTC	cutC copper transporter homolog (E.coli)

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218253_s_at	198	NM_006893	600	5.75E-03	-1.015	LGTN	ligatin
221293_s_at	199	NM_022047	601	2.42E-03	-1.015	DEF6	differentially expressed in FDCP 6 homolog (mouse)
217742_s_at	200	NM_016628	602	5.14E-03	-1.015	WAC	WW domain containing adaptor with coiled-coil
207127_s_at	201	NM_021644	603	4.06E-03	-1.014	HNRPH3	heterogeneous nuclear ribonucleoprotein H3 (2H9)
217850_at	202	NM_014366	604	5.81E-03	-1.014	NS	nucleostemin
212505_s_at	203	AL110250	605	8.73E-03	-1.014	KIAA0892	KIAA0892
211971_s_at	204	AI653608	606	2.30E-05	-1.014	LRPPRC	leucine-rich PPR-motif containing
216387_x_at	205	AL353580	607	5.15E-03	-1.014	---	---
218152_at	206	NM_018200	608	8.22E-03	-1.014	HMG20A	high-mobility group 20A
201788_at	207	NM_007372	609	4.77E-03	-1.014	DDX42	DEAD (Asp-Glu-Ala-Asp) box polypeptide 42
201922_at	208	NM_014886	610	6.30E-03	-1.014	TINP1	TGF beta-inducible nuclear protein 1
212982_at	209	AI621223	611	4.57E-03	-1.014	ZDHHC17	zinc finger, DHHC domain containing 17
214173_x_at	210	AW514900	612	2.20E-04	-1.014	C19orf2	chromosome 19 open reading frame 2
211937_at	211	NM_001417	613	3.24E-03	-1.014	EIF4B	eukaryotic translation initiation factor 4B
216843_x_at	212	U38964	614	6.22E-03	-1.014	---	---
212854_x_at	213	AB051480	615	1.60E-03	-1.013	LOC376745	AG1
212886_at	214	AL080169	616	4.76E-03	-1.013	DKFZP434C171	DKFZP434C171 protein
220607_x_at	215	NM_016397	617	3.15E-03	-1.012	TH1L	TH1-like (Drosophila)
212132_at	216	AL117499	618	5.52E-04	-1.011	C19orf13	chromosome 19 open reading frame 13
214042_s_at	217	AW071997	619	3.11E-03	-1.011	RPL22	ribosomal protein L22
212660_at	218	AI735639	620	5.22E-03	-1.011	PHF15	PHD finger protein 15

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208944_a t	219	D50683	621	5.54E-03	-1.011	---	---
210011_s at	220	BC000527	433	9.77E-03	-1.010	EWSR1	Ewing sarcoma breakpoint region 1
211938_a t	221	BF247371	622	8.55E-03	-1.010	EIF4B	eukaryotic translation initiation factor 4B
220408_x at	222	NM_01756 9	623	6.06E-03	-1.010	FAM48A	family with sequence similarity 48, member A
212114_a t	223	BE967207	624	9.73E-03	-1.010	---	Similar to microtubule- associated proteins 1A/1B light chain 3
212455_a t	224	N36997	625	2.91E-03	-1.009	YT521	splicing factor YT521-B
212299_a t	225	AL117502	626	6.14E-03	-1.009	NEK9	NIMA (never in mitosis gene a)- related kinase 9
200005_a t	226	NM_00375 3	627	4.23E-03	-1.008	EIF3S7	eukaryotic translation initiation factor 3, subunit 7 zeta, 66/67kDa /// eukaryo
200081_s at	227	BE741754	628	1.02E-03	-1.007	RPS6	ribosomal protein S6 /// ribosomal protein S6
213687_s at	228	BE968801	629	8.26E-03	-1.006	RPL35A	ribosomal protein L35a
204102_s at	229	NM_00196 1	630	5.59E-03	-1.006	EEF2	eukaryotic translation elongation factor 2
211666_x at	230	L22453	631	6.17E-03	-1.006	RPL3	ribosomal protein L3 /// ribosomal protein L3
209134_s at	231	BC000524	632	1.77E-03	-1.005	RPS6	ribosomal protein S6
221700_s at	232	AF348700	633	2.98E-04	-1.005	UBA52	ubiquitin A-52 residue ribosomal protein fusion product 1 /// ubiquitin A-52 res
200034_s at	233	NM_00097 0	634	9.34E-03	-1.005	RPL6	ribosomal protein L6 /// ribosomal protein L6
201254_x at	234	NM_00101 0	635	7.37E-03	-1.004	RPS6	ribosomal protein S6
212734_x at	235	AI186735	636	8.06E-03	-1.003	RPL13	ribosomal protein L13

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217718_s _at	236	NM_01405 2	637	6.36E-03	1.005	YWHAB	tyrosine 3- monooxygenase/try ptophan 5- monooxygenase activation protein, beta pol
208678_a t	237	BC004443	638	8.57E-03	1.006	ATP6V1E1	ATPase, H+ transporting, lysosomal 31kDa, V1 subunit E isoform 1
200633_a t	238	NM_01895 5	639	2.94E-03	1.006	UBB	ubiquitin B /// ubiquitin B
201318_s _at	239	NM_00647 1	640	1.52E-03	1.007	MRLC2 /// MRCL3	myosin regulatory light chain MRLC2 /// myosin regulatory light chain MRCL3
202090_s _at	240	NM_00683 0	641	3.41E-03	1.008	UQCR	ubiquinol- cytochrome c reductase (6.4kD) subunit
221474_a t	241	U26162	642	7.75E-03	1.008	MRLC2	myosin regulatory light chain MRLC2
214629_x _at	242	AF320999	643	3.29E-03	1.009	RTN4	reticulon 4
200067_x _at	243	AL078596	644	9.11E-03	1.010	---	---
201899_s _at	244	NM_00333 6	645	8.17E-03	1.010	UBE2A	ubiquitin- conjugating enzyme E2A (RAD6 homolog)
210968_s _at	245	AF333336	646	3.86E-03	1.010	RTN4	reticulon 4
200609_s _at	246	NM_01749 1	647	5.41E-04	1.010	WDR1	WD repeat domain 1
208805_a t	247	BC002979	648	3.97E-03	1.011	PSMA6	proteasome (prosome, macropain) subunit, alpha type, 6
218571_s _at	248	NM_01416 9	649	5.41E-03	1.011	C14orf123	chromosome 14 open reading frame 123
217730_a t	249	NM_02215 2	650	5.26E-03	1.013	PP1201	PP1201 protein
218520_a t	250	NM_01325 4	651	9.23E-03	1.013	TBK1	TANK-binding kinase 1
208908_s _at	251	AF327443	652	7.99E-03	1.014	CAST	calpastatin
207467_x _at	252	NM_00175 0	653	8.54E-03	1.014	CAST	calpastatin

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201975_a t	253	NM_00295 6	654	5.39E-03	1.015	RSN	restin (Reed- Steinberg cell- expressed intermediate filament-associated protein)
200814_a t	254	NM_00626 3	655	2.49E-04	1.015	PSME1	proteasome (prosome, macropain) activator subunit 1 (PA28 alpha)
203090_a t	255	NM_00692 3	656	4.45E-03	1.015	SDF2	stromal cell-derived factor 2
201470_a t	256	NM_00483 2	657	2.35E-03	1.015	GSTO1	glutathione S- transferase omega 1
212586_a t	257	AA195244	658	6.42E-03	1.016	CAST	calpastatin
200703_a t	258	NM_00374 6	659	5.88E-03	1.016	DNCL1	dynein, cytoplasmic, light polypeptide 1
217995_a t	259	NM_02119 9	660	9.02E-03	1.016	SQRDL	sulfide quinone reductase-like (yeast)
201346_a t	260	NM_02455 1	661	3.81E-03	1.016	ADIPOR2	adiponectin receptor 2
201609_x _at	261	AL578502	662	1.36E-03	1.016	ICMT	isoprenylcysteine carboxyl methyltransferase
202000_a t	262	BC002772	663	6.15E-03	1.016	NDUFA6	NADH dehydrogenase (ubiquinone) 1 alpha subcomplex, 6, 14kDa
202001_s _at	263	BC002772	663	7.78E-03	1.017	NDUFA6	NADH dehydrogenase (ubiquinone) 1 alpha subcomplex, 6, 14kDa
203880_a t	264	NM_00569 4	664	6.24E-03	1.017	COX17	COX17 homolog, cytochrome c oxidase assembly protein (yeast)
218525_s _at	265	NM_01790 2	665	9.42E-03	1.017	HIF1AN	hypoxia-inducible factor 1, alpha subunit inhibitor
219798_s _at	266	NM_01960 6	666	3.38E-03	1.017	FLJ20257	hypothetical protein FLJ20257
208398_s _at	267	NM_00486 5	667	7.76E-03	1.017	TBPL1	TBP-like 1
218358_a t	268	NM_02432 4	668	6.57E-03	1.017	---	---

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203097_s_at	269	NM_014247	669	4.66E-03	1.017	RAPGEF2	Rap guanine nucleotide exchange factor (GEF) 2
221598_s_at	270	BC002878	670	7.63E-03	1.017	CRSP8	cofactor required for Sp1 transcriptional activation, subunit 8, 34kDa
209546_s_at	271	AF323540	671	6.71E-03	1.018	APOL1	apolipoprotein L, 1
50400_at	272	AI743990	672	7.35E-03	1.018	PAOX	polyamine oxidase (exo-N4-amino)
217752_s_at	273	NM_018235	673	4.11E-03	1.018	CNDP2	CNDP dipeptidase 2 (metallopeptidase M20 family)
201931_at	274	NM_000126	674	6.90E-03	1.018	ETFA	electron-transfer-flavoprotein, alpha polypeptide (glutaric aciduria II)
218567_x_at	275	NM_005700	675	1.00E-02	1.019	DPP3	dipeptidylpeptidase 3
202056_at	276	AW051311	676	8.93E-03	1.019	KPNA1	Karyopherin alpha 1 (importin alpha 5)
218907_s_at	277	NM_023942	677	2.62E-03	1.019	MGC3036	hypothetical protein MGC3036
213726_x_at	278	AA515698	678	2.22E-03	1.019	TUBB2	tubulin, beta, 2
204448_s_at	279	AF031463	679	7.00E-03	1.019	PDCL	phosducin-like
1405_i_at	280	M21121	680	9.54E-03	1.019	CCL5	chemokine (C-C motif) ligand 5
212864_at	281	Y16521	681	9.17E-03	1.019	CDS2	CDP-diacylglycerol synthase (phosphatidate cytidyltransferase) 2
209444_at	282	BC001851	682	1.27E-04	1.019	RAP1GDS1	RAP1, GTP-GDP dissociation stimulator 1
208898_at	283	AF077614	683	2.37E-03	1.019	ATP6V1D	ATPase, H ⁺ transporting, lysosomal 34kDa, V1 subunit D
202377_at	284	AW026535	684	5.30E-03	1.019	OBRGRP /// LEPR	leptin receptor gene-related protein /// leptin receptor
209547_s_at	285	BC001043	685	4.14E-03	1.020	SF4	splicing factor 4
202922_at	286	BF676980	686	4.19E-03	1.020	GCLC	glutamate-cysteine ligase, catalytic subunit

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201403_s_at	287	NM_004528	687	3.64E-03	1.020	MGST3	microsomal glutathione S-transferase 3
213154_s_at	288	AI934125	688	2.75E-03	1.020	BICD2	bicaudal D homolog 2 (Drosophila)
215676_at	289	N91109	689	8.42E-03	1.021	BRF1	BRF1 homolog, subunit of RNA polymerase III transcription initiation factor IIIB
218927_s_at	290	BC002918	690	8.09E-03	1.021	CHST12	carbohydrate (chondroitin 4) sulfotransferase 12
213648_at	291	AW614427	691	6.11E-03	1.021	EXOSC7	Exosome component 7
209593_s_at	292	AF317129	692	8.82E-03	1.021	TOR1B	torsin family 1, member B (torsin B)
203731_s_at	293	NM_014569	693	1.78E-03	1.022	ZFP95	zinc finger protein 95 homolog (mouse)
201234_at	294	NM_004517	694	2.90E-03	1.022	ILK	integrin-linked kinase
214817_at	295	BE783668	695	8.61E-03	1.022	UNC13A	unc-13 homolog A (C. elegans)
201761_at	296	NM_006636	696	3.83E-03	1.022	MTHFD2	methylene tetrahydrofolate dehydrogenase (NAD ⁺ dependent), methenyltetrahydrofol
205467_at	297	NM_001230	697	6.50E-03	1.022	CASP10	caspase 10, apoptosis-related cysteine protease
222318_at	298	AI744673	698	9.20E-03	1.022	FLJ45850	FLJ45850 protein
209933_s_at	299	AF020314	699	6.24E-03	1.022	CMRF-35H	leukocyte membrane antigen
200734_s_at	300	BG341906	700	4.33E-03	1.023	ARF3	ADP-ribosylation factor 3
219899_x_at	301	NM_014434	701	7.97E-03	1.023	NDOR1	NADPH dependent diflavin oxidoreductase 1
221563_at	302	N36770	702	3.41E-03	1.024	DUSP10	dual specificity phosphatase 10
65517_at	303	AA910946	703	4.00E-03	1.024	AP1M2	adaptor-related protein complex 1, mu 2 subunit
205126_at	304	NM_006296	704	8.89E-03	1.024	VRK2	vaccinia related kinase 2
200696_s_at	305	NM_000177	705	8.30E-03	1.024	GSN	gelsolin (amyloidosis, Finnish type)

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208790_s_at	306	AF312393	706	4.43E-03	1.025	PTRF	polymerase I and transcript release factor
205115_s_at	307	NM_016196	707	8.06E-03	1.025	RBM19	RNA binding motif protein 19
213982_s_at	308	BG107203	708	8.15E-03	1.025	RABGAP1L	RAB GTPase activating protein 1-like
206992_s_at	309	NM_015684	709	2.03E-03	1.025	ATP5S	ATP synthase, H ⁺ transporting, mitochondrial F0 complex, subunit s (factor B)
201060_x_at	310	AI537887	710	9.12E-04	1.025	STOM	stomatin
209278_s_at	311	L27624	711	7.18E-03	1.026	TFPI2	tissue factor pathway inhibitor 2
210542_s_at	312	BC000585	712	4.38E-03	1.026	SLCO3A1	solute carrier organic anion transporter family, member 3A1
202006_at	313	NM_002835	713	3.01E-03	1.026	PTPN12	protein tyrosine phosphatase, non-receptor type 12
210557_x_at	314	M76453	714	2.04E-03	1.026	CSF1	colony stimulating factor 1 (macrophage)
210796_x_at	315	D86359	715	2.95E-03	1.027	SIGLEC6	sialic acid binding Ig-like lectin 6
220825_s_at	316	NM_018240	716	9.74E-03	1.027	KIRREL	kin of IRRE like (Drosophila)
202378_s_at	317	NM_017526	717	2.72E-03	1.027	OBRGRP	leptin receptor gene-related protein
210241_s_at	318	AB007458	718	3.02E-03	1.027	TP53AP1	TP53 activated protein 1
213069_at	319	AI148659	719	2.90E-03	1.027	---	---
216034_at	320	AA558468	720	9.90E-03	1.027	SUHW1	suppressor of hairy wing homolog 1 (Drosophila)
210793_s_at	321	U41815	721	2.64E-03	1.029	NUP98	nucleoporin 98kDa
207375_s_at	322	NM_002189	722	3.71E-03	1.029	IL15RA	interleukin 15 receptor, alpha
206247_at	323	NM_005931	723	2.70E-03	1.029	MICB	MHC class I polypeptide-related sequence B
217078_s_at	324	AJ010102	724	9.65E-03	1.029	CMRF-35H	leukocyte membrane antigen
219257_s_at	325	NM_021972	725	1.62E-03	1.030	SPHK1	sphingosine kinase 1

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
204781_s _at	326	NM_00004 3	726	8.44E-03	1.030	TNFRSF6	tumor necrosis factor receptor superfamily, member 6
221536_s _at	327	AL136897	727	2.28E-03	1.030	FLJ11301	hypothetical protein FLJ11301
209468_a _t	328	AB017498	728	8.91E-03	1.030	LRP5	low density lipoprotein receptor- related protein 5
201061_s _at	329	M81635	729	1.46E-04	1.030	STOM	stomatin
203499_a _t	330	NM_00443 1	730	6.49E-03	1.031	EPHA2	EPH receptor A2
213324_a _t	331	AK024281	731	7.63E-03	1.031	SRC	v-src sarcoma (Schmidt-Ruppin A- 2) viral oncogene homolog (avian)
221535_a _t	332	AL136897	727	3.81E-03	1.031	FLJ11301	hypothetical protein FLJ11301
219938_s _at	333	NM_02443 0	732	3.19E-03	1.032	PSTPIP2	proline-serine- threonine phosphatase interacting protein 2
213787_s _at	334	AV702405	733	1.78E-03	1.032	EBP	emopamil binding protein (sterol isomerase)
219592_a _t	335	NM_02459 6	734	8.43E-03	1.032	MCPH1	microcephaly, primary autosomal recessive 1
200637_s _at	336	AI762627	735	5.52E-03	1.033	PTPRF	protein tyrosine phosphatase, receptor type, F
202193_a _t	337	NM_00556 9	736	4.88E-03	1.033	LIMK2	LIM domain kinase 2
214859_a _t	338	AI635302	737	8.68E-03	1.034	FSTL4	folliculin-like 4
209213_a _t	339	BC002511	738	5.69E-03	1.034	CBR1	carbonyl reductase 1
218945_a _t	340	NM_02410 9	739	3.75E-03	1.034	MGC2654	hypothetical protein MGC2654
220471_s _at	341	NM_02510 7	740	8.29E-03	1.035	MYCT1	myc target 1
59625_at	342	AI912351	741	5.19E-03	1.035	NOL3	nucleolar protein 3 (apoptosis repressor with CARD domain)
207233_s _at	343	NM_00024 8	742	2.93E-03	1.035	MITF	microphthalmia- associated transcription factor
214070_s _at	344	AW006935	743	5.14E-03	1.036	ATP10B	ATPase, Class V, type 10B

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
219785_s_at	345	NM_024735	744	5.84E-03	1.036	FBXO31	F-box protein 31
220684_at	346	NM_013351	745	3.34E-03	1.037	TBX21	T-box 21
202245_at	347	AW084510	746	7.02E-03	1.037	LSS	lanosterol synthase (2,3-oxidosqualene- lanosterol cyclase)
207298_at	348	NM_006632	747	6.68E-03	1.037	SLC17A3	solute carrier family 17 (sodium phosphate), member 3
220201_at	349	NM_018835	748	3.85E-03	1.038	MNAB	membrane associated DNA binding protein
203208_s_at	350	BF214329	749	6.89E-04	1.038	CHPPR	likely ortholog of chicken chondrocyte protein with a poly- proline region
207033_at	351	NM_005142	750	3.42E-04	1.039	GIF	gastric intrinsic factor (vitamin B synthesis)
204929_s_at	352	NM_006634	751	6.38E-03	1.039	VAMP5	vesicle-associated membrane protein 5 (myobrevin)
209735_at	353	AF098951	752	7.05E-03	1.039	ABCG2	ATP-binding cassette, sub-family G (WHITE), member 2
209234_at	354	BF939474	753	3.31E-04	1.039	KIF1B	kinesin family member 1B
210102_at	355	BC001234	754	5.97E-03	1.040	LOH11CR2 A	loss of heterozygosity, 11, chromosomal region 2, gene A
204114_at	356	NM_007361	755	4.45E-03	1.041	NID2	nidogen 2 (osteonidogen)
203344_s_at	357	NM_002894	756	1.33E-03	1.041	RBBP8	retinoblastoma binding protein 8
216891_at	358	U00956	757	5.03E-03	1.042	---	---
221680_s_at	359	AF147782	758	9.08E-03	1.042	ETV7	ets variant gene 7 (TEL2 oncogene)
202087_s_at	360	NM_001912	759	3.63E-03	1.042	CTSL	cathepsin L
218243_at	361	NM_025158	760	6.62E-03	1.043	RUFY1	RUN and FYVE domain containing 1
201980_s_at	362	NM_012425	761	5.59E-03	1.044	RSU1	Ras suppressor protein 1
221280_s_at	363	NM_019619	762	4.65E-04	1.044	PARD3	par-3 partitioning defective 3 homolog (C. elegans)

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
205108_s _at	364	NM_00038 4	763	2.36E-04	1.045	APOB	apolipoprotein B (including Ag(x) antigen)
204241_a _t	365	BF055171	764	6.95E-04	1.045	ACOX3	acyl-Coenzyme A oxidase 3, pristanoyl
201278_a _t	366	N21202	765	2.71E-03	1.049	DAB2	Disabled homolog 2, mitogen- responsive phosphoprotein (Drosophila)
202067_s _at	367	AI861942	766	1.95E-03	1.050	LDLR	low density lipoprotein receptor (familial hypercholesterolemi a)
202085_a _t	368	NM_00481 7	767	3.10E-03	1.051	TJP2	tight junction protein 2 (zona occludens 2)
214157_a _t	369	AA401492	768	4.90E-03	1.053	GNAS	GNAS complex locus
221526_x _at	370	AF196185	769	2.72E-03	1.053	PARD3	par-3 partitioning defective 3 homolog (C. elegans)
207738_s _at	371	NM_01343 6	770	2.00E-03	1.054	NCKAP1	NCK-associated protein 1
204769_s _at	372	NM_00054 4	771	3.15E-03	1.054	TAP2	transporter 2, ATP- binding cassette, sub-family B (MDR/TAP)
216693_x _at	373	AL133102	772	6.34E-03	1.055	HDGFRP3	hepatoma-derived growth factor, related protein 3
202068_s _at	374	NM_00052 7	773	5.72E-03	1.057	LDLR	low density lipoprotein receptor (familial hypercholesterolemi a)
202743_a _t	375	BE622627	774	7.96E-04	1.061	PIK3R3	phosphoinositide-3- kinase, regulatory subunit 3 (p55, gamma)
206638_a _t	376	NM_00086 7	775	7.47E-03	1.061	HTR2B	5- hydroxytryptamine (serotonin) receptor 2B
31874_at	377	Y07846	776	2.32E-03	1.061	GAS2L1	growth arrest- specific 2 like 1
201798_s _at	378	NM_01345 1	777	5.01E-03	1.062	FER1L3	fer-1-like 3, myoferlin (C. elegans)

Affymetrix Probeset ID	SE Q ID NO :	Representa tive Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
219288_a t	379	NM_02068 5	778	6.67E-04	1.063	C3orf14	chromosome 3 open reading frame 14
220167_s at	380	NM_01536 9	779	5.44E-04	1.064	TP53TG3	TP53TG3 protein
211354_s at	381	U52913	780	1.67E-03	1.066	LEPR	leptin receptor
213317_a t	382	AL049313	781	7.15E-03	1.067	CLIC5	Chloride intracellular channel 5
212390_a t	383	AB007923	782	8.15E-03	1.070	PDE4DIP	phosphodiesterase 4D interacting protein (myomegalin)
212062_a t	384	AB014511	783	8.65E-03	1.070	ATP9A	ATPase, Class II, type 9A
203337_x at	385	NM_00476 3	784	6.03E-03	1.071	ITGB1BP1	integrin beta 1 binding protein 1
210029_a t	386	M34455	785	5.97E-03	1.073	INDO	indoleamine-pyrrole 2,3 dioxygenase
206049_a t	387	NM_00300 5	786	8.60E-03	1.074	SELP	selectin P (granule membrane protein 140kDa, antigen CD62)
202949_s at	388	NM_00145 0	787	6.99E-03	1.076	FHL2	four and a half LIM domains 2
211864_s at	389	AF207990	788	7.27E-03	1.076	FER1L3	fer-1-like 3, myoferlin (C. elegans)
206254_a t	390	NM_00196 3	789	5.83E-03	1.077	EGF	epidermal growth factor (beta- urogastrone)
210073_a t	391	L32867	790	5.66E-04	1.077	SIAT8A	sialyltransferase 8A (alpha-N- acetylneuraminate: alpha-2,8- sialyltransferase, GD
209524_a t	392	AK001280	791	1.03E-03	1.080	HDGFRP3	hepatoma-derived growth factor, related protein 3
219759_a t	393	NM_02235 0	792	8.03E-04	1.081	LRAP	leukocyte-derived arginine aminopeptidase
202112_a t	394	NM_00055 2	793	1.22E-03	1.082	VWF	von Willebrand factor
219410_a t	395	NM_01800 4	794	4.81E-03	1.083	FLJ10134	hypothetical protein FLJ10134
203819_s at	396	AU160004	795	3.58E-03	1.088	IMP-3	IGF-II mRNA- binding protein 3
212730_a t	397	AK026420	796	3.19E-03	1.090	DMN	desmuslin

Affymetrix Probeset ID	SE Q ID NO :	Representative Public ID	SE Q ID NO:	p-value BMS vs. typical RRMS	Log Fold Change (BMS vs. typical RRMS)	Gene Symbol	Gene Title
220122_at	398	NM_024717	797	5.82E-03	1.090	MCTP1	multiple C2-domains with two transmembrane regions 1
208782_at	399	BC000055	798	2.14E-03	1.095	FSTL1	follistatin-like 1
202468_s_at	400	NM_003798	799	1.36E-03	1.096	CTNNAL1	catenin (cadherin-associated protein), alpha-like 1
207018_s_at	401	NM_004163	800	4.02E-03	1.101	RAB27B	RAB27B, member RAS oncogene family
210354_at	402	M29383	801	4.60E-03	1.102	---	---
201110_s_at	403	NM_003246	802	9.38E-03	1.111	THBS1	thrombospondin 1
207808_s_at	404	NM_000313	803	2.92E-03	1.114	PROS1	protein S (alpha)
205612_at	405	NM_007351	804	7.97E-03	1.118	MMRN1	multimerin 1
214073_at	406	BG475299	805	2.37E-03	1.121	CTTN	cortactin

Table 1: Presented are the 406 differentiating genes given by the gene name and description, the Affymetrix probeset identification number, and a representative GenBank Accession number between BMS and typical RRMS patients. p-value – statistical significance by ANOVA; Log Fold Change = refers to the logarithms fold change between the expression level of a polynucleotide in a blood sample of a BMS subject as compared to the expression level in a blood sample of a typical RRMS subject: The (-) sign means that the polynucleotide is downregulated (decreased in level) in BMS as compared to typical RRMS subjects; and the (+) sign means that the polynucleotide is upregulated (increased in level) in BMS as compared to typical RRMS subjects.

Table 2 hereinbelow discloses additional polynucleotides (RNA alternative transcripts) of the above identified 406 genes which expression level is differentiating between typical RRMS and BMS.

Table 2

[illegible]

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21962 9_at	2	NM_01 7911 / 997								C22O RF8	Hs.26 5018
21037 9_s_at	3	NM_01 2290 / 948								TLK1	Hs.47 0586
21347 2_at	4	--- /								---	---
21970 0_at	5	NM_02 0405 / 1022								PLXD C1	Hs.12 5036
21107 7_s_at	6	NM_01 2290 / 948								TLK1	Hs.47 0586
21096 9_at	7	--- /								---	---
21629 8_at	8	--- /								---	---
21983 4_at	9	NM_02 4744 / 1051								ALS2 CR8	Hs.44 4982
21526 2_at	10	--- /								---	---
22071 5_at	11	--- /								---	---
20716 4_s_at	12	NM_20 5768 / 1103	NM_0 06352 / 1133							ZNF23 8, ZNF23 8,	Hs.69 997, Hs.69 997,
22023 6_at	13	NM_01 7990 / 999								PDPR	Hs.46 1183
20393 9_at	14	NM_00 2526 / 865								NT5E	Hs.15 3952
21694 5_x_a t	15	NM_01 5148 / 970								PASK	Hs.39 7891
20391 3_s_at	16	NM_00 0860 / 834								HPGD	Hs.77 348
20486 4_s_at	17	NM_17 5767 / 1088	NM_0 02183 / 1189							IL6ST, IL6ST,	Hs.53 2082, Hs.53 2082,
21569 8_at	18	--- /								---	---
21353 4_s_at	19	NM_01 5148 / 970								PASK	Hs.39 7891

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21094 8_s_at	20	NM_01 6269 / 988								LEF1	Hs.12 5132
22045 8_at	21	NM_01 8038 / 1001								FLJ10 246	Hs.27 4274
21209 3_s_at	22	NM_02 0749 / 1025	NM_0 01001 931 / 1114	NM_0 01001 925 / 1182	NM_0 01001 924 / 1216	NM_0 01001 927 / 1234				MTUS 1, MTUS 1, MTUS 1, MTUS 1, MTUS 1, MTUS 1,	Hs.79 46, Hs.79 46, Hs.79 46, Hs.79 46, Hs.79 46, Hs.79 46,
20712 0_at	23	NM_02 2103 / 1033								FLJ14 011	Hs.43 3473
20478 7_at	24	NM_00 7268 / 939								VSIG4	Hs.89 04
21551 2_at	25	--- /								---	---
46947 _at	26	--- /								---	---
21001 2_s_at	27	--- /								---	---
21019 2_at	28	NM_00 6095 / 918								ATP8 A1	Hs.43 5052
20760 3_at	29	--- /								---	---
21858 4_at	30	NM_02 4549 / 1047								FLJ21 127	Hs.21 1511
21347 0_s_at	31	NM_00 5520 / 912								HNRP H1	Hs.20 2166
21046 1_s_at	32	NM_00 6720 / 930	NM_0 02313 / 1120	NM_0 01003 407 / 1187	NM_0 01003 408 / 1232					ABLI M1, ABLI M1, ABLI M1, ABLI M1, ABLI M1,	Hs.43 8236, Hs.43 8236, Hs.43 8236, Hs.43 8236, Hs.43 8236,
21765 6_at	33	--- /								---	---
21517 7_s_at	34	NM_00 0210 / 826								ITGA6	Hs.13 3397

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20657 9_at	53	NM_00 6298 / 922								ZNF19 2	Hs.57 679
20525 4_x_a t	54	NM_20 1632 / 1099	NM_2 01634 / 1173	NM_2 13648 / 1215	NM_0 03202 / 1219					TCF7, TCF7, TCF7, TCF7,	Hs.51 9580, Hs.51 9580, Hs.51 9580, Hs.51 9580,
21034 9_at	55	NM_00 1744 / 856								CAMK 4	Hs.22 0629
20815 7_at	56	NM_00 9586 / 942								SIM2	Hs.14 6186
21497 2_at	57	--- /								---	---
20733 4_s_at	58	NM_00 3242 / 873								TGFB R2	Hs.82 028
21766 6_at	59	--- /								---	---
21577 1_x_a t	60	NM_02 0975 / 1026	NM_0 20630 / 1143							RET, RET,	Hs.35 0321, Hs.35 0321,
21613 9_s_at	61	NM_03 3392 / 1057	NM_0 15133 / 1138							MAPK 8IP3, MAPK 8IP3,	Hs.20 7763, Hs.20 7763,
21690 2_s_at	62	NM_01 8427 / 1010								RRN3	Hs.46 0078
20826 5_at	63	NM_02 0161 / 1019								DKFZ P547H 025	Hs.28 3092
20945 6_s_at	64	NM_03 3645 / 1059	NM_0 33644 / 1148	NM_0 12300 / 1193						FBXW 11, FBXW 11, FBXW 11,	Hs.48 4138, Hs.48 4138, Hs.48 4138,
21723 7_at	65	--- /								---	---
21720 8_s_at	66	NM_00 4087 / 885								DLG1	Hs.29 2549
22175 7_at	67	NM_05 2880 / 1060								MGC1 7330	Hs.26 670

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21348 l_at	68	--- /								---	---
21902 5_at	69	NM_02 0404 / 1021								CD164 L1	Hs.19 5727
21520 l_at	70	--- /								---	---
20537 8_s_at	71	NM_00 0665 / 833	NM_0 15831 / 1139							ACHE, ACHE,	Hs.15 4495, Hs.15 4495,
21982 9_at	72	NM_01 2278 / 947								ITGB1 BP2	Hs.10 9999
21690 8_x_a t	73	NM_01 8427 / 1010	NM_1 45237 / 1156							RRN3, LOC94 431,	Hs.46 0078, Hs.54 6468,
21718 4_s_at	74	NM_20 6961 / 1105	NM_0 02344 / 1121							LTK, LTK,	Hs.43 4481, Hs.43 4481,
21118 1_x_a t	75	NM_00 100189 0 / 843	NM_0 01754 / 1179							RUNX 1, RUNX 1,	Hs.14 9261, Hs.14 9261,
21495 8_s_at	76	NM_00 7267 / 938								EVER 1	Hs.16 165
22036 9_at	77	NM_01 7936 / 998								KIAA2 010	Hs.53 3887
21184 8_s_at	78	NM_00 6890 / 932								CEAC AM7	Hs.74 466
22163 8_s_at	79	NM_00 3763 / 880	NM_0 01001 434 / 842	NM_0 01001 433 / 1181						STX16 , STX16 , STX16 ,	Hs.30 7913, Hs.30 7913, Hs.30 7913,
22172 3_s_at	80	NM_02 1196 / 1028	NM_1 33479 / 1152	NM_1 33478 / 1201	NM_0 33323 / 1223					SLC4 A5, SLC4 A5, SLC4 A5, SLC4 A5,	Hs.46 9033, Hs.46 9033, Hs.46 9033, Hs.46 9033,
20138 0_at	81	NM_00 6371 / 923								CRTA P	Hs.51 7888

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
34063 _at	82	NM_00 100371 5 / 845	NM_0 01003 716 / 1178	NM_0 04259 / 1191						RECQ L5, RECQ L5, RECQ L5,	Hs.51 4480, Hs.51 4480, Hs.51 4480,
21475 7_at	83	--- /								---	---
21817 6_at	84	NM_02 2149 / 1034								MAGE F1	Hs.30 6123
22197 7_at	85	--- /								---	---
21416 1_at	86	--- /								---	---
21485 7_at	87	--- /								---	---
21623 0_x_a t	88	NM_00 0543 / 829	NM_0 01007 593 / 1118							SMPD 1, SMPD 1,	Hs.49 8173, Hs.49 8173,
20501 9_s_at	89	NM_00 4624 / 893								VIPR1	Hs.34 8500
22170 7_s_at	90	NM_01 8289 / 1006								FLJ10 979	Hs.46 1819
20866 4_s_at	91	NM_00 100189 4 / 844	NM_0 03316 / 1180							TTC3, TTC3,	Hs.36 8214, Hs.36 8214,
21516 7_at	92	--- /								---	---
21543 7_x_a t	93	NM_01 3449 / 955								BAZ2 A	Hs.31 4263
21089 2_s_at	94	NM_00 1518 / 854	NM_0 33001 / 1146	NM_0 33000 / 1198	NM_0 32999 / 1222					GTF2I, GTF2I, GTF2I, GTF2I,	Hs.52 0459, Hs.52 0459, Hs.52 0459, Hs.52 0459,
21240 0_at	95	NM_20 3305 / 1101								EEIG1	Hs.46 0208
21513 7_at	96	--- /								---	---
20575 0_at	97	NM_00 4332 / 888								BPHL	Hs.10 136

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21062 8_x_at	98	NM_00 3573 / 877								LTBP4	Hs.46 6766
21821 0_at	99	NM_02 4619 / 1050								FN3K RP	Hs.31 431
21678 4_at	100	--- /								---	---
20181 9_at	101	NM_00 5505 / 911								SCAR B1	Hs.29 8813
21431 2_at	102	--- /								---	---
21555 6_at	103	--- /								---	---
22175 6_at	104	NM_05 2880 / 1060								MGC1 7330	Hs.26 670
21690 9_at	105	--- /								---	---
22024 9_at	106	NM_01 2269 / 946								HYAL 4	Hs.28 673
21127 2_s_at	107	NM_20 1554 / 1098	NM_0 01345 / 1119	NM_2 01445 / 1211	NM_2 01444 / 1229					DGKA , DGKA , DGKA , DGKA ,	Hs.52 4488, Hs.52 4488, Hs.52 4488, Hs.52 4488,
21865 1_s_at	108	NM_01 8357 / 1008								FLJ11 196	Hs.41 6755
21828 5_s_at	109	NM_02 0139 / 1018								DHRS 6	Hs.12 4696
21738 5_at	110	--- /								---	---
20289 4_at	111	NM_00 4444 / 890								EPHB 4	Hs.43 7008
21763 1_at	112	--- /								---	---
22048 7_at	113	NM_01 8968 / 1015								SNTG 2	Hs.14 8819
20357 9_s_at	114	NM_00 3983 / 884								SLC7 A6	Hs.35 1571

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21541 2_x_a t	115	NM_00 5395 / 908	NM_0 01003 686 / 1115	NM_0 01003 687 / 1183	NM_1 74930 / 1086	NM_0 02679 / 867	NM_0 00535 / 828			PMS2 L3, PMS2 L3, PMS2 L3, PMS2 L5, POM1 21, PMS2,	Hs.54 9057, Hs.54 9057, Hs.54 9057, Hs.39 7073, Hs.48 8624, Hs.48 7470,
21221 7_at	116	--- /								---	---
22160 2_s_at	117	NM_00 5449 / 909								TOSO	Hs.58 831
21784 7_s_at	118	NM_00 5119 / 905								THRA P3	Hs.16 0211
21527 5_at	119	--- /								---	---
22195 1_at	120	NM_17 4940 / 1087								LOC28 3232	Hs.44 8664
21322 4_s_at	121	--- /								---	---
21896 7_s_at	122	NM_03 0664 / 1055	NM_0 01001 484 / 1113							PTER, PTER,	Hs.44 4321, Hs.44 4321,
20844 2_s_at	123	NM_13 8292 / 1069	NM_0 00051 / 1109							ATM, ATM,	Hs.43 5561, Hs.43 5561,
20367 5_at	124	NM_00 5013 / 900								NUCB 2	Hs.12 8686
20710 5_s_at	125	NM_00 5027 / 901								PIK3R 2	Hs.37 1344
21914 4_at	126	NM_02 4025 / 1042								MGC1 136	Hs.87 19
21246 3_at	127	NM_00 0611 / 831	NM_2 03331 / 1175	NM_2 03329 / 1212	NM_2 03330 / 1230					CD59, CD59, CD59, CD59,	Hs.27 8573, Hs.27 8573, Hs.27 8573, Hs.27 8573,

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20204 0_s_at	128	NM_00 5056 / 902								JARID 1A	Hs.76 272
64432 _at	129	NM_01 6534 / 991								FLJ39 616	Hs.33 3120
20096 5_s_at	130	NM_00 6720 / 930	NM_0 02313 / 1120	NM_0 01003 407 / 1187	NM_0 01003 408 / 1232					ABLI M1, ABLI M1, ABLI M1, ABLI M1,	Hs.43 8236, Hs.43 8236, Hs.43 8236, Hs.43 8236,
20535 3_s_at	131	NM_00 2567 / 866								PBP	Hs.43 3863
21317 7_at	132	NM_03 3392 / 1057	NM_0 15133 / 1138							MAPK 8IP3, MAPK 8IP3,	Hs.20 7763, Hs.20 7763,
21837 3_at	133	NM_02 2476 / 1040								FTS	Hs.38 0897
20833 3_at	134	NM_02 2363 / 1039								LHX5	Hs.30 2029
20615 0_at	135	NM_00 1242 / 851								TNFR SF7	Hs.35 5307
20243 1_s_at	136	NM_00 2467 / 863								MYC	Hs.20 2453
20783 8_x_a t	137	NM_02 0524 / 1023								PBXIP 1	Hs.50 5806
21832 5_s_at	138	NM_08 0797 / 1062	NM_0 80796 / 1149	NM_0 22105 / 1197						DATF 1, DATF 1, DATF 1,	Hs.55 1527, Hs.55 1527, Hs.55 1527,
21847 5_at	139	NM_18 2984 / 1090	NM_0 22727 / 1144							HTF9 C, HTF9 C,	Hs.54 9133, Hs.54 9133,
21867 0_at	140	NM_02 5215 / 1053								PUS1	Hs.50 7295
20559 4_at	141	NM_01 4897 / 967								ZNF65 2	Hs.46 3375

[illegible]

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20283 0_s_at	155	NM_00 1467 / 853								SLC37 A4	Hs.13 2760
20983 5_x_a t	156	NM_00 100139 1 / 841	NM_0 01001 390 / 1177	NM_0 01001 389 / 1186	NM_0 00610 / 1231	NM_0 01001 392 / 1233				CD44, CD44, CD44, CD44, CD44,	Hs.50 2328, Hs.50 2328, Hs.50 2328, Hs.50 2328, Hs.50 2328,
21225 7_s_at	157	NM_13 9045 / 1071	NM_0 03070 / 1124							SMAR CA2, SMAR CA2,	Hs.29 8990, Hs.29 8990,
21207 1_s_at	158	--- /								---	---
21332 2_at	159	NM_14 5063 / 1075								C6OR F130	Hs.54 9281
21064 5_s_at	160	NM_00 100189 4 / 844	NM_0 03316 / 1180							TTC3, TTC3,	Hs.36 8214, Hs.36 8214,
20251 4_at	161	--- /								---	---
21712 2_s_at	162	NM_01 4854 / 964								SLC35 E2	Hs.55 1612
21844 0_at	163	NM_02 0166 / 1020								MCCC 1	Hs.47 649
21070 7_x_a t	164	NM_17 4930 / 1086	NM_0 02679 / 867	NM_0 05395 / 908	NM_0 01003 686 / 1115	NM_0 01003 687 / 1183	NM_0 00535 / 828			PMS2 L5, POM1 21, PMS2 L3, PMS2 L3, PMS2 L3, PMS2 L3, PMS2,	Hs.39 7073, Hs.48 8624, Hs.54 9057, Hs.54 9057, Hs.54 9057, Hs.54 9057, Hs.48 7470,
61734 _at	165	NM_02 0650 / 1024								RCN3	Hs.43 9184

[illegible]

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21798 8_at	179	NM_02 1178 / 1027	NM_1 82852 / 1169	NM_1 82851 / 1208	NM_1 82849 / 1227					CCNB 1IP1, CCNB 1IP1, CCNB 1IP1, CCNB 1IP1,	Hs.10 7003, Hs.10 7003, Hs.10 7003, Hs.10 7003,
22198 1_s_at	180	NM_03 0581 / 1054								FLJ12 270	Hs.28 0951
21220 1_at	181	--- /								---	---
20879 5_s_at	182	NM_00 5916 / 916	NM_1 82776 / 1168							MCM7 , MCM7 ,	Hs.43 8720, Hs.43 8720,
20833 6_s_at	183	NM_00 4868 / 898	NM_1 38501 / 1153							GPSN 2, GPSN 2,	Hs.51 5642, Hs.51 5642,
22149 9_s_at	184	NM_00 100143 4 / 842	NM_0 03763 / 880	NM_0 01001 433 / 1181						STX16 , STX16 , STX16 ,	Hs.30 7913, Hs.30 7913, Hs.30 7913,
20216 7_s_at	185	NM_02 2362 / 1038								MMS1 9L	Hs.50 0721
20106 5_s_at	186	NR_002 206 / 1107	NM_0 01518 / 854	NM_0 33001 / 1146	NM_0 33000 / 1198	NM_0 32999 / 1222				---, GTF2I, GTF2I, GTF2I, GTF2I,	---, Hs.52 0459, Hs.52 0459, Hs.52 0459, Hs.52 0459,
21117 2_x_a t	187	NM_00 4842 / 896	NM_1 38633 / 1154	NM_0 16377 / 1194						AKAP 7, AKAP 7, AKAP 7,	Hs.48 6483, Hs.48 6483, Hs.48 6483,
21870 7_at	188	NM_01 8337 / 1007								ZNF44 4	Hs.24 545
20677 0_s_at	189	NM_01 2243 / 945								SLC35 A3	Hs.44 8979

<i>PRO BE S E T</i>	<i>SEQ ID NO:</i>	<i>Refseq Ids / SEQ ID NO:</i>								<i>REFS EQ_U NIGE NE</i>	<i>REFS EQ_B AND</i>
20089 8_s_at	190	NM_01 2215 / 943								MGEA 5	Hs.50 0842
21551 9_x_at	191	NM_01 5705 / 981								RUTB C3	Hs.47 4914
21240 6_s_at	192	NM_01 8257 / 1005								C20O RF36	Hs.47 3317
21082 6_x_at	193	NM_13 3341 / 1066	NM_1 33339 / 1151	NM_1 33340 / 1200	NM_1 33344 / 1224	NM_1 33342 / 1237	NM_1 33343 / 1239	NM_1 33338 / 1243	NM_0 02873 / 1246	RAD1 7, RAD1 7, RAD1 7, RAD1 7, RAD1 7, RAD1 7, RAD1 7, RAD1 7,	Hs.16 184, Hs.16 184, Hs.16 184, Hs.16 184, Hs.16 184, Hs.16 184, Hs.16 184, Hs.16 184, Hs.16 184,
21844 4_at	194	NM_02 4105 / 1043								ALG1 2	Hs.52 6711
22182 2_at	195	NM_13 8414 / 1070								LOC11 2869	Hs.46 0487
22075 5_s_at	196	NM_01 6947 / 992								C6OR F48	Hs.10 9798
21897 0_s_at	197	NM_01 5960 / 983								CUTC	Hs.16 606
21825 3_s_at	198	NM_00 6893 / 933								LGTN	Hs.49 7581
22129 3_s_at	199	NM_02 2047 / 1032								DEF6	Hs.15 476
21774 2_s_at	200	NM_10 0486 / 1063	NM_0 16628 / 1140	NM_1 00264 / 1199						WAC, WAC, WAC,	Hs.43 5610, Hs.43 5610, Hs.43 5610, Hs.43 5610,
20712 7_s_at	201	NM_02 1644 / 1030	NM_0 12207 / 1136							HNRP H3, HNRP H3,	Hs.49 9891, Hs.49 9891,

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21785 0_at	202	NM_20 6826 / 1104	NM_0 14366 / 1137	NM_2 06825 / 1213						GNL3, GNL3, GNL3,	Hs.31 3544, Hs.31 3544, Hs.31 3544,
21250 5_s_at	203	NM_01 5329 / 974								KIAA0 892	Hs.11 2751
21197 1_s_at	204	NM_13 3259 / 1065								LRPP RC	Hs.36 8084
21638 7_x_a t	205	NM_01 3269 / 952	NM_0 01004 420 / 1117	NM_0 01004 419 / 1185	NM_0 02520 / 1218					OCIL, OCIL, OCIL, NPM1,	Hs.26 8326, Hs.26 8326, Hs.26 8326, Hs.51 9452,
21815 2_at	206	NM_01 8200 / 1002								HMG2 0A	Hs.69 594
20178 8_at	207	NM_20 3499 / 1102	NM_0 07372 / 1135							DDX4 2, DDX4 2,	Hs.87 65, Hs.87 65,
20192 2_at	208	NM_01 4886 / 966								TINP1	Hs.48 2526
21298 2_at	209	NM_01 5336 / 975								ZDHH C17	Hs.40 14
21417 3_x_a t	210	NM_13 4447 / 1068	NM_0 03796 / 1128							C19O RF2, C19O RF2,	Hs.46 6391, Hs.46 6391,
21193 7_at	211	NM_00 1417 / 852								EIF4B	Hs.51 2629

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PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21245 5_at	224	NM_13 3370 / 1067								YT521	Hs.17 5955
21229 9_at	225	--- /								---	---
20000 5_at	226	NM_00 3753 / 879								EIF3S 7	Hs.55 682
20008 1_s_at	227	NM_00 1010 / 846								RPS6	Hs.40 8073
21368 7_s_at	228	NM_00 0996 / 840								RPL35 A	Hs.52 9631
20410 2_s_at	229	NM_00 1961 / 859								EEF2	Hs.51 5070
21166 6_x_a t	230	NM_00 0967 / 836								RPL3	Hs.11 9598
20913 4_s_at	231	NM_00 1010 / 846								RPS6	Hs.40 8073
22170 0_s_at	232	NM_00 3333 / 876								UBA5 2	Hs.53 08
20003 4_s_at	233	NM_00 0970 / 837								RPL6	Hs.52 8668
20125 4_x_a t	234	NM_00 1010 / 846								RPS6	Hs.40 8073
21273 4_x_a t	235	NM_00 0977 / 838	NM_0 33251 / 1147							RPL13 , RPL13 ,	Hs.41 0817, Hs.41 0817,
21771 8_s_at	236	NM_13 9323 / 1073	NM_0 03404 / 1126							YWH AB, YWH AB,	Hs.27 9920, Hs.27 9920,
20867 8_at	237	NM_00 1696 / 855								ATP6 V1E1	Hs.51 7338
20063 3_at	238	NM_01 8955 / 1014								UBB	Hs.35 6190
20131 8_s_at	239	NM_03 3546 / 1058	NM_0 06471 / 1134							MRLC 2, MRCL 3,	Hs.46 4472, Hs.19 0086,

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20209 0_s_at	240	NM_00 6830 / 931								UQCR	Hs.83 72
22147 4_at	241	NM_03 3546 / 1058								MRLC 2	Hs.46 4472
21462 9_x_a t	242	NM_15 3828 / 1080	NM_2 07520 / 1176	NM_2 07521 / 1214	NM_0 20532 / 1221	NM_0 07008 / 1236				RTN4, RTN4, RTN4, RTN4, RTN4,	Hs.42 9581, Hs.42 9581, Hs.42 9581, Hs.42 9581, Hs.42 9581, Hs.42 9581,
20006 7_x_a t	243	NM_15 2827 / 1078	NM_0 03795 / 1127	NM_1 52828 / 1203						SNX3, SNX3, SNX3,	Hs.12 102, Hs.12 102, Hs.12 102,
20189 9_s_at	244	NM_18 1762 / 1089	NM_0 03336 / 1125	NM_1 81777 / 1207						UBE2 A, UBE2 A, UBE2 A,	Hs.37 9466, Hs.37 9466, Hs.37 9466,
21096 8_s_at	245	NM_15 3828 / 1080	NM_2 07520 / 1176	NM_2 07521 / 1214	NM_0 20532 / 1221	NM_0 07008 / 1236				RTN4, RTN4, RTN4, RTN4, RTN4,	Hs.42 9581, Hs.42 9581, Hs.42 9581, Hs.42 9581, Hs.42 9581,
20060 9_s_at	246	NM_00 5112 / 904	NM_0 17491 / 1142							WDR1 , WDR1 ,	Hs.12 8548, Hs.12 8548,
20880 5_at	247	NM_00 2791 / 868								PSMA 6	Hs.44 6260
21857 1_s_at	248	NM_01 4169 / 957								C14O RF123	Hs.27 9761
21773 0_at	249	NM_02 2152 / 1035								PP120 1	Hs.98 475

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
218520_at	250	NM_013254 / 951								TBK1	Hs.505874
208908_s_at	251	NM_173061 / 1085	NM_173062 / 1165	NM_173060 / 1084	NM_001750 / 1217					CAST, CAST, CAST, CAST,	Hs.440961, Hs.440961, Hs.440961, Hs.440961,
207467_x_at	252	NM_173061 / 1085	NM_173062 / 1165	NM_173060 / 1084	NM_001750 / 1217					CAST, CAST, CAST, CAST,	Hs.440961, Hs.440961, Hs.440961, Hs.440961,
201975_at	253	NM_198240 / 1093	NM_002956 / 1123							RSN, RSN,	Hs.524809, Hs.524809,
200814_at	254	NM_006263 / 920	NM_176783 / 1167							PSME1, PSME1,	Hs.75348, Hs.75348,
203090_at	255	NM_006923 / 934								SDF2	Hs.514036
201470_at	256	NM_004832 / 895								GSTO1	Hs.190028
212586_at	257	NM_173060 / 1084								CAST	Hs.440961
200703_at	258	NM_003746 / 878								DNCL1	Hs.5120
217995_at	259	NM_021199 / 1029								SQRDL	Hs.511251
201346_at	260	NM_024551 / 1048								ADIPOR2	Hs.371642
201609_x_at	261	NM_012405 / 949	NM_170705 / 1162							ICMT, ICMT,	Hs.515688, Hs.515688,
202000_at	262	NM_002490 / 864								NDUF A6	Hs.274416

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20200 1_s_at	263	NM_00 2490 / 864								NDUF A6	Hs.27 4416
20388 0_at	264	NM_00 5694 / 914								COX1 7	Hs.53 4383
21852 5_s_at	265	NM_01 7902 / 995								HIF1A N	Hs.50 0788
21979 8_s_at	266	NM_01 9606 / 1017								FLJ20 257	Hs.17 8011
20839 8_s_at	267	NM_00 4865 / 897								TBPL1	Hs.48 6507
21835 8_at	268	NM_02 4324 / 1045								MGC1 1256	Hs.21 1282
20309 7_s_at	269	--- /								---	---
22159 8_s_at	270	NM_00 4269 / 887								CRSP8	Hs.37 4262
20954 6_s_at	271	AF3235 40 / 811								GenBa nk	Hs.11 4309
50400 _at	272	AI7439 90 / 812								GenBa nk	Hs.24 859
21775 2_s_at	273	NM_01 8235 / 1003								GenBa nk	Hs.27 3230
20193 1_at	274	NM_00 0126 / 825								GenBa nk	Hs.16 9919
21856 7_x_a t	275	NM_00 5700 / 915								GenBa nk	Hs.22 880
20205 6_at	276	AW051 311 / 816								GenBa nk	Hs.16 9149
21890 7_s_at	277	NM_02 3942 / 1041								GenBa nk	Hs.28 4135
21372 6_x_a t	278	AA515 698 / 806								GenBa nk	Hs.25 1653
20444 8_s_at	279	AF0314 63 / 808								GenBa nk	Hs.93 02
1405_ i_at	280	M21121 / 823								GenBa nk	---
21286 4_at	281	Y16521 / 1108								GenBa nk	Hs.24 812

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20944 4_at	282	BC0018 51 / 819								GenBa nk	Hs.79 40
20889 8_at	283	AF0776 14 / 809								GenBa nk	Hs.27 2630
20237 7_at	284	AW026 535 / 815								GenBa nk	Hs.23 581
20954 7_s_at	285	BC0010 43 / 818								GenBa nk	Hs.15 075
20292 2_at	286	BF6769 80 / 821								GenBa nk	Hs.15 1393
20140 3_s_at	287	NM_00 4528 / 892								GenBa nk	Hs.11 1811
21315 4_s_at	288	AI9341 25 / 814								GenBa nk	Hs.17 411
21567 6_at	289	N91109 / 824								GenBa nk	Hs.32 935
21892 7_s_at	290	NM_01 8641 / 1012								GenBa nk	Hs.25 204
21364 8_at	291	AW614 427 / 817								GenBa nk	Hs.18 2877
20959 3_s_at	292	AF3171 29 / 810								GenBa nk	Hs.25 2682
20373 1_s_at	293	NM_01 4569 / 960								GenBa nk	Hs.11 0839
20123 4_at	294	NM_00 4517 / 891								GenBa nk	Hs.61 96
21481 7_at	295	BE7836 68 / 820								GenBa nk	Hs.17 5780
20176 1_at	296	NM_00 6636 / 929								GenBa nk	Hs.15 4672
20546 7_at	297	NM_00 1230 / 850								GenBa nk	Hs.53 53
22231 8_at	298	AI7446 73 / 813								GenBa nk	Hs.18 6970
20993 3_s_at	299	AF0203 14 / 807								GenBa nk	Hs.96 88
20073 4_s_at	300	BG3419 06 / 822								GenBa nk	Hs.11 9177
21989 9_x_a t	301	NM_01 4434 / 959								NDOR 1	Hs.51 2564

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
22156 3_at	302	NM_14 4729 / 1074	NM_1 44728 / 1155	NM_0 07207 / 1192						DUSP 10, DUSP 10, DUSP 10,	Hs.49 7822, Hs.49 7822, Hs.49 7822,
65517 _at	303	NM_00 5498 / 910								AP1M 2	Hs.18 894
20512 6_at	304	NM_00 6296 / 921								VRK2	Hs.46 8623
20069 6_s_at	305	NM_19 8252 / 1094	NM_0 00177 / 1110							GSN, GSN,	Hs.52 2373, Hs.52 2373,
20879 0_s_at	306	NM_01 2232 / 944								PTRF	Hs.43 7191
20511 5_s_at	307	NM_01 6196 / 987								RBM1 9	Hs.74 82
21398 2_s_at	308	NM_01 4857 / 965								RABG APIL	Hs.49 5391
20699 2_s_at	309	NM_01 5684 / 980	NM_0 01003 805 / 1116	NM_0 01003 803 / 1184						ATP5S , ATP5S , ATP5S ,	Hs.43 8489, Hs.43 8489, Hs.43 8489,
20106 0_x_a t	310	NM_19 8194 / 1092	NM_0 04099 / 1129	NM_0 17723 / 1195						STOM, STOM, FLJ20 245,	Hs.25 3903, Hs.25 3903, Hs.49 5541,
20927 8_s_at	311	NM_00 6528 / 924								TFPI2	Hs.43 8231
21054 2_s_at	312	--- /								---	---
20200 6_at	313	NM_00 2835 / 869								PTPN1 2	Hs.61 812
21055 7_x_a t	314	NM_17 2211 / 1082	NM_0 00757 / 1112	NM_1 72210 / 1205	NM_1 72212 / 1226					CSF1, CSF1, CSF1, CSF1,	Hs.17 3894, Hs.17 3894, Hs.17 3894, Hs.17 3894,

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21079 6_x_a t	315	NM_19 8846 / 1097	NM_1 98845 / 1171	NM_0 01245 / 1188						SIGLE C6, SIGLE C6, SIGLE C6,	Hs.39 7255, Hs.39 7255, Hs.39 7255,
22082 5_s_at	316	NM_01 8240 / 1004								KIRRE L	Hs.27 2234
20237 8_s_at	317	NM_01 7526 / 993								LEPR	Hs.23 581
21024 1_s_at	318	NM_00 7233 / 936								TP53A P1	Hs.27 4329
21306 9_at	319	--- /								---	---
21603 4_at	320	NM_08 0740 / 1061								SUHW 1	Hs.17 8665
21079 3_s_at	321	NM_13 9131 / 1072	NM_0 05387 / 1131							NUP98 , NUP98 ,	Hs.52 4750, Hs.52 4750,
20737 5_s_at	322	NM_00 2189 / 860	NM_1 72200 / 1163							IL15R A, IL15R A,	Hs.52 4117, Hs.52 4117,
20624 7_at	323	NM_00 5931 / 917								MICB	Hs.21 1580
21707 8_s_at	324	NM_00 7261 / 937								CMRF -35H	Hs.96 88
21925 7_s_at	325	NM_02 1972 / 1031	NM_1 82965 / 1170							SPHK 1, SPHK 1,	Hs.68 061, Hs.68 061,

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20478 1_s_at	326	NM_15 2876 / 1079	NM_1 52877 / 1161	NM_1 52874 / 1204	NM_1 52875 / 1225	NM_1 52873 / 1238	NM_1 52871 / 1240	NM_1 52872 / 1244	NM_0 00043 / 1245	FAS, FAS, FAS, FAS, FAS, FAS, FAS, FAS, FAS,	Hs.24 4139, Hs.24 4139, Hs.24 4139, Hs.24 4139, Hs.24 4139, Hs.24 4139, Hs.24 4139, Hs.24 4139, Hs.24 4139,
22153 6_s_at	327	NM_01 8385 / 1009								FLJ11 301	Hs.51 8505
20946 8_at	328	NM_00 2335 / 861								LRP5	Hs.63 47
20106 1_s_at	329	NM_19 8194 / 1092	NM_0 04099 / 1129							STOM, STOM,	Hs.25 3903, Hs.25 3903,
20349 9_at	330	NM_00 4431 / 889								EPHA 2	Hs.17 1596
21332 4_at	331	NM_19 8291 / 1095	NM_0 05417 / 1132							SRC, SRC,	Hs.19 5659, Hs.19 5659,
22153 5_at	332	NM_01 8385 / 1009								FLJ11 301	Hs.51 8505
21993 8_s_at	333	NM_02 4430 / 1046								PSTPI P2	Hs.36 8623
21378 7_s_at	334	NM_00 6579 / 926								EBP	Hs.52 2636
21959 2_at	335	NM_02 4596 / 1049								MCPH 1	Hs.55 0532
20063 7_s_at	336	NM_13 0440 / 1064	NM_0 02840 / 1122							PTPRF , PTPRF ,	Hs.27 2062, Hs.27 2062,
20219 3_at	337	NM_00 5569 / 913	NM_0 16733 / 1141							LIMK 2, LIMK 2,	Hs.47 4596, Hs.47 4596,

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21485 9_at	338	NM_01 5082 / 969								FSTL4	Hs.48 3390
20921 3_at	339	NM_00 1757 / 857								CBR1	Hs.88 778
21894 5_at	340	NM_02 4109 / 1044								MGC2 654	Hs.30 6380
22047 1_s_at	341	NM_02 5107 / 1052								MYCT 1	Hs.18 160
59625 _at	342	NM_00 3946 / 883								NOL3	Hs.51 3667
20723 3_s_at	343	NM_19 8158 / 1091	NM_0 00248 / 1111	NM_1 98178 / 1210	NM_1 98177 / 1228	NM_0 06722 / 1235	NM_1 98159 / 1241			MITF, MITF, MITF, MITF, MITF, MITF,	Hs.16 6017, Hs.16 6017, Hs.16 6017, Hs.16 6017, Hs.16 6017, Hs.16 6017, Hs.16 6017,
21407 0_s_at	344	--- /								---	---
21978 5_s_at	345	--- /								---	---
22068 4_at	346	NM_01 3351 / 954								TBX21	Hs.27 2409
20224 5_at	347	NM_00 2340 / 862								LSS	Hs.51 7366
20729 8_at	348	NM_00 6632 / 927								SLC17 A3	Hs.32 7179
22020 1_at	349	NM_01 8835 / 1013								MNAB	Hs.53 3499
20320 8_s_at	350	NM_01 4637 / 961								CHPP R	Hs.52 1608
20703 3_at	351	NM_00 5142 / 906								GIF	Hs.11 0014
20492 9_s_at	352	NM_00 6634 / 928								VAMP 5	Hs.17 2684

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20973 5_at	353	NM_00 4827 / 894								ABCG 2	Hs.48 0218
20923 4_at	354	NM_01 5074 / 968								KIF1B	Hs.97 858
21010 2_at	355	NM_19 8315 / 1096								LOH1 1CR2 A	Hs.15 2944
20411 4_at	356	NM_00 7361 / 941								NID2	Hs.36 9840
20334 4_s_at	357	NM_20 3292 / 1100	NM_2 03291 / 1174	NM_0 02894 / 1190						RBBP 8, RBBP 8, RBBP 8, RBBP 8,	Hs.54 6282, Hs.54 6282, Hs.54 6282, Hs.54 6282,
21689 1_at	358	--- /								---	---
22168 0_s_at	359	NM_01 6135 / 986								ETV7	Hs.27 2398
20208 7_s_at	360	NM_00 1912 / 858	NM_1 45918 / 1159							CTSL, CTSL,	Hs.41 8123, Hs.41 8123,
21332 4_at	331	NM_19 8291 / 1095	NM_0 05417 / 1132							SRC, SRC,	Hs.19 5659, Hs.19 5659,
22153 5_at	332	NM_01 8385 / 1009								FLJ11 301	Hs.51 8505
21993 8_s_at	333	NM_02 4430 / 1046								PSTPI P2	Hs.36 8623
21378 7_s_at	334	NM_00 6579 / 926								EBP	Hs.52 2636
21959 2_at	335	NM_02 4596 / 1049								MCPI 1	Hs.55 0532
20063 7_s_at	336	NM_13 0440 / 1064	NM_0 02840 / 1122							PTPRF , PTPRF ,	Hs.27 2062, Hs.27 2062,
20219 3_at	337	NM_00 5569 / 913	NM_0 16733 / 1141							LIMK 2, LIMK 2,	Hs.47 4596, Hs.47 4596,

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
21485 9_at	338	NM_01 5082 / 969								FSTL4	Hs.48 3390
20921 3_at	339	NM_00 1757 / 857								CBR1	Hs.88 778
21894 5_at	340	NM_02 4109 / 1044								MGC2 654	Hs.30 6380
22047 1_s_at	341	NM_02 5107 / 1052								MYCT 1	Hs.18 160
59625 _at	342	NM_00 3946 / 883								NOL3	Hs.51 3667
20723 3_s_at	343	NM_19 8158 / 1091	NM_0 00248 / 1111	NM_1 98178 / 1210	NM_1 98177 / 1228	NM_0 06722 / 1235	NM_1 98159 / 1241			MITF, MITF, MITF, MITF, MITF, MITF,	Hs.16 6017, Hs.16 6017, Hs.16 6017, Hs.16 6017, Hs.16 6017, Hs.16 6017, Hs.16 6017,
21407 0_s_at	344	--- /								---	---
21978 5_s_at	345	--- /								---	---
22068 4_at	346	NM_01 3351 / 954								TBX21	Hs.27 2409
20224 5_at	347	NM_00 2340 / 862								LSS	Hs.51 7366
20729 8_at	348	NM_00 6632 / 927								SLC17 A3	Hs.32 7179
22020 1_at	349	NM_01 8835 / 1013								MNAB	Hs.53 3499
20320 8_s_at	350	NM_01 4637 / 961								CHPP R	Hs.52 1608
20703 3_at	351	NM_00 5142 / 906								GIF	Hs.11 0014
20492 9_s_at	352	NM_00 6634 / 928								VAMP 5	Hs.17 2684

[illegible]

PRO BESE T	SEQ ID NO:	Refseq Ids / SEQ ID NO:								REFS EQ_U NIGE NE	REFS EQ_B AND
20878 2_at	399	NM_00 7085 / 935								FSTL1	Hs.26 9512
20246 8_s_at	400	NM_00 3798 / 882								CTNN AL1	Hs.58 488
20701 8_s_at	401	NM_00 4163 / 886								RAB2 7B	Hs.25 318
21035 4_at	402	NM_00 0619 / 832								IFNG	Hs.85 6
20111 0_s_at	403	NM_00 3246 / 874								THBS 1	Hs.16 4226
20780 8_s_at	404	NM_00 0313 / 827								PROS1	Hs.64 016
20561 2_at	405	NM_00 7351 / 940								MMR N1	Hs.26 8107
21407 3_at	406	--- /								---	---

Table 2.

The genes encoding RNA polymerase I transcription factor (RRN3) and leucine-rich PPR-motif containing protein (LRPRC) were found as most significantly down-regulated genes in BMS signature (Table 1, hereinabove). RRN33 (transcription initiation factor TIF-IA), a 72-kDa protein, is essential for ribosomal DNA (rDNA) transcription and acts as a bridge between RNA pol I and the committed rDNA promoter (Hirschler-Laszkiewicz I, et al., 2003; Miller G, et al., 2001;20:1373-1382). The suppression of polymerase I regulation mechanism is confirmed by down-regulation of polymerase (RNA) I polypeptide D (POLR1D).

LRPPRC is a candidate gene for the French-Canadian type of Leigh syndrome, a form of cytochrome c oxidase deficiency, and plays a role in translation or stability of mitochondrially encoded cytochrome c oxidase (COX) subunits (Mootha VK, et al., 2003). The LRPPC together with POLR1D molecules comprise a complex with NFkBIB protein (Bouwmeester T, et al., 2004) that inhibits proinflammatory NFkB pathway.

Without being bound by any theory, the suppression of molecules involved in polymerase I related mechanism, COXI and NFkB regulation could account for the differences between BMS and typical RRMS patients. In addition, the polymerase I related mechanism can be potential drug targets for the treatment of RRMS aimed to switch RRMS to the BMS variant. One of commercially available drug that has proven effects on polymerase I mechanism is a diterpenoid triepoxide Triptolide (TPT), isolated from the Chinese herb *Tripterygium wilfordii* (Leuenroth SJ and Crews CM. Triptolide-induced transcriptional arrest is associated with changes in nuclear substructure. *Cancer Res.* 2008; 68:5257-5266). Triptolide has various anti-inflammatory effects (Liu Y, et al. Triptolide, a component of Chinese herbal medicine, modulates the functional phenotype of dendritic cells. *Transplantation.* 2007; 84:1517-1526), it modulates T-cell inflammatory responses and ameliorates Experimental Autoimmune Encephalomyelitis (Wang Y, et al. Triptolide modulates T-cell inflammatory responses and ameliorates experimental autoimmune encephalomyelitis. *J Neurosci Res.* 2008; 86:2441-2449).

More specifically TPT demonstrated to suppress of T lymphocyte function including T cell apoptosis induction, inhibition of lymphocyte proliferation and IFN γ production (Chen BJ. 2001. Triptolide, a novel immunosuppressive and anti-inflammatory agent purified from a Chinese herb *Tripterygium wilfordii* Hook f. *Leuk Lymphoma* 42:253–265; Qiu D, Kao PN. 2003. Immunosuppressive and anti-inflammatory mechanisms of triptolide, the principal active diterpenoid from the Chinese medicinal herb *Tripterygium wilfordii* Hook f. *Drugs R D* 4:1–18; Yang Y, Liu Z, Tolosa E, Yang J, Li L. 1998. Triptolide induces apoptotic death of T lymphocyte. *Immunopharmacology* 40:139–149; Chan MA, Kohlmeier JE, Branden M, Jung M, Benedict SH. 1999. Triptolide is more effective in preventing T cell proliferation and interferon- γ production than is FK506. *Phytother Res* 13:464–467). The TPT decreased IL2 and IL2 receptor expression by inhibiting activation of the purine box regulator of the NFkB of activated T cells (Qiu 1999). Additionally, it was demonstrated that TPT can inhibit the maturation, antigen processing, and presentation of dendritic cells and can suppress tumor necrosis factor (TNF)- α and IL-6 production by activated macrophages (Zhu KJ, Shen QY, Cheng H, Mao XH, Lao LM, Hao GL. 2005. Triptolide affects the differentiation, maturation and function of human dendritic

cells. *Int Immunopharmacol* 5:1415–1426; Wu Y, Cui J, Bao X, Chan S, Young DO, Liu D, Shen P. 2006. Triptolide attenuates oxidative stress, NF-kappaB activation and multiple cytokine gene expression in murine peritoneal macrophage. *Int J Mol Med* 17:141–150).

- 5 Table 3, hereinbelow, discloses the genes involved in the RNA polymerase I pathway, which are likely to be involved in typical RRMS or BMS pathology.

Table 3
Genes involved in the RNA polymerase I pathway

10

<i>Affymetrix ProbSet</i>	<i>SEQ ID NO:</i>	<i>Representative Public ID / SEQ ID NO:</i>	<i>Gene Symbol</i>	<i>Gene Title</i>
216902_s_at	62	AF001549 / 1284 ; NM_018427 / 1285	RRN3	RRN3 RNA polymerase I transcription factor homolog
211971_s_at	204	AI653608 / 1286 ; NM_133259 / 1287	LRPPRC	leucine-rich PPR-motif containing
220113_x_at	1248	NM_019014 / 1288	POLR1B	polymerase (RNA) I polypeptide B, 128kDa
207515_s_at	1249	NM_004875 / 1289	POLR1C	polymerase (RNA) I polypeptide C, 30kDa
209317_at	1250	AF008442 / 1290	POLR1C	polymerase (RNA) I polypeptide C, 30kDa
218258_at	173	NM_015972 / 1291	POLR1D	polymerase (RNA) I polypeptide D, 16kDa
202725_at	1251	NM_000937 / 1292	POLR2A	polymerase (RNA) II (DNA directed) polypeptide A, 220kDa
217420_s_at	1252	M21610 / 1293	POLR2A	polymerase (RNA) II (DNA directed) polypeptide A, 220kDa
201803_at	1253	NM_000938 / 1294	POLR2B	polymerase (RNA) II (DNA directed) polypeptide B, 140kDa
208996_s_at	1254	BC000409 / 1295	POLR2C	polymerase (RNA) II (DNA directed) polypeptide C, 33kDa
214263_x_at	1255	AI192781 / 1296	POLR2C	polymerase (RNA) II (DNA directed) polypeptide C, 33kDa
216282_x_at	1256	AJ224143 / 1297	POLR2C	polymerase (RNA) II (DNA directed) polypeptide C, 33kDa
203664_s_at	1257	NM_004805 / 1298	POLR2D	polymerase (RNA) II (DNA directed) polypeptide D
214144_at	1258	BF432147 / 1299	POLR2D	polymerase (RNA) II (DNA directed) polypeptide D
213887_s_at	1259	AI554759 / 1300	POLR2E	polymerase (RNA) II (DNA directed) polypeptide E, 25kDa
217854_s_at	1260	BC004441 / 1301	POLR2E	polymerase (RNA) II (DNA directed) polypeptide E, 25kDa
209511_at	1261	BC003582 / 1302	POLR2F	polymerase (RNA) II (DNA directed) polypeptide F
202306_at	1262	NM_002696 / 1303	POLR2G	polymerase (RNA) II (DNA directed) polypeptide G
209302_at	1263	U37689 / 1304	POLR2H	polymerase (RNA) II (DNA directed) polypeptide H

<i>Affymetrix ProbSet</i>	<i>SEQ ID NO:</i>	<i>Representative Public ID / SEQ ID NO:</i>	<i>Gene Symbol</i>	<i>Gene Title</i>
212955_s_at	1264	AL037557 / 1305	POLR2I	polymerase (RNA) II (DNA directed) polypeptide I, 14.5kDa
212782_x_at	1265	BG335629 / 1306	POLR2J	polymerase (RNA) II (DNA directed) polypeptide J, 13.3kDa
216242_x_at	1266	AW402635 / 1307	POLR2J2	DNA directed RNA polymerase II polypeptide J-related gene
214740_at	1267	BE676209 / 1308	POLR2J2 /// MGC13098	DNA directed RNA polymerase II polypeptide J-related gene /// hypothetical prote
202634_at	1268	AL558030 / 1309	POLR2K	polymerase (RNA) II (DNA directed) polypeptide K, 7.0kDa
202635_s_at	1269	NM_005034 / 1310	POLR2K	polymerase (RNA) II (DNA directed) polypeptide K, 7.0kDa
202586_at	1270	AA772747 / 1311	POLR2L	polymerase (RNA) II (DNA directed) polypeptide L, 7.6kDa
211730_s_at	1271	BC005903 / 1312	POLR2L	polymerase (RNA) II (DNA directed) polypeptide L, 7.6kDa /// polymerase (RNA) II
219459_at	1272	NM_018082 / 1313	POLR3B	polymerase (RNA) III (DNA directed) polypeptide B
209382_at	1273	U93867 / 1314	POLR3C	polymerase (RNA) III (DNA directed) polypeptide C (62kD)
210573_s_at	1274	BC004424 / 1315	POLR3C	polymerase (RNA) III (DNA directed) polypeptide C (62kD)
208361_s_at	1275	NM_001722 / 1316	POLR3D	polymerase (RNA) III (DNA directed) polypeptide D, 44kDa
218016_s_at	1276	NM_018119 / 1317	POLR3E	polymerase (RNA) III (DNA directed) polypeptide E (80kD)
205218_at	1277	NM_006466 / 1318	POLR3F	polymerase (RNA) III (DNA directed) polypeptide F, 39 kDa
206653_at	1278	BF062139 / 1319	POLR3G	Polymerase (RNA) III (DNA directed) polypeptide G (32kD)
206654_s_at	1279	NM_006467 / 1320	POLR3G	polymerase (RNA) III (DNA directed) polypeptide G (32kD)
218866_s_at	1280	AF060223 / 1321	POLR3K	polymerase (RNA) III (DNA directed) polypeptide K, 12.3 kDa
203782_s_at	1281	NM_005035 / 1322	POLRMT	polymerase (RNA) mitochondrial (DNA directed)
203783_x_at	1282	BF057617 / 1323	POLRMT	polymerase (RNA) mitochondrial (DNA directed)
202466_at	1283	NM_006999 / 1324	POLS	polymerase (DNA directed) sigma

Table 3.

The measurement of RRN3, LRPPRC, POLR1D and other polymerase I mechanism related biomarkers could be used for diagnosis and prediction of BMS.

5 Additionally those markers could be useful for typical RRMS patients to monitor the

efficacy of various immunomodulatory drugs for assessment of patients with good response to treatment.

Measurement of BMS biomarkers can be performed on the mRNA level by the quantitative reverse-transcriptase polymerase chain reaction (QRT-PCR) method and on
5 protein level by LUMINEX technology. Possible modification of the invention is developing biomarkers on protein level (e.g., using ELISA, Western Blot analysis and the like) in PBMC and serum.

Multiple sclerosis (MS) is a heterogeneous disease. To better diagnose and treat MS patients the various types of disease have to be distinguished. The teachings of the
10 invention enable, for the first time, to distinguish between BMS and typical RRMS patients using molecular tools, which when combined with accurate clinical information enables to dissect the biological complexity of MS.

In the current study the use of gene expression profiling enabled to diagnose benign MS using a phenotypic approach to differentiate subtypes of the disease. The
15 identified gene expression phenotypes also enable to better understand the biology of benign MS and to develop therapeutics strategies to treat MS.

The gene expression signature generated herein of benign MS enables refining MS to diagnose low risk patients versus high risk patients and accordingly suggest appropriate treatment.

20 The benign MS patients that represent low risk would not be treated while the high risk relapsing-remitting MS patients will be treated. In addition, the teachings of the invention enable to monitor response to treatment and better use of current approved medications.

The teachings of the invention can be used to develop a kit or device for
25 diagnosis and prediction of typical RRMS clinical outcome, improving medical decision support systems and individualizing patient care. In addition, the teachings of the invention can be used to develop new drugs that will imitate BMS gene expression signature and will result in silencing of the active RRMS.

EXAMPLE 2***Study subject and methods***

Subjects – 31 patients (age 44.5 ± 1.5 ; female to male ratio 24:7) with BMS were characterized by mean EDSS 1.95 ± 0.15 , disease duration 17.0 ± 1.3 years, annual EDSS rate 0.13 ± 0.01 , annual relapse rate 0.23 ± 0.04 . 36 patients (age 40.3 ± 1.8 ; female to male ratio 8:3) with RRMS were characterized by mean EDSS 3.54 ± 0.23 , disease duration 10.9 ± 1.4 years, annual EDSS rate 0.45 ± 0.06 , annual relapse rate 0.64 ± 0.09 .

RNA isolation and microarray expression profiling - Peripheral blood mononuclear cells (PBMC) were separated on ficoll-hypaque gradient. Total RNA was isolated using the TRIzol Reagent (Invitrogen, Carlsbad, CA), and cDNA was synthesized, labeled and hybridized to HG-U133A-2 array (Affymetrix, Inc, Santa Clara, CA) containing 22,215 gene-transcripts, washed and scanned (Hewlett Packard, GeneArray-TM scanner G2500A) according to manufacturer's protocol Affymetrix (Inc, Santa Clara, CA).

Data Analysis - Data analysis was performed using the Partek Genomics Solution software [World Wide Web (dot) partek (dot) com]. Expression values were computed from raw CEL (cells) files by applying the Robust Multi-Chip Average (RMA) background correction algorithm. The RMA correction included: 1) values background correction; 2) quantile normalization; 3) log2 transformation; 4) median polish summarization. The gene transcripts were filtered using Affymetrix MAS5 Present/Absent Detection. Thereafter, 9987 transcript that were detected as Present in 100% microarrays were used for analysis. In order to avoid the noise caused by variable set effects each set was normalized to pre-saved distribution pattern of a well balanced set used as a reference distribution. To reduce batch effect ANOVA multiple model analysis was applied. Source of variation was analyzed; nuisance batches effects such as working batch, patient age, gender and treatment were eliminated. Most informative genes (MIGs) were defined as genes that passed Falls Discovery Rate (FDR) correction with $p < 0.05$ by ANOVA linear contrasts model. Thereafter, predictive algorithm based on two level cross validation method, Super Vector Machine (SVM) and K-Nearest Neighbor algorithms were applied to calculate MIGs

classification rates. Only genes which were included in classifiers (from 1 to 10 genes) with more than 70% correct classification rates were analyzed.

Experimental Results

Identification of classification rates of genes involved in the RNA polymerase

- 5 ***I pathway between typical RRMS and BMS disease course*** – Tables 4A-C presents the corrected classification rates for all combinations of RRN3, LRPPRC and POLR1D genes of the RNA polymerase I pathway for the entire BMS and typical RRMS group (Table 4A), for the BMS group (Table 4B) and for the typical RRMS group (Table 4C). The best predictive performance for each classifier is presented.

Table 4A

<i>Classifiers</i>	<i>Aver. % Correct</i>	<i>St. Err (%)</i>
<i>RRN3</i>	<i>63.6</i>	<i>11.2</i>
<i>LRPPRC</i>	<i>73.3</i>	<i>9.5</i>
<i>POLR1D</i>	<i>72.7</i>	<i>9.5</i>
<i>RRN3, POLR1D</i>	<i>72.7</i>	<i>9.5</i>
<i>RRN3, LRPPRC</i>	<i>63.6</i>	<i>10.2</i>
<i>POLR1D, LRPPRC</i>	<i>81.8</i>	<i>8.2</i>
<i>RRN3, LRPPRC, POLR1D</i>	<i>77.2</i>	<i>8.9</i>

Table 4A: Average % correct = Average percent of correct classification between BMS and typical RRMS patients using specific classifier; St. Err. = standard error;

Table 4B

<i>Classifiers</i>	<i>BMS % corr</i>
<i>RRN3</i>	<i>67</i>
<i>LRPPRC</i>	<i>80</i>
<i>POLR1D</i>	<i>60</i>
<i>RRN3, POLR1D</i>	<i>60</i>
<i>RRN3, LRPPRC</i>	<i>30</i>
<i>POLR1D, LRPPRC</i>	<i>80</i>
<i>RRN3, LRPPRC, POLR1D</i>	<i>70</i>

Table 4B: BMS % Corr. = Average percent of correct classification for BMS patients using specific classifier;

Table 4C

<i>Classifiers</i>	<i>Typical RRMS % corr</i>
<i>RRN3</i>	<i>60</i>
<i>LRPPRC</i>	<i>67</i>
<i>POLR1D</i>	<i>80</i>
<i>RRN3, POLR1D</i>	<i>83</i>

<i>Classifiers</i>	<i>Typical RRMS % corr</i>
<i>RRN3, LRPPRC</i>	<i>90</i>
<i>POLR1D, LRPPRC</i>	<i>83</i>
<i>RRN3, LRPPRC, POLR1D</i>	<i>83</i>

Table 4C: RRMS % corr. – Average percent of correct classification for typical RRMS patients using specific classifier.

The results presented in Table 4B demonstrate that for classification of BMS, each of the genes of the RNA polymerase I pathway, *i.e.*, RRN3, LRPPRC and POLR1D exhibits a correct classification rate of 67%, 80% and 60%, respectively. The results presented in Table 4C demonstrate that for classification of typical RRMS, each of RRN3, LRPPRC and POLR1D exhibits a correct classification rate of 60%, 67% and 80%, respectively. In addition, an increased rate of correct classification of typical RRMS, which is a more complex and heterogenous condition, can be achieved using a combination of 2 or 3 genes of the RNA polymerase I pathway. For example, a correct classification rate of 83% is obtained using the combination of RRN3 and POLR1D; a correct classification rate of 83% is obtained using the combination of POLR1D and LRPPRC; and a correct classification rate of 90% is obtained using the combination of RRN3 and LRPPRC (Table 4C, above).

The classification for the combination of RRN3 and LRPPRC resulted in outstanding classification rate of 90% of typical RRMS (TMS) (Table 4C), while showing a low 30% classification for these 2 genes for BMS (Table 4B), further supporting the differentiations between the two disease patterns.

To conclude, correct classification for both groups would be achieved using more than one combination of genes of the RNA polymerase I pathway, for example LRPPRC alone or with POLR1D reached 80% classification rate for BMS and RRN3 and LRPPRC correctly classify 90% of TMS patients.

Identification of biomarkers for differentiation of patients with BMS or typical RRMS course of disease - BMS patients differentiated from typical RRMS by 177 MIGs (Table 5). The 17 genes with higher classification performance (Table 6) were identified from MIGs by applying predictive algorithms.

Table 5
MIGs discriminating between BMS and typical RRMS patients

<i>Affymetrix Probe Set ID</i>	<i>SEQ NO ID:</i>	<i>Representative Public ID</i>	<i>SEQ ID NO:</i>	<i>p-value BMS vs typical RRMS</i>	<i>Log Fold Change (BMS vs typical RRMS)</i>	<i>Gene Symbol</i>	<i>Gene description</i>
216902_s_at	62	AF001549	468	2.12E-08	-2.01	RRN3	RRN3 RNA polymerase I transcription factor homolog (yeast)
210502_s_at	1410	AF042386	1534	2.30E-08	-1.28	PPIE	peptidylprolyl isomerase E (cyclophilin E)
211615_s_at	1418	M92439	1512	2.44E-08	-1.20	LRPPRC	leucine-rich PPR-motif containing /// leucine-rich PPR-motif containing
37950_at	1500	X74496	1526	1.05E-06	1.19	PREP	prolyl endopeptidase
218258_at	173	NM_015972	984	1.92E-06	-1.18	POLR1D	polymerase (RNA) I polypeptide D, 16kDa
214439_x_at	1453	AF043899	1535	2.24E-06	-1.62	BIN1	bridging integrator 1
214450_at	1454	NM_001335	1629	3.82E-06	-1.45	CTSW	cathepsin W (lymphopain) /// cathepsin W (lymphopain)
214470_at	1455	NM_002258	1615	3.83E-06	-1.77	KLRB1	killer cell lectin-like receptor subfamily B, member 1 /// killer cell lectin-li
205789_at	1373	NM_001766	1627	4.32E-06	1.62	CD1D	CD1D antigen, d polypeptide /// CD1D antigen, d polypeptide
206584_at	1378	NM_015364	1650	5.84E-06	1.59	LY96	lymphocyte antigen 96
212252_at	1425	AA181179	1660	6.14E-06	1.62	CAMKK2	calcium/calmodulin-dependent protein kinase kinase 2, beta
212748_at	1431	AB037859	1583	7.92E-06	1.18	MKL1	megakaryoblastic leukemia (translocation) 1
211654_x_at	1419	M17565	1519	9.99E-06	2.75	HLA-DQB1	major histocompatibility complex, class II, DQ beta 1 /// major histocompatibili
204860_s_at	1370	AI817801	1673	1.21E-05	1.62	BIRC1	baculoviral IAP repeat-containing 1
211971_s_at	204	AI653608	606	1.62E-05	-1.14	LRPPRC	leucine-rich PPR-motif containing
202832_at	1347	NM_014635	1601	2.21E-05	-1.33	GCC2	GRIP and coiled-coil domain containing 2

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40446_at	1502	AL021366	1537	2.62E-05	-1.39	PHF1	PHD finger protein 1
210136_at	1404	AW070431	1676	2.71E-05	-1.76	MBP	Myelin basic protein
202441_at	1342	AL568449	1691	3.92E-05	1.37	C10orf69	chromosome 10 open reading frame 69
213241_at	1442	AF035307	1532	4.12E-05	1.55	PLXNC1	Plexin C1
212978_at	1435	AU146004	1686	4.13E-05	1.56	TA-LRRP	T-cell activation leucine repeat-rich protein
220005_at	1490	NM_023914	1591	4.55E-05	2.43	P2RY13	purinergic receptor P2Y, G-protein coupled, 13 /// purinergic receptor P2Y, G-pr
218304_s_at	1473	NM_022776	1645	4.89E-05	1.58	OSBP-L11	oxysterol binding protein-like 11
218932_at	1480	NM_017953	1652	5.00E-05	-1.27	FLJ20729	hypothetical protein FLJ20729
213106_at	1440	AI769688	1670	5.57E-05	-1.51	ATP8A1	ATPase, aminophospholipid transporter (APLT), Class I, type 8A, member 1
219892_at	1489	NM_023003	1648	5.63E-05	1.86	TM6SF1	transmembrane 6 superfamily member 1
219666_at	1486	NM_022349	1585	6.07E-05	1.81	MS4A6A	membrane-spanning 4-domains, subfamily A, member 6A
206120_at	1376	NM_001772	1631	6.26E-05	1.58	CD33	CD33 antigen (gp67)
209970_x_at	1403	M87507	1523	6.51E-05	1.33	CASP1	caspase 1, apoptosis-related cysteine protease (interleukin 1, beta, convertase)
200980_s_at	1330	NM_000284	1654	8.59E-05	-1.27	PDHA1	pyruvate dehydrogenase (lipoamide) alpha 1
200610_s_at	1326	NM_005381	1613	8.97E-05	-1.17	NCL	nucleolin
213418_at	1445	NM_002155	1609	8.99E-05	2.73	HSPA6	heat shock 70kDa protein 6 (HSP70B')
212421_at	1429	AB023147	1545	9.22E-05	1.66	C22orf9	chromosome 22 open reading frame 9
210201_x_at	1407	AF001383	1531	9.59E-05	-1.59	BIN1	bridging integrator 1
207000_s_at	1381	NM_005605	1600	9.72E-05	-1.32	PPP3CC	protein phosphatase 3 (formerly 2B), catalytic subunit, gamma isoform (calcineur

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203139_at	1352	NM_004938	1623	0.000102879	1.78	DAPK1	death-associated protein kinase 1
211368_s_at	1416	U13700	1527	0.000107311	1.36	CASP1	caspase 1, apoptosis-related cysteine protease (interleukin 1, beta, convertase)
212820_at	1433	AB020663	1582	0.000108529	1.70	RC3	rabconnectin-3
216945_x_at	15	U79240	421	0.000111457	-1.86	PASK	PAS domain containing serine/threonine kinase
209337_at	1395	AF063020	1540	0.000112781	-1.35	PSIP1	PC4 and SFRS1 interacting protein 1
201756_at	1337	NM_002946	1599	0.000114388	-1.41	RPA2	replication protein A2, 32kDa
221565_s_at	1493	BC000039	1602	0.00011473	1.50	FAM26B	family with sequence similarity 26, member B
117_at	1325	X51757cds	#N/A	0.000116998	2.01	HSPA6	heat shock 70kDa protein 6 (HSP70B')
43544_at	1503	AA314406	1661	0.000118986	-1.65	THRAP5	thyroid hormone receptor associated protein 5
219132_at	1483	NM_021255	1621	0.000121966	1.36	PELI2	pellino homolog 2 (Drosophila)
57715_at	1507	W72694	1657	0.000123909	1.35	FAM26B	family with sequence similarity 26, member B
220066_at	1491	NM_022162	1569	0.000125076	1.53	CARD15	caspase recruitment domain family, member 15
212414_s_at	1428	D50918	1528	0.000139145	-1.53	SEPT6	septin 6
213902_at	1448	AI379338	1666	0.000139994	1.32	ASAH1	N-acylsphingosine amidohydrolase (acid ceramidase) 1
212998_x_at	1436	AI583173	1667	0.000143667	2.26	HLA-DQB1	major histocompatibility complex, class II, DQ beta 1 /// major histocompatibili
202931_x_at	1350	NM_004305	1584	0.000148834	-1.55	BIN1	bridging integrator 1
204112_s_at	1361	NM_006895	1619	0.00014954	1.81	HNMT	histamine N-methyltransferase
205467_at	297	NM_001230	850	0.000150498	1.44	CASP10	caspase 10, apoptosis-related cysteine protease

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209199_s_at	1394	N22468	1656	0.0001 57534	1.51	MEF2C	MADS box transcription enhancer factor 2, polypeptide C (myocyte enhancer factor)
211676_s_at	1421	AF056979	1578	0.0001 57969	1.40	IFNGR1	interferon gamma receptor 1 /// interferon gamma receptor 1
203492_x_at	1355	AA918224	1664	0.0001 63582	-1.22	KIAA0092	translokin
211776_s_at	1423	BC006141	1577	0.0001 64075	3.18	EPB41L3	erythrocyte membrane protein band 4.1-like 3 /// erythrocyte membrane protein ba
56919_at	1505	AI806628	1672	0.0001 73847	1.32	KIAA1449	WD repeat endosomal protein
202521_at	1344	NM_006565	1616	0.0001 784	-1.14	CTCF	CCCTC-binding factor (zinc finger protein)
211727_s_at	1422	BC005895	1576	0.0001 79886	-1.49	COX11	COX11 homolog, cytochrome c oxidase assembly protein (yeast) /// COX11 homolog,
204222_s_at	1363	NM_006851	1628	0.0001 8487	1.54	GLIPR1	GLI pathogenesis-related 1 (glioma)
204839_at	1369	NM_015918	1603	0.0001 85062	-1.21	POP5	processing of precursor 5, ribonuclease P/MRP subunit (S. cerevisiae)
39729_at	1501	L19185	1524	0.0001 85886	-1.62	PRDX2	peroxiredoxin 2
221078_s_at	1492	NM_018084	1646	0.0001 88529	1.49	KIAA1212	KIAA1212
35156_at	1499	AL050297	1548	0.0001 96739	-1.24	LOC203069	Hypothetical protein LOC203069
219630_at	1485	NM_005764	1606	0.0001 98793	1.94	MAP17	membrane-associated protein 17
202662_s_at	1345	NM_002223	1625	0.0002 0557	1.34	ITPR2	inositol 1,4,5-triphosphate receptor, type 2
210212_x_at	1409	BC002600	1571	0.0002 19441	-1.27	MTCP1	mature T-cell proliferation 1
217925_s_at	1471	NM_022758	1610	0.0002 24904	-1.21	C6orf106	chromosome 6 open reading frame 106
218739_at	1477	NM_016006	1643	0.0002 28499	1.75	ABHD5	abhydrolase domain containing 5
56197_at	1504	AI783924	1671	0.0002 31008	-1.24	PLSCR3	phospholipid scramblase 3
208653_s_at	1388	AF263279	1561	0.0002 3174	1.46	CD164	CD164 antigen, sialomucin
213292_s_at	1444	AA908770	1663	0.0002 3191	1.54	SNX13	sorting nexin 13

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201194_at	1331	NM_003009	1608	0.0002 40228	-1.36	SEPW1	selenoprotein W, 1
201619_at	1336	NM_006793	1596	0.0002 50445	1.27	PRDX3	peroxiredoxin 3
203569_s_at	1356	NM_003611	1640	0.0002 72476	-1.45	OFD1	oral-facial-digital syndrome 1
213979_s_at	1450	BF984434	1689	0.0002 7541	-3.22	---	---
203814_s_at	1359	NM_000904	1635	0.0002 81273	1.67	NQO2	NAD(P)H dehydrogenase, quinone 2
215118_s_at	1462	AW519168	1680	0.0003 02493	-2.77	MGC27165	Hypothetical protein MGC27165
203624_at	1357	NM_005088	1607	0.0003 30513	-1.38	DXYS155E	DNA segment on chromosome X and Y (unique) 155 expressed sequence
206999_at	1380	NM_001559	1588	0.0003 44019	2.07	IL12RB2	interleukin 12 receptor, beta 2
205842_s_at	1374	AF001362	1538	0.0003 51718	1.85	JAK2	Janus kinase 2 (a protein tyrosine kinase)
210166_at	1405	AF051151	1536	0.0003 53968	1.63	TLR5	toll-like receptor 5
219714_s_at	1487	NM_018398	1612	0.0003 64622	2.39	CACNA2D3	calcium channel, voltage-dependent, alpha 2/delta 3 subunit
212311_at	1426	AA522514	1662	0.0003 70292	-1.36	KIAA0746	KIAA0746 protein
213830_at	1447	AW007751	1675	0.0003 70999	-1.69	TRD@	T-cell receptor delta chain HE/801 /// T cell receptor delta locus
213005_s_at	1438	D79994	1614	0.0003 76513	2.26	ANKRD15	ankyrin repeat domain 15
202944_at	1351	NM_000262	1639	0.0003 76931	1.68	NAG A	N-acetylgalactosaminidase, alpha-
206011_at	1375	AI719655	1668	0.0003 79898	1.49	CASP1	caspase 1, apoptosis-related cysteine protease (interleukin 1, beta, convertase)
204332_s_at	1365	M64073	1516	0.0003 82893	-1.22	AGA	aspartylglucosaminidase
215592_at	1463	AU147620	1687	0.0003 85413	-1.91	---	Transcribed locus, weakly similar to XP_375099.1 hypothetical protein LOC283585
201798_s_at	378	NM_013451	777	0.0003 87903	2.24	FER1L3	fer-1-like 3, myoferlin (C. elegans)
212069_s_at	1424	AK026025	1566	0.0003 88281	-1.17	KIAA0515	KIAA0515

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206255_at	1377	NM_001715	1597	0.0003 95152	-1.72	BLK	B lymphoid tyrosine kinase
221932_s_at	1497	AA133341	1658	0.0003 96589	-1.42	C14orf87	chromosome 14 open reading frame 87
203246_s_at	1353	NM_006545	1611	0.0004 02598	-1.33	TUSC4	tumor suppressor candidate 4
213534_s_at	19	D50925	425	0.0004 08073	-1.73	PASK	PAS domain containing serine/threonine kinase
219045_at	1481	NM_019034	1604	0.0004 12198	-1.33	RHOF	ras homolog gene family, member F (in filopodia)
202347_s_at	1340	AB022435	1550	0.0004 14497	-1.16	HIP2	huntingtin interacting protein 2
212636_at	1430	AL031781	1543	0.0004 20707	1.61	QKI	quaking homolog, KH domain RNA binding (mouse)
202392_s_at	1341	NM_014338	1598	0.0004 23758	1.26	PISD	phosphatidylserine decarboxylase
216950_s_at	1467	X14355	1509	0.0004 37652	2.62	FCGR1A	Fc fragment of IgG, high affinity Ia, receptor for (CD64)
214511_x_at	1457	L03419	1515	0.0004 47516	2.83	LOC40607 /// FCGR1A	Fc-gamma receptor I B2 /// Fc fragment of IgG, high affinity Ia, receptor for (C
219316_s_at	1484	NM_017791	1641	0.0004 72922	1.68	C14orf58	chromosome 14 open reading frame 58
220122_at	398	NM_024717	797	0.0004 73634	3.36	MCTP1	multiple C2-domains with two transmembrane regions 1
64883_at	1508	AI744083	1669	0.0004 78581	1.51	MOSPD2	motile sperm domain containing 2
203279_at	1354	NM_014674	1644	0.0004 79669	-1.24	EDEM1	ER degradation enhancer, mannosidase alpha-like 1
208891_at	1390	BC003143	1573	0.0004 83009	1.98	DUSP6	dual specificity phosphatase 6
205715_at	1372	NM_004334	1636	0.0004 90107	1.60	BST1	bone marrow stromal cell antigen 1
214085_x_at	1451	AI912583	1674	0.0005 10466	1.67	HRB2	HIV-1 rev binding protein 2
215000_s_at	1461	AL117593	1553	0.0005 15959	1.24	FEZ2	fasciculation and elongation protein zeta 2 (zyglin II)
202194_at	1339	AL117354	1556	0.0005 23169	1.36	CGI-100	CGI-100 protein
210202_s_at	1408	U87558	1530	0.0005 32952	-1.46	BIN1	bridging integrator 1

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218181_s_at	1472	NM_017792	1560	0.0005 47382	1.13	MAP4K4	mitogen-activated protein kinase kinase kinase kinase 4
204023_at	1360	NM_002916	1595	0.0005 53752	-1.48	RFC4	replication factor C (activator 1) 4, 37kDa
207872_s_at	1384	NM_006863	1551	0.0005 61178	1.58	LILRA1	leukocyte immunoglobulin-like receptor, subfamily A (with TM domain), member 1
202878_s_at	1349	NM_012072	1622	0.0005 83921	1.42	C1QR1	complement component 1, q subcomponent, receptor 1
201285_at	1333	NM_013446	1649	0.0005 93976	-1.21	MKRN1	makorin, ring finger protein, 1 /// makorin, ring finger protein, 1
211612_s_at	1417	U62858	1529	0.0006 13797	1.80	IL13RA1	interleukin 13 receptor, alpha 1 /// interleukin 13 receptor, alpha 1
219117_s_at	1482	NM_016594	1647	0.0006 16534	-1.61	FKBP11	FK506 binding protein 11, 19 kDa
215761_at	1464	AK000156	1557	0.0006 18215	2.26	RC3	rabconnectin-3
200800_s_at	1328	NM_005345	1642	0.0006 25103	1.65	HSPA1A /// HSPA1B	heat shock 70kDa protein 1A /// heat shock 70kDa protein 1B
202816_s_at	1346	AW292882	1679	0.0006 41212	1.44	SS18	synovial sarcoma translocation, chromosome 18
209440_at	1397	BC001605	1572	0.0006 46328	-1.25	PRPS1	phosphoribosyl pyrophosphate synthetase 1
204221_x_at	1362	U16307	1587	0.0006 54803	1.51	HRB2	HIV-1 rev binding protein 2
57082_at	1506	AA169780	1659	0.0006 58787	-1.41	ARH	LDL receptor adaptor protein
201887_at	1338	NM_001560	1589	0.0006 79598	1.52	IL13RA1	interleukin 13 receptor, alpha 1
215933_s_at	1465	Z21533	1510	0.0006 92626	1.47	HHEX	hematopoietically expressed homeobox
208774_at	1389	AV700224	1685	0.0006 99738	1.20	CSNK1D	Casein kinase 1, delta
201478_s_at	1334	U59151	1533	0.0007 07429	-1.18	DKC1	dyskeratosis congenita 1, dyskerin
208918_s_at	1391	AI334128	1665	0.0007 08781	1.38	FLJ13052	NAD kinase

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208158_s_at	1386	NM_018030	1581	0.0007 26407	1.37	OSBP L1A	oxysterol binding protein-like 1A /// oxysterol binding protein-like 1A
202838_at	1348	NM_000147	1651	0.0007 3801	1.43	FUCA 1	fucosidase, alpha-L- 1, tissue
205039_s_at	1371	NM_006060	1633	0.0007 54874	2.05	ZNFN 1A1	zinc finger protein, subfamily 1A, 1 (Ikaros)
204834_at	1368	NM_006682	1626	0.0007 60238	1.74	FGL2	fibrinogen-like 2
200701_at	1327	NM_006432	1592	0.0007 63959	1.41	NPC2	Niemann-Pick disease, type C2
204254_s_at	1364	NM_000376	1617	0.0007 7458	1.64	VDR	vitamin D (1,25-dihydroxyvitamin D3) receptor
217922_at	1470	AL157902	1562	0.0007 79841	-1.34	MAN 1A2	Mannosidase, alpha, class 1A, member 2
218888_s_at	1479	NM_018092	1586	0.0008 01694	1.39	NETO 2	neuropilin (NRP) and tolloid (TLL)-like 2
210947_s_at	1414	J04810	1514	0.0008 10544	1.30	MSH3	mutS homolog 3 (E. coli)
208923_at	1392	BC005097	1575	0.0008 16109	1.41	CYFI P1	cytoplasmic FMR1 interacting protein 1
209429_x_at	1396	AF112207	1555	0.0008 18155	-1.22	---	---
204566_at	1366	NM_003620	1593	0.0008 23091	1.67	PPM1 D	protein phosphatase 1D magnesium-dependent, delta isoform
218854_at	1478	NM_013352	1630	0.0008 23847	1.47	SART 2	squamous cell carcinoma antigen recognized by T cells 2
213198_at	1441	AL117643	1554	0.0008 36	1.33	ACV R1B	activin A receptor, type IB
218642_s_at	1476	NM_024300	1618	0.0008 49263	-1.21	CHC HD7	coiled-coil-helix-coiled-coil-helix domain containing 7
203645_s_at	1358	NM_004244	1632	0.0008 51877	1.60	CD16 3	CD163 antigen
208117_s_at	1385	NM_031206	1653	0.0008 52389	-1.32	FLJ12 525	hypothetical protein FLJ12525 /// hypothetical protein FLJ12525
218519_at	1474	NM_017945	1594	0.0008 54014	1.30	SLC3 5A5	solute carrier family 35, member A5
217764_s_at	1469	AF183421	1563	0.0008 71159	1.30	RAB3 1	RAB31, member RAS oncogene family
214765_s_at	1459	AK024677	1564	0.0009 19505	1.50	ASAH L	N-acylsphingosine amidohydrolase (acid ceramidase)-like

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212799_at	1432	BE217875	1682	0.00093697	1.31	---	Clone 23570 mRNA sequence
204744_s_at	1367	NM_013417	1624	0.000961709	-1.24	IARS	isoleucine-tRNA synthetase
218526_s_at	1475	NM_014185	1559	0.000964651	-1.22	RAN GNRF	RAN guanine nucleotide release factor
211139_s_at	1415	AF045452	1539	0.000972219	1.40	NAB1	NGFI-A binding protein 1 (EGR1 binding protein 1)
213958_at	1449	AW134823	1677	0.000977101	-1.39	CD6	CD6 antigen /// CD6 antigen
221695_s_at	1494	AF239798	1558	0.000979214	1.37	MAP3 K2	mitogen-activated protein kinase kinase 2 /// mitogen-activated protein k
210176_at	1406	AL050262	1549	0.000989413	1.62	TLR1	toll-like receptor 1
212314_at	1427	AB018289	1542	0.00099	-1.34	KIAA 0746	KIAA0746 protein
209882_at	1402	AF084462	1544	0.000992617	1.42	RIT1	Ras-like without CAAX 1
214500_at	1456	AF044286	1541	0.00102256	1.51	H2AF Y	H2A histone family, member Y
213088_s_at	1439	BE551340	1684	0.00104481	-1.17	DNAJ C9	DnaJ (Hsp40) homolog, subfamily C, member 9
200821_at	1329	NM_013995	1638	0.00104551	1.22	LAM P2	lysosomal-associated membrane protein 2
210732_s_at	1411	AF342816	1574	0.00106818	1.84	LGAL S8	lectin, galactoside-binding, soluble, 8 (galectin 8)
201224_s_at	1332	AU147713	1688	0.00107129	-1.23	SRR M1	serine/arginine repetitive matrix 1
202444_s_at	1343	NM_006459	1634	0.00107236	1.72	C10orf69	chromosome 10 open reading frame 69
206707_x_at	1379	NM_015864	1580	0.00107472	-1.24	C6orf32	chromosome 6 open reading frame 32
221839_s_at	1496	AK026088	1567	0.00107916	-1.34	UBAP 2	ubiquitin associated protein 2
213279_at	1443	AL050217	1547	0.00108194	1.21	DHRS 1	dehydrogenase/reductase (SDR family) member 1
214974_x_at	1460	AK026546	1568	0.00109223	2.67	CXCL 5	chemokine (C-X-C motif) ligand 5
209583_s_at	1399	AF063591	1570	0.00109644	1.65	CD20 0	CD200 antigen
209870_s_at	1401	AW571582	1681	0.00112322	-1.39	APBA 2	amyloid beta (A4) precursor protein-binding, family A, member 2 (X11-like)

<i>Affymetrix ProbeSet ID</i>	<i>SEQ NO ID:</i>	<i>Representative Public ID</i>	<i>SEQ ID NO:</i>	<i>p-value BMS vs typical RRMS</i>	<i>Log Fold Change (BMS vs typical RRMS)</i>	<i>Gene Symbol</i>	<i>Gene description</i>
201494_at	1335	NM_005040	1605	0.0011371	1.27	PRCP	prolylcarboxypeptidase (angiotensinase C)
219806_s_at	1488	NM_020179	1637	0.00113985	1.36	FN5	FN5 protein
208651_x_at	1387	M58664	1513	0.00119019	-1.38	CD24	CD24 antigen (small cell lung carcinoma cluster 4 antigen)
216191_s_at	1466	X72501	1522	0.00132593	-1.84	TRDD3 /// TRD @	T cell receptor delta diversity 3 /// T cell receptor delta locus
217143_s_at	1468	X06557	1511	0.00187286	-1.83	TRDD3 /// TRD @	T cell receptor delta diversity 3 /// T cell receptor delta locus

Table 5. vs. = versus.

Table 6

Genes which when included in classifiers of no more than 10 genes exhibit at least 70% correct classification rates between BMS and typical RRMS patients

<i>Probeset ID</i>	<i>SEQ ID NO:</i>	<i>Gene Symbol</i>	<i>P value (min)</i>	<i>BMS vs RRMS</i>	<i>Sequence Derived From</i>	<i>SEQ ID NO:</i>	<i>Gene Title</i>	<i>Full Length Ref. Sequences</i>	<i>SEQ ID NO:</i>
222204_s_at	1498	RRN3	2.1*10-8	(-)	AL110238	1552	RNA polymerase I transcription factor RRN3	NM_018427	1010
221714_s_at	1495	LOC94431			BC006441	1579	similar to RNA polymerase I transcription factor RRN3	NM_145237	1156
211615_s_at	1418	LRPPRC	2.4*10-8	(-)	M92439	1512	leucine-rich PPR-motif containing protein	NM_133259	1065
211971_s_at	204	LRPPRC			AI653608	606	leucine-rich PPR-motif containing protein	NM_133259	1065
218258_at	173	POLR1D	1.9*10-6	(-)	NM_015972	984	hypothetical protein MGC9850	NM_152705	1695
205789_at	1373	CD1D	4.3*10-6	(+)	NM_001766	1627	CD1D antigen, d polypeptide	NM_001766	1627

<i>Probeset ID</i>	<i>SEQ ID NO:</i>	<i>Gene Symbol</i>	<i>P value (min)</i>	<i>BMS vs RRM S</i>	<i>Sequence Derived From</i>	<i>SEQ ID NO:</i>	<i>Gene Title</i>	<i>Full Length Ref. Sequences</i>	<i>SEQ ID NO:</i>
212999_x_at	1437	HLA-DQB1	9.9*10 ⁻⁶	(+)	AW276186	1678	major histocompatibility complex, class II, DQ beta 1 precursor	NM_002123	1699
209823_x_at	1400	HLA-DQB1			M17955	1518	major histocompatibility complex, class II, DQ beta 1 precursor	NM_002123	1699
209480_at	1398	HLA-DQB1			M16276	1521	major histocompatibility complex, class II, DQ beta 1 precursor	NM_002123	1699
211656_x_at	1420	HLA-DQB1			M32577	1520	major histocompatibility complex, class II, DQ beta 1 precursor	NM_002123	1699
211654_x_at	1419	HLA-DQB1			M17565	1519	major histocompatibility complex, class II, DQ beta 1 precursor	NM_002123	1699
210747_at	1412	HLA-DQB1			M24364	1525	major histocompatibility complex, class II, DQ beta 1 precursor	NM_002123	1699
206584_at	1378	LY96	5.8*10 ⁻⁶	(+)	NM_015364	1650	lymphocyte antigen 96; // MD-2 protein	NM_015364	1650
207359_at	1383	CAMKK2	6.1*10 ⁻⁶	(+)	NM_006549	1590	calcium/calmodulin-dependent protein kinase kinase 2, beta isoform 1	NM_006549	1590
214209_s_at	1452	CAMKK2			BE504895	1683	calcium/calmodulin-dependent protein kinase kinase 2, beta isoform 1	NM_006549	1590
213812_s_at	1446	CAMKK2			AK024748	1565	calcium/calmodulin-dependent protein kinase kinase 2, beta isoform 1	NM_006549	1590

<i>Probeset ID</i>	<i>SEQ ID NO:</i>	<i>Gene Symbol</i>	<i>P value (min)</i>	<i>BMS vs RRM S</i>	<i>Sequence Derived From</i>	<i>SEQ ID NO:</i>	<i>Gene Title</i>	<i>Full Length Ref. Sequences</i>	<i>SEQ ID NO:</i>
210787_s_at	1413	CAMKK2			AF140507	1546	calcium/calmodulin-dependent protein kinase 2, beta isoform 1	NM_006549	1590
212252_at	1425	CAMKK2			AA181179	1660	calcium/calmodulin-dependent protein kinase 2, beta isoform 1	NM_006549	1590
214643_x_at	1458	BIN1	2.2*10-6	(-)	BG034080	1690	bridging integrator 1 isoform 8	NM_004305 // NM_139343	1693 //1694
210202_s_at	1408	BIN1			U87558	1530	bridging integrator 1 isoform 8	NM_004305 // NM_139343	1693 //1694
214439_x_at	1453	BIN1			AF043899	1535	bridging integrator 1 isoform 8	NM_004305 // NM_139343	1693 //1694
202931_x_at	1350	BIN1			NM_004305	1584	bridging integrator 1 isoform 8	NM_004305 // NM_139343	1693 //1694
210201_x_at	1407	BIN1			AF001383	1531	bridging integrator 1 isoform 8	NM_004305 // NM_139343	1693 //1694
214450_at	1454	CTSW	3.8*10-6	(-)	NM_001335	1629	cathepsin W preproprotein	NM_001335	1629
214470_at	1455	KLRB1	3.8*10-6	(-)	NM_002258	1615	killer cell lectin-like receptor subfamily B, member 1	NM_002258	1615
212748_at	1431	MKL1	7.9*10-9	(+)	AB037859	1583	megakaryoblastic leukemia (translocation) 1	NM_020831	1696
209072_at	1393	MBP	2.7*10-5	(-)	M13577	1517	myelin basic protein	NM_002385	1620
210136_at	1404	MBP			AW070431	1676	myelin basic protein	NM_002385	1620
207323_s_at	1382	MBP			NM_002385	1620	myelin basic protein	NM_002385	1620
212978_at	1435	TALRRP	4.1*10-5	(+)	AU146004	1686	T-cell activation leucine repeat-rich protein	NM_015350	1700
212976_at	1434	TALRRP			R41498	1655	T-cell activation leucine repeat-rich protein	NM_015350	1700

<i>Probeset ID</i>	<i>SEQ ID NO:</i>	<i>Gene Symbol</i>	<i>P value (min)</i>	<i>BMS vs RRMS</i>	<i>Sequence Derived From</i>	<i>SEQ ID NO:</i>	<i>Gene Title</i>	<i>Full Length Ref. Sequences</i>	<i>SEQ ID NO:</i>
117_at	1325	HSPA6	8.9*10 ⁻⁵	(+)	X51757	1692	heat shock 70kDa protein 6 (HSP70B')	NM_002155	1609
213418_at	1445	HSPA6			NM_002155	1609	heat shock 70kDa protein 6 (HSP70B')	NM_002155	1609
200610_s_at	1326	NCL	8.9*10 ⁻⁵	(-)	NM_005381	1613	nucleolin	NM_005381	1613
216191_s_at	1466	TRD@	3.7*10 ⁻⁴	(-)	X72501	1522	T cell receptor delta locus	---	#N/A
217143_s_at	1468	TRD@			X06557	1511	T cell receptor delta locus	---	#N/A
213830_at	1447	TRD@			AW007751	1675	T cell receptor delta locus	---	#N/A
213005_s_at	1438	KANK	3.7*10 ⁻⁴	(+)	D79994	1614	kidney ankyrin repeat-containing protein	NM_015158 // NM_153186	1698 // 1697

Table 6. Presented are 38 gene transcripts which correspond for 17 human genes which classify BMS and typical RRMS patients. Genes given by the gene name and description, the Affymetrix probeset identification number, and a representative GenBank Accession numbers. The (-) sign means that the polynucleotide is downregulated (decreased in level) in BMS as compared to typical RRMS subjects; and the (+) sign means that the polynucleotide is upregulated (increased in level) in BMS as compared to RRMS subjects.

Examples of classifiers are presented as following:

- 10 NCL, MKL1, CTSW, KLRB1 and LRPPRC which results in correct classification rate of 79% of BMS and typical RRMS; and NCL, MKL1, CTSW, KLRB1, POLR1D, CD1D, and CAMKK2 which results in correct classification rate of 77% of BMS and typical RRMS.

15 **EXAMPLE 3**

Testing the effect of an anti MS drug in Experimental allergic encephalomyelitis (EAE) - animal MS model

- 20 ***Animal model for multiple sclerosis*** - EAE is induced in female Lewis rats (N = 15; 6-8 weeks old, body weight 180-200 g) by hind footpad subcutaneous inoculation with emulsion of 25 mg guinea-pig MBP (myelin basic protein) in CFA containing 40 mg of *Mycobacterium tuberculosis* (Difco, Detroit, MI) in 0.1 ml oil. Control rats (N =

15) are injected with the same emulsion where saline solution replaces MBP. The EAE is scored as follows: 0 - No obvious changes in motor function on of the rats in comparison to non-immunized rats; 1 - Limp tail; 2 - Limp tail and weakness of hind legs; 3 - Limp tail and complete paralysis of hind legs, or limp tail with paralysis of one front and one hind leg. Or all of: walking only along the edges of the cage, pushing against the cage wall, pushing against the cage wall, spinning when picked up by the tail; 4 - Limp tail, complete hind leg and partial front leg paralysis; 5 - Complete hind and complete front leg paralysis, no movement around the cage, or mouse is spontaneously rolling in the cage, mouse is found dead due to paralysis.

Treatment with an anti MS drug or with an agent which downregulates at least one gene of the RNA polymerase pathway - A therapeutically effective amount of a drug or an agent which downregulates the expression level of a gene involved in the RNA polymerase I pathway (e.g., TPT) is administered to the animal on the day of EAE induction and blood samples are drawn before and after treatment, at predetermined time points which include baseline = Time 0, Day 12 - peak disease, Day 17 and day 21.

Testing the level of expression of genes involved in the RNA polymerase I pathway - Blood samples, obtained from the control and treated animals, are tested using Q-RT-PCR for bio-markers of benign multiple sclerosis (e.g., RRN₃, POLR_{1D} and LRPPRC). Control animals are compared to animals on the peak of EAE disease and to animals treated by the anti MS drug (e.g., TPT).

EXAMPLE 4

IN VITRO TESTING EFFICACY OF A DRUG IN VITRO USING CELLS OF A MULTIPLE SCLEROSIS SUBJECT

Peripheral blood samples are obtained from female subjects with typical RRMS disease course. All patients are free of immunomodulatory or corticosteroid treatments at least 30 days before blood withdrawal. PBMC are extracted from peripheral blood, separated by Ficoll-Hypaque gradient. 15 ml of peripheral blood from patients is diluted 1:1 with Phosphate Buffered Saline (PBS) (without Ca²⁺/Mg²⁺). Blood samples are underlied with 10 ml of Ficoll- Lymphoprep (*Axis Chield, Norway*) and spinned (*Eppendorf centrifuge, Germany*) at 2300 RPM for 30 minutes. PBMCs are collected,

washed with PBS and counted and incubated at 37 °C in a humidified CO₂ incubator with or without anti MS drugs. After incubation, total RNA is extracted using both Trizol (Invitrogen, USA) and Phase-Lock-Gel columns (Eppendorf, Germany) including a DNase digestion step. RNA integrity is assessed by RNA Experion automated electrophoresis system.

Treatment with an anti MS drug or with an agent which downregulates at least one gene of the RNA polymerase pathway – A therapeutically effective amount of a drug or an agent which downregulates the expression level of a gene involved in the RNA polymerase I pathway (e.g., TPT) is incubated during 3 hours with the PBMC (of the subject having typical RRMS) and compared with PBMC (of the subject having typical RRMS) incubated during 3 hours without drug or with placebo excluding therapeutic component.

Testing the level of expression of genes involved in the RNA polymerase I pathway – Total RNA samples, obtained from PBMC of a subject having typical RRMS, are tested in-vitro with anti-RNA polymerase I agents (e.g. TPT), are tested using Q-RT-PCR for bio-markers of benign multiple sclerosis (e.g., RRN₃, POLR₁D and LRPPRC). The results compared with PBMC from same patients incubated without drug or with placebo.

Although the invention has been described in conjunction with specific embodiments thereof, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, it is intended to embrace all such alternatives, modifications and variations that fall within the spirit and broad scope of the appended claims.

All publications, patents and patent applications mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present invention. To the extent that section headings are used, they should not be construed as necessarily limiting.

REFERENCES*(Additional references are cited in text)*

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WHAT IS CLAIMED IS:

1. A method of classifying a subject as being more likely to have benign multiple sclerosis (BMS) or as being more likely to have typical relapsing remitting multiple sclerosis (RRMS), the method comprising comparing a level of expression of at least one gene involved in the RNA polymerase I pathway in a cell of the subject to a reference expression data of said at least one gene obtained from a cell of at least one subject pre-diagnosed as having BMS and/or from a cell of at least one subject pre-diagnosed as having typical RRMS, thereby classifying the subject as being more likely to have BMS or as being more likely to have typical RRMS.

2. A method of diagnosing a subject pre-diagnosed with multiple sclerosis (MS) as having benign multiple sclerosis (BMS) or typical relapsing remitting multiple sclerosis (RRMS), the method comprising:

(a) classifying the subject as being more likely to have BMS or as being more likely to have typical RRMS according to the method of claim 1,

(i) wherein when the subject is classified as being more likely to have said BMS then the subject is diagnosed as having BMS;

(ii) wherein when the subject is classified as being more likely to have said typical RRMS, then the subject is diagnosed as having typical RRMS; and

(c) informing the subject of the diagnosis,
thereby diagnosing the subject pre-diagnosed with the MS as having the BMS or the typical RRMS.

3. A method of monitoring an efficiency of an anti multiple sclerosis (MS) drug in treating a subject diagnosed with a typical relapsing remitting multiple sclerosis (RRMS) course, the method comprising:

(a) treating the subject with the anti MS drug; and
(b) comparing a level of expression of least one gene involved in the RNA polymerase I pathway in a cell of the subject following said treating with the anti MS drug to a level of expression of said at least one gene in a cell of the subject prior to said treating the subject with the anti MS drug,

(i) wherein a decrease above a predetermined threshold in said level of expression of said at least one gene following said treating with the anti MS drug relative to said level of expression of said at least one gene prior to said treating with the anti MS drug indicates that the anti MS drug is efficient for treating the subject;

(ii) wherein an increase above a predetermined threshold in said level of expression of said at least one gene following said treating with the anti MS drug relative to said level of expression of said at least one gene prior to said treating with the anti MS drug indicates that the anti MS drug is not efficient for treating the subject; or

(iii) wherein when a level of expression of said at least one gene following said treating with the anti MS drug is identical or changed below a predetermined threshold as compared to prior to said treating with the anti MS drug then the treatment is not efficient for treating the subject

thereby monitoring the efficiency of the anti multiple sclerosis (MS) drug in treating the subject diagnosed with the typical RRMS course.

4. An *in vitro* method of predicting an efficiency of an anti multiple sclerosis (MS) drug for treatment of a subject diagnosed with a typical relapsing remitting multiple sclerosis (RRMS), the method comprising:

(a) contacting cells of the subject with a therapeutically effective amount of the anti MS drug; and

(b) comparing a level of expression in said cells of at least one gene involved in the RNA polymerase I pathway following said contacting with the anti MS drug to a level of expression of said at least one gene in said cells prior to said contacting with the anti MS drug,

(i) wherein a decrease above a predetermined threshold in said level of expression of said at least one gene following said contacting with the anti MS drug relative to said level of expression of said at least one gene prior to said contacting with the anti MS drug indicates that the treatment is efficient for treating the subject;

(ii) wherein an increase above a predetermined threshold in said level of expression of said at least one gene following said contacting with the anti MS drug relative to said level of expression of said at least one gene prior to said contacting with the anti MS drug indicates that the treatment is not efficient for treating the subject; or

(iii) wherein when a level of expression of said at least one gene following said contacting with the anti MS drug is identical or changed below a predetermined threshold as compared to prior to said contacting with the anti MS drug then the treatment is not efficient for treating the subject

thereby predicting the efficiency of the anti MS drug for treatment of the subject diagnosed with the typical RRMS.

5. A method of treating a subject diagnosed with multiple sclerosis, the method comprising

(a) classifying the subject as being more likely to have BMS or typical RRMS according to the method of claim 1,

(b) selecting a treatment regimen based on classification results of step (a);
thereby treating the subject diagnosed with multiple sclerosis.

6. A method of treating a subject diagnosed with multiple sclerosis, the method comprising:

(a) diagnosing a typical relapsing remitting multiple sclerosis (RRMS) according to the method of claim 2,

(b) administering to the subject a therapeutically effective amount of diterpenoid triepoxide Triptolide (TPT) or a derivative thereof, thereby treating the subject.

7. A probeset comprising a plurality of oligonucleotides and no more than 50 oligonucleotides, wherein an oligonucleotide of said plurality of oligonucleotides specifically recognizes a polynucleotide of at least one gene involved in the RNA polymerase pathway.

8. A kit for classifying a disease course in a subject diagnosed with multiple sclerosis (MS), comprising the probeset of claim 7.

9. A method of selecting a drug for treating a typical relapsing remitting multiple sclerosis (RRMS) in a subject, the method comprising:

contacting cells of a subject classified as having a typical RRMS with a plurality of drug molecules,

identifying at least one drug molecule which downregulates a level of expression of at least one gene involved in the RNA polymerase I pathway, said at least one drug molecule is suitable for treating the typical RRMS in the subject,

thereby selecting the drug for treating the typical RRMS in the subject.

10. The method of claim 1, 2 or 5, wherein a decrease above a predetermined threshold in said level of expression of said at least one gene in said cell of the subject relative to said reference expression data of said at least one gene obtained from said at least one subject having said typical RRMS classifies the subject as being more likely to have the BMS.

11. The method of claim 1, 2 or 5, wherein an increase above a predetermined threshold in said level of expression of said at least one gene in said cell of the subject relative to said reference expression data of said at least one gene obtained from said at least one subject having said BMS classifies the subject as being more likely to have the typical RRMS.

12. The method of claim 1, 2 or 5, wherein when a level of expression of said at least one gene in said cell of the subject is identical or changed below a predetermined threshold as compared to said reference expression data of said at least one gene obtained from said at least one subject having said BMS, then the subject is classified as being more likely to have the BMS.

13. The method of claim 1, 2 or 5, wherein when a level of expression of said at least one gene in said cell of the subject is identical or changed below a predetermined threshold as compared to said reference expression data of said at least one gene obtained from said at least one subject having said typical RRMS, then the subject is classified as being more likely to have the typical RRMS.

14. The method of claim 5, wherein when the subject being more likely to

have typical RRMS then said treatment regimen comprises administering to the subject an agent which downregulates the level of expression of said at least one gene involved in said RNA polymerase I pathway.

15. The method of any of claims 1-6 and 9-14, the probeset of claim 7 or the kit of claim 8, wherein said at least one gene involved in said RNA polymerase 1 pathway is selected from the group consisting of POLR1D, LRPPRC, RRN3 and NCL.

16. The method of any of claims 1-6 and 9-14, the probeset of claim 7 or the kit of claim 8, wherein said at least one gene involved in said RNA polymerase 1 pathway comprises the POLR1D, LRPPRC, RRN3 and NCL genes.

17. The method of claim 6 or 14, wherein said agent is selected from the group consisting of an siRNA, an antisense, an antibody and a small molecule.

18. The method of claim 17, wherein said small molecule is Cycloheximide.

19. The method of claim 6 or 14, wherein said at least one gene is RRN3, and whereas said downregulating is effected using diterpenoid triepoxide Triptolide (TPT) or a derivative thereof.

20. The method of claim 6 or 14, wherein said at least one gene is RRN3, and whereas said downregulating is effected using Cycloheximide.

21. The kit of claim 8, further comprising a positive control for an expression level of said at least one gene involved in the RNA polymerase pathway.

22. The probeset of claim 7 or the kit of claim 8, wherein each of said plurality of oligonucleotides is bound to a solid support.

23. The probeset or the kit of claim 22, wherein said plurality of oligonucleotides are bound to said solid support in an addressable location.

24. The method of any of claims 1-6 and 9-14, wherein said level of expression is determined using an RNA detection method.

25. The method of any of claims 1-6 and 9-14, wherein said level of expression is determined using a protein detection method.

26. The method of any of claims 1-4 and 9-14, wherein said cell is a blood cell.

27. The method of claim 5 or 6, further comprising administering to the subject a therapeutically effective amount of an anti MS agent.

28. The method of claim 3, 4 or 27, wherein said anti-MS agent is selected from the group consisting of Diterpenoid triepoxide Triptolide (TPT), Adderall, Ambien, Avonex, Baclofen, Beta interferon, Betaseron, Celexa, Clonazepam, Copaxone, Corticosteroids, Cymbalta, Cytoxan, Dexamethasone, Effexor, Elavil, Gabapentin, Hydrocodone, Lexapro, Lyrica, Mitoxantrone, Naltrexone, Neurontin, Novantrone, Prednisone, Provigil, Rebif, Solumedrol, Symmetrel, Topamax, Tysabri, Wellbutrin, Xanax, Zanaflex, Zoloft, fingolimod, laquinimod, Mylinax (cladribine), teriflunomide, BG-12 (Biogen Idec's), firtagrast (GSK/Mitsubishi Tanabe Pharma), ibudilast (MediciNova's), and CDP323 (Biogen/UCB).

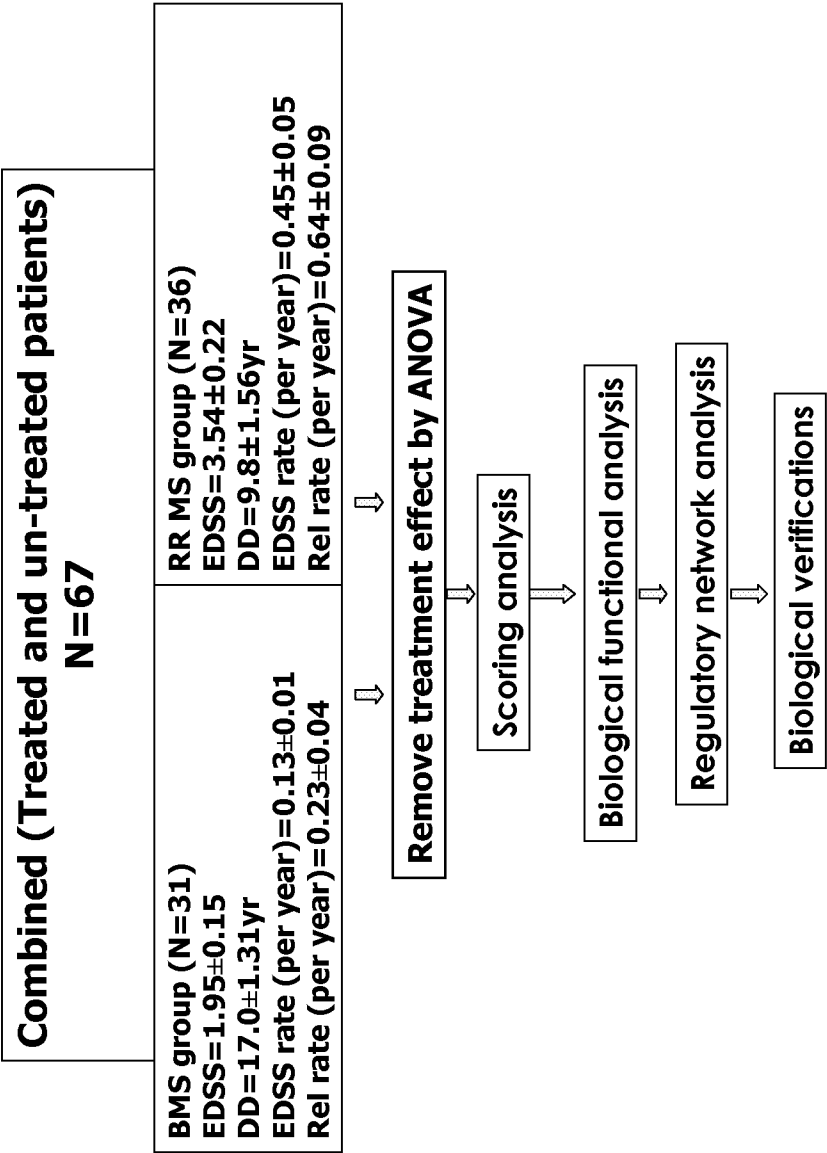


FIG. 1

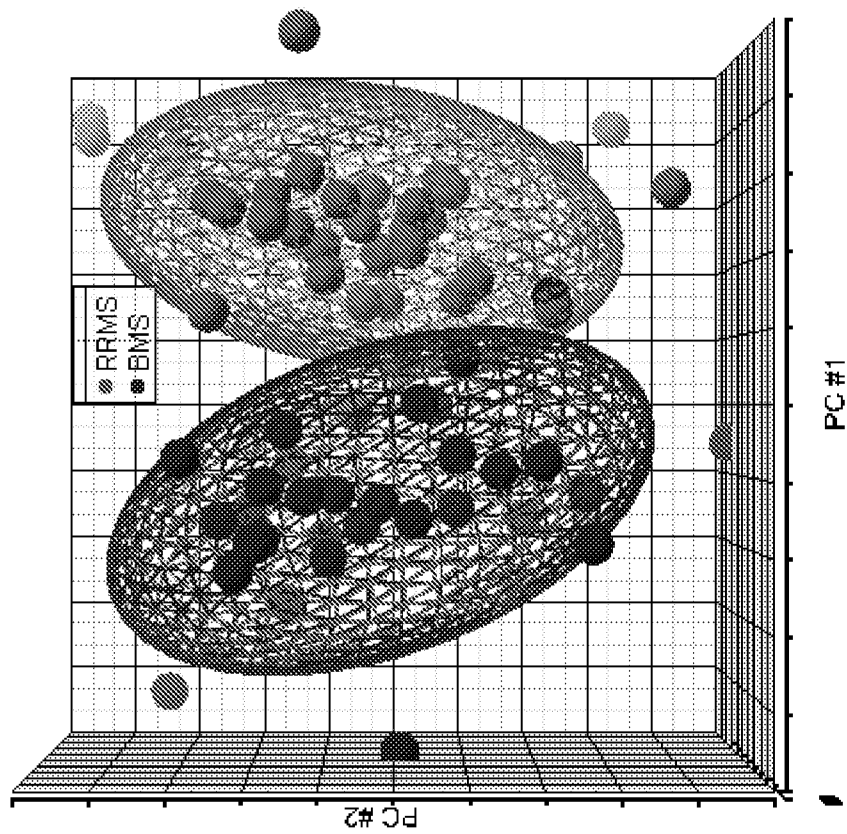


FIG. 2

INTERNATIONAL SEARCH REPORT

International application No

PCT/IB2010/051344

A. CLASSIFICATION OF SUBJECT MATTER

INV. C12Q1/68

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

C12Q

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practical, search terms used)

EPO-Internal, BIOSIS, Sequence Search, EMBASE, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2008/081435 A2 (TEL HASHOMER MEDICAL RES INFRA [IL]; ACHIRON ANAT [IL]; GUREVICH MICHA) 10 July 2008 (2008-07-10) cited in the application claim 21; example 2 ----- -/--	1,2,5,7, 8,10-15, 21-26



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier document but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art.

"&" document member of the same patent family

Date of the actual completion of the international search

4 June 2010

Date of mailing of the international search report

18/06/2010

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Authorized officer

Santagati, Fabio

INTERNATIONAL SEARCH REPORT

International application No

PCT/IB2010/051344

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
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